

Program overview

12-Sep-2023 13:20

Year 2023/2024
Organization Technology, Policy and Management
Education Master Complex Systems Engineering and Management

Code	Omschrijving	ECTS	p1	p2	p3	p4	p5
Master CoSEM 2023							
MSc CoSEM 2023							
1e jaar CoSEM 2023							
SEN111A	Introduction to Design in Complex Systems	2					
SEN1121	Complex Systems Engineering	5					
SEN1131	Institutional Economics for Designing in Socio-technical Systems	3					
SEN1141	Managing Multi-actor Decision-making	5					
SEN115B	Law and Institutions	5					
SEN1161	Design Project	5					
SEN131B	CoSEM Research Challenges	5					
Compulsory Choice							
SEN1211	Agent-based Modelling	5					
SEN1221	Statistical Analysis of Choice Behaviour	5					
Compulsory Choice							
SEN1231	Mixed Research Methods for Multi-actor Systems	5					
SEN124A	Design in Networked Systems	5					
Track E							
SEN1511	Engineering Optimization and Integrating Renewables in Electricity Markets	5					
SEN1522	Electricity and Gas: Market Design and Policy Issues	5					
SEN1531	Design of Integrated Energy Systems	5					
SEN1541	Sociotechnology of Future Energy Systems	5					
Track I&C							
SEN1611	I&C Architecture Design	5					
SEN1622	I&C Service Design	5					
SEN163B	Responsible Data Analytics	5					
SEN1641	Digital Platform Design	5					
Track T&L							
SEN171A	Advanced Evaluation Methods for Transport Policy Decision-making	5					
SEN1721	Travel Behaviour Research	5					
SEN173A	Analysis and Design of Freight and Logistic Systems	5					
SEN1741	Innovations in Transport and Logistics	5					
2e jaar CoSEM 2023							
SEN232A	Master Thesis Research Design	5					
SEN2331	CoSEM Master Thesis	30					
Track Electives CoSEM (10 EC)							
TPM593A	Internship TPM	10					
Master Specialisaties							
Emerging Technology-based Innovation and Entrepreneurship							
TPM029A	Responsible Innovation	3					
TPM033A	Sustainable Innovation and Transitions	3					
TPM959A	Technology Battles	3					
Electives							
TPM406A	Corporate Entrepreneurship and Start-ups	6					
TPM416A	Turning Technology into Business	6					
ICT Design and Management							
SEN1611	I&C Architecture Design	5					
SEN1622	I&C Service Design	5					
TPM409A	Digital technology entrepreneurship and management	5					
Economics and Finance							
TPM021A	Economics and Finance	6					
TPM022A	Intermediate Economics	5					
TPM023B	Cost-Benefit Analysis: Theory and Application	4					
Advanced Modelling Gaming and Design							
SEN9110	Simulation Masterclass	5					
SEN9115	Participatory Systems	5					
SEN9235	Game Design Project	5					
Supply Chain Management							
SEN9720	Logistics and Supply Chain Innovations	5					

SEN9725	Supply Chain Gaming		
TPM028A	Decision Making in Multimodal Transport Systems	5	
TPM040A	Logistics Systems Engineering	5	
Cyber Security			
Compulsory (three courses of the following)		This specialization includes three courses, which can be chosen from the following	
TPM010A	Cyber Crime Science	5	
TPM020B	Economics of Cybersecurity	5	
TPM025A	User-Centred Security	5	
TPM027A	Cyber Risk Management	5	
TPM030A	Introduction to Cloud as Infrastructure: The effects of the new business of computing on practice	5	
AI Specialisation			
EPA and CoSEM students			
SEN9110	Simulation Masterclass	5	
TPM034A	Machine learning for socio-technical systems	5	
Electives			
TPM014B	Ethics of AI	5	
TPM038A	Artificial Intelligence and Cybernetic Justice	5	
MOT students			
EPA122A	Spatial Data Science	5	
EPA123A	Introduction to TPM Modelling	5	
Electives			
TPM014B	Ethics of AI	5	
TPM038A	Artificial Intelligence and Cybernetic Justice	5	
Water and Delta Systems			
TPM003A	Water Ethics	5	
TPM026A	System Reliability in Quantitative Risk Assessment	4	
TPM801A	Introduction to Water Governance in Deltas	5	
TPM802A	Model-based Water Systems Assessment	5	
TPM803A	Building with Nature and Beyond	5	
TPM804A	Adaptive Planning for Management of Water Systems in Deltas	5	

Year	2023/2024
Organization	Technology, Policy and Management
Education	Master Complex Systems Engineering and Management

Master CoSEM 2023	
Director of Studies	Dr.ir. G. Bekebrede
Program Title	Complex Systems Engineering & Management
Director of Education	I. Bouwmans (Ivo)
Prerequisites	For additional information check: https://cosem.nl/
Introduction 1	https://www.tudelft.nl/studenten/faculteiten/tbm-studentenportal/onderwijs/master/welcome-new-msc-students

Year	2023/2024
Organization	Technology, Policy and Management
Education	Master Complex Systems Engineering and Management

MSc CoSEM 2023

Year	2023/2024
Organization	Technology, Policy and Management
Education	Master Complex Systems Engineering and Management

1e jaar CoSEM 2023	
Director of Studies	Dr.ir. G. Bekebrede
Director of Studies	Dr.ir. I. Bouwmans

SEN111A	Introduction to Design in Complex Systems	2
Module Manager	Dr.ir. M.P.M. Ruijgh-van der Ploeg	
Instructor	Dr. P.W.G. Bots	
Instructor	Dr.ir. I. Bouwmans	
Instructor	Dr. M.L.C. de Bruijne	
Instructor	Prof.dr. M.E. Warnier	
Instructor	mr. J.M. Kooijman	
Instructor	Dr.ir. R. van Bergem	
Instructor	Dr. B.J. Pearce	
Responsible for assignments	Dr.ir. M.P.M. Ruijgh-van der Ploeg	
Co-responsible for assignments	Dr.ir. I. Bouwmans	
Contact Hours / Week x/x/x/x	40/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	English	
Required for	This is the first course in the Fundamentals learning line that consists of courses SEN 1121, 1131, 1141, 1151, 1161.	
Expected prior knowledge	Must have been admitted to COSEM programme	
Course Contents	This is an intensive, structured one-week programme. The aim of the activities in this course is to engage you in forming a learning community with your peers and instructors, to introduce you to the TPM mindset, and to support you in using and enlarging the skill set that is needed for designing in socio-technical systems. As such, this course is an introduction to the other design-related courses in the COSEM MSc programme: SEN1121, 1131, 1141, 1151 and 1161. The course consists of a mix of lectures and collaborative project work or assignments. Students build on knowledge and skills from their Bachelor education and enrich these. Lectures introduce the basic vocabulary, concepts, analytical and design methods that are useful for designing technical structures, institutions and stakeholder processes. Daily assignments are related to a real-world design problem and challenge students to apply these concepts and methods. On Friday teams present their final assignment to peers and instructors.	
Study Goals	After completion of the course, the student is able to: a) Explain what socio-technical systems are and what designing in a socio-technical system entails; b) Use concepts such as structure, process, context and driver in relation to systems design and the system life cycle; c) Apply basic techniques for conceptual design, especially identifying and formulating objectives, functions and requirements, generating alternatives and evaluating alternatives; d) Explain what an institution is, what institutional design entails, and how the designing of institutions differs from designing technical structures; e) Explain what a stakeholder process is, what process design entails, and how the designing of processes differs from designing technical and institutional structures; f) Can reflect on the complexity of the case that was used to practice designing.	
Education Method	This course requires full attendance during all scheduled hours (8:30-17:30 , WET/GST +1). Students work in teams and under supervision. Hands-on learning through a mix of lectures and collaborative assignments. Attendance of the on-campus social functions is expected. The course has been set up for on-campus education. We will switch to online teaching and collaboration in TEAMS if TU Delft regulations would limit or prohibit access to the campus classrooms.	
Computer Use	Each student must have a lap top available that is suited for online education and online collaboration, including audio-visual communication in online environments such as Skype, Zoom, Teams or Discord.	
Literature and Study Materials	1) C.L. Dym, P. Little, E.J. Orwin (2014). Engineering Design: A Project Based Introduction, 4th edition, John Wiley & Sons, Inc.* 2) Course handout "Designing in socio-technical systems" dated 2020 3) Course slides and texts published in Brightspace. *Please take note of the correct edition. This book is also available in a digital version.	
Assessment	A final grade is based for 75% on results of an individual, written examination and for 25% on the results of team work assignments. These are administered in Osiris as Exam grade and Bootcamp Project grade. The grade for team work is based on the results of a cumulative, final assignment named Assignment 4. We administer the Bootcamp Project grade in Osiris only if the team has received a pass for assignments 1A, 1B, 2 and 3. In the case of an insufficient grade for group work (<5 for the graded assignment or a fail on pass/fail assignments 1A, 1B, 2 and 3), teams must resubmit the failed or missed assignment that was given to the team. The deadline for this is Friday of week 1.3. If the resubmitted pass/fail assignments 1, 2 or 3 are of sufficient quality, then a pass grade will be given in gradebook. If the resubmitted assignment 4 is of sufficient quality, then a grade of 5,5 will be awarded to the team and administered as the Bootcamp Project grade in Osiris. The group work is mandatory. The attendance policy is published on Brightspace. In case of absenteeism: Some students may be asked to make an additional assignment to make up for begin absent during the week. The Bootcamp Project grade will be administered in Osiris for this individual student if the additional assignment has been given a 'pass.' A closed book exam will be administered on-campus. If the exam must be administered as an on-line exam (due to TU wide decision), the exam will be an open book exam. Exam will be administered in week 1.4 on Tuesday afternoon The retake will be administered in week 1.10.	
Exam Hours	2 hrs. Exam is administered in week 4 and retake in week 10.	
Module Material	These include the literature and study materials. Except for the book, all materials will be posted on Brightspace.	
Targetgroup	Obligatory, introductory course for COSEM programme. This course is open only to students who have been admitted to the CoSEM programme before the start of the course.	

SEN1121	Complex Systems Engineering	5
Module Manager	Prof.dr. F.M. Brazier	
Instructor	Y. Huang	
Instructor	Prof.dr. F.M. Brazier	
Instructor	Dr. P.H.G. van Langen	
Responsible for assignments	Prof.dr. F.M. Brazier	
Co-responsible for assignments	Y. Huang	
Contact Hours / Week x/x/x/x	4/0/0/0 and interactive lab sessions	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	The course introduces design thinking in complex sociotechnical systems in line with the core orientation of the CoSem MSc programme. Students learn about designing complex technological systems in multi-actor environments. Confronted with a challenge students explore the context, values and requirements of the different stakeholders involved (both directly and indirectly). Students learn to structure and coordinate a design process: creating, exploring and evaluating potential designs in relation to requirements given a mission statement. The course provides the foundation for further design-oriented courses. Students learn to - explore the context of a challenge in a multi-actor setting - understand stakeholder values and requirements - formulate a mission statement for a complex sociotechnical system - structure a design process, creating and evaluating potential design options - manage their design project Students participate in online interactive lectures and work together in groups (and with each other) on a specific challenge to put the knowledge into practice: learning by experience.	
Study Goals	After completion of the course, a student is able to - use and differentiate between concepts and terminology related to the design of complex sociotechnical systems - analyze challenges in complex sociotechnical systems in multi-actor environments - structure and coordinate a design process for a complex sociotechnical system - use and combine methods for requirements acquisition and analysis - use and combine approaches for conceptual design - use and compare methods for systems design and engineering - use and compare methods for project and risk management	
Education Method	Learning by participation: online interactive lectures (for which participation is essential), and labs on campus in which groups work together and with each other supported by TAs and teaching staff.	
Literature and Study Materials	The main content of this course is provided on the slides used during the lecture, together with papers, articles and book chapters as appropriate. These reading materials are made available on Brightspace.	
Assessment	The overall grade for this course consists of 2 sub grades: 1) Exam: The exam focuses both on the theoretical knowledge and the ability to apply and reflect on the theory taught. The digital exam is open book and contributes to 70% of the final grade.. 2) Assignments: During the labs, theory from the lectures is applied to a practical case. The assignments contribute to 30% of the final grade. Both parts must be completed with a passing grade (at least 5.75).	
Tags	Analysis Challenging Design Group Dynamics/Project Organisation Group work Modelling Projects Sustainability	

SEN1131	Institutional Economics for Designing in Socio-technical Systems	3
Module Manager	Dr. A.F. Correlje	
Instructor	Dr. A.F. Correlje	
Instructor	Dr.ir. R. van Bergem	
Instructor	G. Tran	
Instructor	Dr. C.C.R. Penna	
Co-responsible for assignments	S. Renes	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	This course provides fundamental concepts from the field of Institutional Economics that are instrumental for the design of institutions in socio-technical systems. Relevant frameworks and theories will be elaborated in relation to the design of socio-technical systems. The fundamentals of institutional economics will be applied to concrete design challenges, like the transition to sustainable infrastructures. It will be demonstrated how to relate Institutional Economics to interdisciplinary research challenges, including engineering, political science, law and philosophy. This provides interesting perspectives to other courses in the curriculum. Typical questions that are addressed in this module include: What is the specific contribution of Institutional Economics to the design of socio-technical systems? How to frame and interpret design challenges from an Institutional Economic perspective? What is an institutional design? How can institutional designs be related to the technical design of complex systems? How to operationalize concepts from institutional economics and apply them to empirical design challenges?	
Study Goals	After the completion of the course, the student is able to: - to understand and analyse complex institutional arrangements from an Institutional Economic perspective - reflect on the possible contribution of Institutional Economics in addressing different design challenges - apply approaches from Institutional Economics to concrete design challenges - position and apply different approaches from Institutional Economics to interdisciplinary design challenges. - establish a coherent institutional design given the features of the technical system(s) embedded in their socio-political environment	
Education Method	Lectures and workshops	
Literature and Study Materials	Reader of relevant study material, including cases to be discussed in the workshops	
Assessment	On-campus: Digital exam. In case of unforeseen circumstances or measures resulting from COVID-19, the prescribed assessment will be modified as follows: online: timed take-home exam (open book)	

SEN1141	Managing Multi-actor Decision-making	5
Module Manager	Dr. M.L.C. de Bruijne	
Instructor	Prof.mr.dr. J.A. de Bruijn	
Instructor	Dr. M. Leijten	
Instructor	Dr.ir. E. Minkman	
Instructor	F. Hirschhorn Zonana	
Instructor	Dr. F.S. Gürses	
Instructor	B. Wagner	
Responsible for assignments	Dr. M.L.C. de Bruijne	
Co-responsible for assignments	B. Wagner	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	Previous courses have focused on, first, diagnosing the complexity of socio-technical systems and identifying problems and, then, designing for solutions (which oftentimes have a large engineering component) to be introduced and their accompanying institutions in the midst of this complexity. This course on multi-actor decision-making goes one step further: (reflecting on) realizing change. We focus on the actors in the system. How do they behave, individually and collectively? What does the system look like from an actor perspective? Why does intentional change in socio-technical systems often appear so hard? What limits the designability of systems? By studying characteristics of sociotechnical systems and identifying the complexity of the challenges they pose for engineering (e.g. the increased importance of IT for design, the increased interdependencies between systems) as well as the specific actor characteristics (e.g. the global character of the manufacturers and supply chains, issues of plurality and inclusiveness) we identify key factors that affect intentional change (i.e. the wickedness of problems, public values, strategic behaviour of actors and dynamics of the system). How do these factors affect decision-making, design and change of complex engineering systems? How can we assess the consequences of this decision-making complexity and how does this affect the potential for interventions for (re)design in these systems in deliberate and responsible ways? In this course, we identify key characteristics which affect multi-actor decision making in socio-technical systems. Furthermore, we assess how and under which conditions process managerial tactics may help to manage multi-actor decision-making with its plural rationalities and dynamics of negotiation and design.	
Study Goals	At the end of this course, students will be able to - explain why actors in networks behave as they do, and how their behaviour evolves - describe different network structures and their practical implications - assess the designability of institutions, systems, policies and technologies - argue why certain process managerial tactics might work under which conditions - create case-specific process managerial tactics to anticipate complex decision-making. - assess the quality of process designs - reflect on multi-actor decision-making in light of broader scientific and public policy trends and debates	
Education Method	Lectures Assignments (group and individual) Readings	
Assessment	Assessments group assignment analysing a multi-actor decision making process (1/2) individual assignments which enable students to reflect on prescribed literature (1/2)	

SEN115B	Law and Institutions	5
Module Manager	mr. J.M. Kooijman	
Instructor	Dr. A.F. Correlje	
Instructor	Dr. R.S. van Wegberg	
Co-responsible for assignments	Dr. A.F. Correlje	
Contact Hours / Week x/x/x/x	0/0/2x45 min/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	Suppose we want to switch to a hydrogen based economy in which fuel cell cars are used as power plants. Or suppose we want a fully automated inland terminal. And what about the construction of multifunctional dikes? You cannot just realize that but you have to be aware of the relevant legal and institutional framework. That framework will enable you to make things happen, but it can also contain constraints. This course focuses on the legal and institutional framework to take into account with that kind of projects. Are there for instance legal instruments to force people to switch to fuel cell cars? And to what institutional layer do those instruments belong? Are they easy to change? Or: can the government subsidize the transition to a fully automated inland terminal? What are the constraints? And can we work around those constraints? Or: what exactly has to be studied for the environmental impact assessment regarding the dike? Does an environmental impact assessment have to be conducted at all? Students form groups and pick a case. Then, the group will present its case in an elevator pitch. On the basis of those pitches, we will dive into the world of law in order to see what defines the design space for those projects, what areas of law are relevant, whether the law forms a barrier or not and how the law can be used to realize those projects. Next to the group assignments there will be an individual exam.	
Study Goals	At the end of the course, students are able to: - Relate legal instruments to the institutional framework - Work with the law as a system - Relate the legal instruments to the institutional framework - Identify the legal borders of the design space regarding a socio technical system - Identify relevant law for a socio technical system - Apply relevant law to a socio technical system and identify barriers and opportunities - Deal with legal language	
Education Method	Interactive lectures.	
Assessment	In order to pass the course, the students write a group paper and take an individual exam. The paper is based on the institutional and legal framework, the legal barriers and the legal opportunities regarding a specific case. The individual exam will be (if possible) conducted in ANS. The final grade is based on: - a group elevator pitch (10%) - two peer reviews (P/F) - a final paper (40%) - an individual exam (50%) There will be a resit possibility for the pitch and final paper when the obtained grade is lower than 6. There will be a general resit possibility for the individual exam. Every partial result will be registered in Osiris and can therefore be taken to the next academic year.	

SEN1161	Design Project	5
Responsible Instructor	Dr. B.J. Pearce	
Module Manager	Dr.ir. M.P.M. Ruijgh-van der Ploeg	
Module Manager	Dr. B.J. Pearce	
Instructor	Dr. M.L.C. de Bruijne	
Instructor	Dr. A.F. Correlje	
Instructor	Dr.ir. R. van Bergem	
Instructor	Dr. B.J. Pearce	
Responsible for assignments	Dr.ir. M.P.M. Ruijgh-van der Ploeg	
Co-responsible for assignments	Dr. M.L.C. de Bruijne	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	This course is a ten-week, capstone project; students work in teams of 5 persons. The module-manager forms the teams based on students preferences and in Q3 (week 3.8). Teams work under supervision during weekly workshops and meet at other times during the week also. Students are encouraged to also consult with experts and stakeholders from the field. Student teams receive a case: the implementation of a recent policy decision that is related to energy transition, mobility, or digitization in the Metropole the Hague-Rotterdam. Starting in week 1, the design team analyses the problem situation, frames the design problem, and then ideates, prototypes, evaluates and communicates suitable interventions. The final product comprises the lay-out of an intervention that contributes substantially to policy implementation. This includes basic but well-aligned technical and institutional designs, a design for a relevant multi-actor process (including a roadmap and process rules), and products needed to communicate these designs. Deliverables are (1) a team report that documents all steps in the design process, including products that inform stakeholders about the results of the design process. (2) an individual, written assignment with questions that prompt reflection on the design process.	
Study Goals	After completion of the course, the student should be able to design interventions that solve (socio-technical) problems which are incompletely stated, as well as: LO 1. Frame and define a complex design problem in terms of a much needed intervention, with a whole systems- and multi-stakeholder perspective LO 2. Ideate interventions and generate alternative, conceptual designs for the technical, institutional and multi-actor process aspects of an intervention LO 3. Prototype and evaluate the impact and unintended consequences of an intervention LO 4. Create materials for communicating the intervention to stakeholders LO 5. Evaluate the teams strategies for phasing of the design process, including choice and use of available methods, tools, theory LO 6. Evaluate how the team made decisions based on acquired knowledge and theory throughout the design process LO 7. Communicate the process of analysis and designing orally, in writing and graphically	
Education Method	Project-based learning. Students work in teams of 5-6 people. Their process of problem analysis, framing, ideating prototyping, evaluating and communicating is supported with weekly workshops and meetings with supervisors.	
Course Relations	This is the capstone project in which students are asked to apply knowledge and skills from the 4 courses in the Foundations Line: SEN111a, SEN1121, SEN1131, SEN1141, and SEN1151	
Literature and Study Materials	Study materials include all materials published on Brightspace. Students may also use the materials that were provided in the courses SEN111, SEN112, SEN113, SEN114 and SEN115.	
Prerequisites	SEN1121, SEN1131 en SEN1141 (grades >5)	
Assessment	The guidelines and deadlines for submission of assignments will be published on Brightspace. The final grade is calculated based on the grades that were awarded to two assignments: a group grade for a team report (80% of final grade, >5,5) and a grade for an individual assignment (20% of final grade, >5,5). The team report will be evaluated on learning objectives L1-5. If the team report is of insufficient quality (not all learning objectives are met and/or guidelines were not followed), then a repair assignment is given. This is an individual assignment. The maximum grade given to a repair assignments is 6. Students must retake the course in the next study year if (a) they fail the repair assignment or (b) if they did not succeed to participate in the team effort and make a fair contribution to the report. An individual assignment will be evaluated with respect to L6-L8 (20% of final grade, >5,5). This assignment will be given during the exam week and can be considered as take-home exam with reflective questions about the design process. No retakes.	

SEN131B	CoSEM Research Challenges	5
Module Manager	Dr. J. Ubacht	
Instructor	Prof.dr. G.P. van Wee	
Instructor	S. Steinert	
Instructor	Dr. J.J. Casey	
Co-responsible for assignments	Prof.dr. G.P. van Wee	
Contact Hours / Week x/x/x/x	0/0/0/2-4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Required for	Mandatory course for first year CoSEM students. International visiting students are welcome to join too, please contact the module manager to discuss the match between your interests and the course.	
Expected prior knowledge	For this course, the student is expected to: 1. know how to find scientific articles, using e.g. Web of Science or Scopus (these are library information skills); 2. know how to store references, e.g. by making use of a reference database such as EndNote, Mendeley or Zotero; 3. know how to include a bibliography (in which a reference databases can support); 4. know how to include references to other people's work correctly (citations or paraphrasing); 5. know how to acknowledge other people's input; (see http://tulib.tudelft.nl); 6. be proficient in writing in academic English (if you wish to improve: consult the courses provided by ITAV)	
Course Contents	This module provides students with a profound overview of the state of the art in a research challenge related to socio-technical systems design in the three departments of the TPM Faculty: Multi Actor Systems (MAS), Engineering Systems & Services (ESS) and Values, Technology & Innovation (VTI). Apart from raising academic curiosity and teaching an academic attitude by exploring a research challenge, the module also addresses scientific integrity issues. The module can inspire students in selecting electives and their topic of the master thesis project in the second year of the CoSEM programme, in relation to the research themes at the Faculty.	
Study Goals	After taking this module the student should be able to: 1) have in-depth knowledge of a research theme that represents the state of the art in research related to socio-technical systems; 2) conduct an individual literature review and present the findings in an academic article; 3) identify issues of scientific integrity/misconduct and to critically assess its practical and intellectual consequences.	
Education Method	In the module the following types of educational methods are applied: 1) lectures on conducting a literature review and writing an academic article; 2) peer reviewing of concept versions of an academic article; 3) groups sessions with the research theme leader; 4) interactive discussion workshops on scientific integrity; 5) writing an individual academic article 6) conducting an individual literature review within domain of the research theme 7) writing a critical assessment on aspects of scientific integrity applied to the research theme of the literature review. These educational methods address the following ways of learning within the networked learning process: - by distribution; - by interaction; - by network-based cooperation and - by learning by leading.	
Computer Use	For conducting the literature review, a VPN connection to the TU Library is required.	
Literature and Study Materials	All study material will be provided via Brightspace	
Prerequisites	See Expected Prior Knowledge.	
Assessment	The assessment of the module consists of two partial marks: 1. writing an individual academic article in which a literature review on a topic within the research theme is presented (individual assignment): 80% 2. writing an individual reflection on aspects of scientific integrity within the research theme topic (individual assignment): 20% These assignments need to be scored at least 5; the final mark needs to be minimal 5.75. In case of a sufficient mark for these assignments, no retake is allowed.	
Enrolment / Application	Before the start of the course the students will be asked to prioritize their preference for the research theme on which they will be working during the course. Each research theme group is supervised by a research theme leader and consists of ~5 students.	
Self Test	Highly recommended to take this test: https://plagiarism.arts.cornell.edu/tutorial/exercises.cfm	

Year

Organization

Education

2023/2024

Technology, Policy and Management

Master Complex Systems Engineering and Management

Compulsory Choice	
Director of Studies	Dr.ir. G. Bekebrede
Director of Studies	Dr.ir. I. Bouwmans

SEN1211	Agent-based Modelling	5
Module Manager	Dr.ir. I. Nikolic	
Instructor	Dr.ir. I. Nikolic	
Instructor	Dr.ir. E.J.L. Chappin	
Instructor	Dr. O. Kammouh	
Instructor	Dr. C.N. van der Wal	
Responsible for assignments	Dr.ir. I. Nikolic	
Co-responsible for assignments	Dr.ir. E.J.L. Chappin	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	Our human society consists of many intertwined Large Scale Socio-Technical Systems (LSSTS), such as infrastructures, industrial networks, the financial and legal systems etc. Environmental pressures created by these systems on Earth's carrying capacity are leading to exhaustion of natural resources, loss of habitats and biodiversity, and are causing a resource and climate crisis. To avoid this sustainability crisis, we urgently need to transform our production and consumption patterns. Given that we, as inhabitants of this planet, are part of a complex and integrated global system where and how should we begin this transformation? And how can we also ensure that our transformation efforts will lead to a sustainable world? Society, with its industrial, social, economic and institutional systems, embedded in the planetary ecosystems are Complex Adaptive Systems (CAS). According to John Holland CAS are "...a dynamic network of many agents (which may represent cells, species, individuals, firms, nations) acting in parallel, constantly acting and reacting to what the other agents are doing. The control of a CAS tends to be highly dispersed and decentralized. If there is to be any coherent behavior in the system, it will have to arise from competition and cooperation among the agents themselves. The overall behavior of the system is the result of a huge number of decisions made every moment" by many individual agents. Understanding Complex Adaptive Systems requires tools that themselves are complex to create and understand. Shalizi defines Agent Based Modeling as "An agent is a persistent thing which has some state we find worth representing, and which interacts with other agents, mutually modifying each others states. The components of an agent-based model are a collection of agents and their states, the rules governing the interactions of the agents and the environment within which they live." This course has two focus points: 1-Complex Adaptive Systems Theory and 2- Agent Based Modelling. We will explore Complex Adaptive Systems theory, across many different types of systems. Lectures will be quite philosophical in nature, and are meant to provide a coherent, high level understanding of systems, their properties and their interactions. Lecture materials provide further reading to broaden your view and deepen your understanding. Second focus point is developing Agent Based Modelling skills. This course is not meant as an introduction to basic programming, or a thorough introduction to NetLogo / Python, but will rather focus on understand the concepts of ABM and how model design choices affect the model created and how it can be used. Additional reading materials are provided for self-study in getting up to speed on basic programming, NetLogo and Python.	
Study Goals	The two goal of the course are, first, to understand Complex Adaptive Systems theory and its relation to the socio-technical systems around us. Second goal is for the student to learn about the the basics of Agent Based Modeling. More formally, there are 5 course goals: 1. Students should know the definitions of CAS properties, as listed in the course topics 2. Students should be able to identify relevant properties of a system and determine whether or not the system can be classified as CAS. 3. Students should be able to create a coherent description of a system from both top down and bottom up 4. Students should be able to understand, modify and create Agent Based Simulations in NetLogo or Python. 5. Students should be able to reflect on the traditional engineering systems thinking from the CAS perspective and understand the implications of changing the traditional perspective.	
Education Method	Course consists of a series of theoretical lectures and a modeling project. Lectures Lectures introduce theoretical concepts and their application, and form the basis for the knowledge for the theoretical exam. Modeling project The second part of the course is focused on creating an Agent Based Model in small groups. The project deliverable consists of the simulation code and report documenting the problem statement, model design, performed experiments and data analysis. This is a compulsory project, contributing 50% to the final grade.	
Computer Use	The course involves developing and experiment with simulations in NetLogo or Python.	
Assessment	The final grade of the course is based 50% on the written theory exam and 50% on the modeling project (simulation + report). Both parts must be completed with a passing grade (at least 5.75) in order to pass the course.	

SEN1221	Statistical Analysis of Choice Behaviour	5
Module Manager	Dr. E.J.E. Molin	
Instructor	Dr. E.J.E. Molin	
Instructor	Ir. S. van Cranenburgh	
Responsible for assignments	Dr. E.J.E. Molin	
Co-responsible for assignments	Ir. S. van Cranenburgh	
Contact Hours / Week x/x/x/x	0/6/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	If one wants to design effective (transport) systems, one needs to understand the behavior of -future- users of these systems, e.g. the market share of a new public transport service. If one wants to design effective (transport) policies, one needs to know how people will respond to these policies, e.g. impact of a tax incentive on electric vehicle sales. This course sets out to provide students with in-depth knowledge and hands-on experience with behavioral theories, mathematical models and statistical methods that have proved very effective in understanding and predicting human choice behaviour. Importantly, this course takes a quantitative (i.e., mathematical and statistical) perspective on choice behaviour; this is quite different from the usual perspective in behavioral science-courses, which tends to be more qualitative. More specifically, we formalize behavioral theories into mathematical models, which we then estimate on empirical datasets using cutting edge statistical methods. In combination, the estimated econometric 'choice models' support rigorous analysis and forecasts of quantitative phenomena, thereby playing a crucial role in private and public sector policy development. The theories, models and methods that are taught in this course are applicable -and have been used- throughout a wide variety of domains, including but not limited to Transportation, Health, Marketing, Environment & Energy, and Political sciences. As such, our course optimally prepares students for a wide range of (quantitative, empirical) topics for their graduation projects. The course will cover both canonical (and Nobel Prize winning) methods which are based on the notion of rational decision making, as well as more recently proposed ones which are based on notions of bounded rationality (economic psychology) and moral decision-making. Both lecturers are highly active in the international research community, and their work forms part of the academic state of the art.	
Study Goals	Students will acquire an in-depth knowledge concerning the below mentioned theories, models and methods, as well as the skills to use them in their own research. Finally, they will acquire an understanding of the usefulness and limitations of these theories, models and methods; and an understanding of how they can and should be used in combination. Knowledge transfer will take the form of a series of lectures, whereas skills will be (further) developed in tailor made computer exercises. Specifically, during the computer exercises, students will learn how to design a statistically optimal choice experiment, and how to analyze data obtained from such experiments. In addition, they will learn how to interpret the outcomes of such analyses and how to use them to evaluate and design policies. - Choice behaviour modelling methods (van Cranenburgh) - o Discrete choice theory and the Logit model - o Advanced (Mixed) Logit models - o Choice models for bounded rationality and morality - Stated Choice data collection methods (Molin) - o Data-collection paradigms - o Stated choice experiments - o Efficient design approaches.	
Education Method	Each method will we covered by means of: - o Lecture series - o Exercises, in which concepts taught during the lectures will be used in hands on exercises. In case covid-regulations require this, a hybrid form of teaching will be adopted, where on campus lectures are livestreamed and computer exercises will be discussed online. IMPORTANT INFORMATION ON KNOWLEDGE REQUIREMENTS: We expect you to have substantial background in mathematics (calculus and statistics), including knowledge of mathematical modeling, and of statistical metrics and testing procedures such as variance, correlation, normal distribution, standard errors, t-test, and statistical testing of hypotheses (including the use of p-values). We also expect you to have knowledge of survey data collection concepts and methods (sample, population, sampling procedures, questionnaires). Note that students with insufficient background in mathematics and statistics will struggle to keep up with the pace of the course.	
Assessment	The final grade is determined by two partial exams which are held during the course and by a regular exam (in the exam week). The partial exams focus primarily on skills and practices, i.e. the analyses, modeling, and experimental design (which are practiced in the computer practice sessions). In contrast, the regular exam focuses mainly on theory and conceptual thinking. Note, however, that both exam types pose questions on theory and practical skills. Furthermore, Mobius or ANS examination software is used for both exams. Exams consist of multiple choice, true/false, and numerical questions and, to a lesser extent, open-ended (essay) questions. Since no re-sit will be offered for the partial exams, the way in which the final grade is determined depends on the period the exam is taken: regular exam period: Final grade = 7.5% * grade partial exam 1 + 7.5% * grade partial exam 2 + 85% * grade regular exam re-sit exam period: Final grade = 100% * grade re-sit exam Thus, the grades of the partial exams are only used in combination with the grade of the regular exam period. After this, the grades of the partial exams lose their validity. Also, the grades of the partial exams are not transferred to the next year and are not registered in Osiris.	
Elective	Yes	
Tags	Analysis Calculus Economics	

Involved
Modelling
Research Methods
Stochastics
Transport & Logistics

Year	2023/2024
Organization	Technology, Policy and Management
Education	Master Complex Systems Engineering and Management

Compulsory Choice	
Director of Studies	Dr.ir. G. Bekebrede
Director of Studies	Dr.ir. I. Bouwmans

SEN1231	Mixed Research Methods for Multi-actor Systems	5
Module Manager	Dr.ir. G. Bekebrede	
Instructor	Dr. M.L.C. de Bruijne	
Instructor	Dr. N. Goyal	
Responsible for assignments	Dr.ir. G. Bekebrede	
Co-responsible for assignments	Dr. M.L.C. de Bruijne	
Contact Hours / Week x/x/x/x	0/0/8/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Required for	This course prepares you for doing a master thesis where you want to use mixed research methods. More in general, you will experience an entire research project, including the uncertainties, practical issues and iterative process.	
Expected prior knowledge	We expect that students have basic research skills such as interviewing, survey research, document analysis, literature review.	
Course Contents	The course Mixed research methods for multi-actor systems focuses on doing research of a multi-actor system. In the course, we will follow an inductive research cycle, including reporting. The design of multi-actor systems requires an in-depth understanding of the way technologies and innovations are both shaped and received by public and private actors. Next to the actors involved, insights into the role of the institutional context are necessary. A mixed research design approach can support in getting this understanding as in this way you can deal with advantage and disadvantage of qualitative and quantitative methods. In this course, students do a mixed research method study as input for a design process of a topic given by the teachers. In groups of 2/3 students, they follow an inductive research cycle and they will focus specifically on two research methods: case study to uncover various perspectives in a multi-actor system, and gaming to understand the relationships between actors and the physical system. By using these methods and by practicing with a number of research instruments that are key to this type of research (e.g. interviews, surveys, document analysis), students will learn when and how to apply these methods. The final results have to be written down in a research paper. We will also demonstrate how using both methods as part of a mixed-methods research design can greatly strengthen the findings and thus, ultimately, the design of multi-actor systems. As this course focuses on making choices in the research design, interpretation of results, combination of applying a mixed method approach, we simulate a research process in the controlled environment of the classroom.	
Study Goals	Students who have followed the course are able to: - Appraise the differences between different qualitative, quantitative and mixed-methods research designs - Develop and justify the mixed method approach that fits their research problem and research question (and subsequently (re)align their research problem and/or question) - Explain the strengths and weaknesses of gaming simulation as research method. - Design and develop a research set up with a given game - Conduct a simulation/game research and synthesize the data and draw conclusions - Explain the strengths and weaknesses of case study research for examining specific real-world data - Design and develop a research set up for case study research - Analyse real-world situation with case study - Evaluate the findings from both research approached and reflect on the (mixed-methods) research approach.	
Education Method	Lectures, assignments, executing an analysis, organizing, facilitating and evaluating a game session, writing a scientific paper	
Literature and Study Materials	Reading materials will be available on Brightspace.	
Prerequisites	We expect that students have basic research skills such as research design, interviewing, survey research, document analysis, literature review.	
Assessment	deliverables are required assignments and a research paper. Grade is based on the final research paper. As this course is project based there is no retake opportunity.	
Enrolment / Application	Enrollment is via enrollment in Brightspace.	
Tags	Analysis Challenging Design Group work Project Research Methods	
Targetgroup	MSc CoSEM, also other MSc students can follow this course as elective.	

SEN124A	Design in Networked Systems	5
Module Manager	Prof.dr. M.E. Warnier	
Instructor	Dr.ir. P.W. Heijnen	
Responsible for assignments	Prof.dr. M.E. Warnier	
Co-responsible for assignments	Dr.ir. P.W. Heijnen	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Non-CoSEM students who wish to follow the course as an elective are expected to have basic knowledge of mathematical subjects such as calculus, linear algebra, probability theory, and discrete maths, corresponding roughly to the level of a Bachelor engineering degree. In addition, some prior experience with programming, e.g. in Python or MatLab will be useful.	
Course Contents	In our highly connected world, networks exist in a multitude of contexts and domains, for example, in infrastructure, supply chains, transport, information, and social networks. These are socio-technical systems that consist of a large number of components and actors organized into a network. Analysing and managing such systems is a complex endeavour. This course provides an introduction to the analysis of networked systems and the design of interventions that aim to improve their efficiency. The analysis is based on graph theory, which studies structural network properties. Topics covered include, amongst other material, the following: - Network metrics - Social networks - Formation and evolution of networks - Complex networks - Optimal trees - Community detection - Epidemics	
Study Goals	Networked systems often have a high degree of complexity, both in terms of structure and in terms of their social and institutional environment. In this course, we aim to provide you with theoretical knowledge and practical skills that enable you to reason about networked systems and design mechanisms for improving their efficiency. After taking the module the student will be able to: - Analyze structural properties of networked systems using graph theory. - Apply a range of tools (Gephi, various Python libraries) for analysis of networked systems. - Design a new mechanism that can be used as a basis for interventions in networked systems.	
Education Method	Mix of lectures, which discuss relevant examples within a CoSEM context, and labs, which provide the opportunity to apply theory discussed in lectures and the book. 4 contact hours per week.	
Course Relations	Book: Easley and Kleinberg (2010) Networks, Crowds and Markets Reasoning about a Highly connected World see: https://www.cs.cornell.edu/home/kleinber/networks-book/ for a (free) pdf of the book. The course also provides various research papers that are used to illustrate interventions. All covered material will be posted on BrightSpace.	
Assessment	The assessment consists of an exam of the covered material and a group project about a relevant case. Both need to be passed, i.e. you need to have a grade > 5.75 for both parts in order to pass the course. Both parts count for 50% towards the end grade. The exam is an open book exam on paper. During the exam you will have access to a computer with all course material. The computer will not have access to internet nor will python be installed. The project will be done in groups of (max) 4 students. If you pass an individual part of the course (either exam or project) the grade stays valid for next year(s). In case of unforeseen circumstances or measures resulting from COVID-19, the prescribed assessment will be modified as follows: - the exam will be a timed take home exam (open book) - the assessment for the project will stay the same	

Year	2023/2024
Organization	Technology, Policy and Management
Education	Master Complex Systems Engineering and Management

Track E	
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Director of Studies	Dr.ir. I. Bouwmans
Director of Studies	Dr.ir. G. Bekebrede

SEN1511	Engineering Optimization and Integrating Renewables in Electricity Markets	5
Module Manager	Dr. S.J. Pfenninger-Lee	
Instructor	Prof.dr.ir. Z. Lukszo	
Instructor	Dr. S.J. Pfenninger-Lee	
Instructor	Dr.ir. K. Bruninx	
Co-responsible for assignments	Prof.dr.ir. Z. Lukszo	
Co-responsible for assignments	Dr.ir. K. Bruninx	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	A basic knowledge of analysis and matrix algebra.	
Course Contents	This course is an introduction to mathematical optimisation modelling applied to diverse problems in planning and operating energy systems, in particular energy systems with a high share of variable (weather-dependent) wind and solar generation. We will go from a solid theoretical grounding in formulating different kinds of optimisation problems, to practical strategies to solve them under real-world conditions. To do so, we will formulate problems in Python using the Pyomo optimisation package.	
Study Goals	After completing the course, you should be able to: - LO1: Explain the challenges in planning and operating renewable energy systems and the role of mathematical optimisation in addressing these challenges - LO2: Formulate linear, mixed-integer, and non-linear optimisation problems - LO3: Apply different methods to deal with parametric uncertainty in optimisation - LO4: Determine and justify the choice of an optimisation approach to formulate an energy system decision problem - LO5: Formulate and solve optimisation problems using Python	
Education Method	Lectures and instructions	
Course Relations	This course cannot be followed together with SET3061.	
Assessment	The final grade of the course is based 100% on an individual written exam.	
Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).		

SEN1522	Electricity and Gas: Market Design and Policy Issues	5
Module Manager	Prof.dr.ir. L.J. de Vries	
Instructor	Dr. A.F. Correlje	
Responsible for assignments	Prof.dr.ir. L.J. de Vries	
Co-responsible for assignments	Dr. A.F. Correlje	
Contact Hours / Week x/x/x/x	0/6/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Economics of Infrastructures SPM2230 or EPA1233) or an equivalent introduction to micro economics is required.	
Course Contents	Introduction, models of liberalization, network regulation Conceptual framework of market design Gas system & actor description Electricity system & actor description Long-term development of electricity markets Long-term development of gas markets Energy network regulation Policy options for intervention in electricity and gas markets Perspectives on long-term, sustainable energy market design Energy market modeling Electricity market simulation game.	
Study Goals	At the end of this course, students should be able to analyze energy policy issues from a technical, economic, and institutional perspective, to make informed choices regarding market design and policy options and to develop improvements to these options. In this course, the students technical, economic, legal and policy making skills are applied to the complex reality of regulation and market design in electricity and gas sectors.	
Education Method	The course makes use of interactive lectures. The reading materials are partly developed by ourselves and consist partly of relevant, recent policy-related documents. Quantitative modeling exercises are used to learn how electricity prices come about, the influence of policy instruments and renewable energy on these prices and to learn about investment decisions. An elaborate simulation game is used to apply this knowledge and gain insights in the dynamic, long-term development of an electricity market. Finally, an excursion provides in inside view in an energy company.	
Computer Use	An exercise requires the development of a simulation model. Excel may be used for this purpose. In addition, an on-line market simulation game is part of this course.	
Literature and Study Materials	Readers and other reading materials such as scientific papers will be provided through the online learning environment.	
Assessment	Digital exam (70%), modeling exercise (20%) and simulation game (10%). Each of these elements must be passed with a grade of 5,75 or higher.	
Permitted Materials during Tests	Calculator.	
Remarks	This course overlaps in part with SET3055 and is therefore not available to M.Sc. SET students.	
Elective	Yes	
Tags	Economics Energy Policy Analysis	
Targetgroup	CoSEM Energy Track Students	

SEN1531	Design of Integrated Energy Systems	5
Module Manager	Dr.ir. E.J.L. Chappin	
Instructor	Dr.ir. P.W. Heijnen	
Instructor	Dr.ir. G. Korevaar	
Responsible for assignments	Dr.ir. E.J.L. Chappin	
Co-responsible for assignments	Dr.ir. G. Korevaar	
Contact Hours / Week x/x/x/x	0/0/x/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	The SEN Energy Track modules taught in Q1 and Q2.	
Course Contents	Students are challenged to tackle an integrated energy system design problem in the form of a modelling project. Students are expected to come up with a creative (set of) design(s). Afterwards, they evaluate the expected performance of one or more of these designs as a future system or improvement to existing energy systems. The design challenge is complex and can only be met: 1) by taking into account dynamics on the short and long term, 2) by tapping into relevant knowledge on technological developments, 3) by using various quantitative analyses, 4) by considering various parts of the value chain, in terms of energy, mass, financial and information flows and 6) by relating and addressing the multi-actor and multi-level complexities. The quantitative analyses are to be executed with a set of quantitative tools, some used in earlier courses in the Energy Track, some developed by research institutes and companies, and others used in ongoing academic research at the faculty, made available for this course.	
Study Goals	Understand and execute the main steps in the design of integrated energy systems Is able to provide an overview of and explore the design space of integrated energy systems, consisting of multiple carriers, multiple relevant levels, multiple time-scales, multiple sectors Is able to identify and conceptualize core processes that affect the performance of an integrated energy system that play out on short, medium and long term Is able to select, develop and use quantitative simulation models for designing integrated energy systems Is able to reflect on aspects that are placed outside the design space, i.e. the relationship to food, water, built environment, etc.	
Education Method	The core task within this module is to carry out a group project in which an integrated energy system is designed and evaluated by using various models. Groups of students use various models and interact with model experts: teachers, researchers but also various companies are involved. There are lectures and visiting hours for debating modelling choices and details, but also for reflection and debate on modelling and simulation. Students do peer review and present their results for each other and for model experts. This enables academic reflection as well as direct interaction with the potential work field.	
Literature and Study Materials	A reader is provided with the approach applied and descriptions of all the models available. Various energy transition models and simulations are made available for use in the projects. Some were introduced in earlier courses, some are extracted from ongoing energy research at the TU Delft, and others come from various companies in the field that do their own energy system modelling and analysis.	
Assessment	An individual grade will be assigned on the basis of the following inputs: - The group project report that describes the integrated energy systems design including an assessment of its expected performance - An individual appendix with a personal reflection and description of contribution to the project - The contribution to peers, through peer review and discussions	

SEN1541	Sociotechnology of Future Energy Systems	5
Module Manager	Dr.ir. I. Bouwmans	
Instructor	Dr.ir. I. Bouwmans	
Instructor	Dr. A.F. Correlje	
Responsible for assignments	Dr.ir. I. Bouwmans	
Co-responsible for assignments	Dr. A.F. Correlje	
Contact Hours / Week x/x/x/x	0/0/0/x	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	This course is only open to CoSEM students. See under Target Group.	
Course Contents	In this module the changing technological and societal characteristics of future energy systems are dealt with.	
Study Goals	Contemporary energy systems are confronted by technical developments and socio-economic concerns, such as maturing renewable technologies and ICT-based control innovations or energy scarcity and environmental concerns. These changes will reflect on operational roles and responsibilities, market structures and actor interests, and business models for energy services. Moreover, they may necessitate new forms of governance to ensure a reliable operation that meets socio-economic performance criteria. In this light, the course discusses the technical and socio-economic aspects of future, mostly sustainable, energy systems, touching upon issues like the integration of new forms of distributive solar PV or large-scale off-shore wind into existing networks, various forms of new motor fuels, other more experimental renewable energy sources and storage methods, and the dynamics of transition processes. At the end of this module the students know: - Which technical and societal developments can be expected in the energy sector. - How new energy technologies and societal developments may reshape operational roles and responsibilities, energy markets and business models, and governance arrangements. - How to compare and analyze future energy systems, develop plausible transition paths towards them, and estimate the potential of new energy technologies within various scenarios. At the end of this module the students can: - Formulate descriptions of the technical and societal characteristics of future energy systems. - Make a conceptual design of an energy system, given certain demands. - Formulate a balanced advice on issues in the energy sector.	
Education Method	Interactive sessions, self-study, online & class exercises, class discussion. Reading material: scientific papers.	
Assessment	Individual written exam (on the computer, on campus) consisting of essay questions. Any reference material on paper and/or in pdf format on memory stick is allowed; internet access other than to the examination site is not allowed. Sufficient participation in the online exercises (one exercise may be missed) is a prerequisite for taking the exam, but not for the resit.	
Targetgroup	CoSEM students. Students from other studies should follow the elective TPM001B (Q1).	

Year	2023/2024
Organization	Technology, Policy and Management
Education	Master Complex Systems Engineering and Management

Track I&C	
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Director of Studies	Dr.ir. I. Bouwmans
Director of Studies	Dr.ir. G. Bekebrede

SEN1611	I&C Architecture Design	5
Module Manager	Prof.dr.ir. M.F.W.H.A. Janssen	
Instructor	N. Bharosa	
Instructor	Dr. F. Kleiman	
Instructor	Dr.ir. J. De Stefani	
Responsible for assignments	Prof.dr.ir. M.F.W.H.A. Janssen	
Co-responsible for assignments	G.A. De Reuver	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	· Basic knowledge of databases, including Entity-Relationship Diagram (ERD) · Basic programming skills (such as JavaScript) · Basic knowledge of Unified Modelling Language (UML) and software engineering (agile development) · Basic understanding of multi-actor system and stakeholder analyses methods This prior knowledge are part of the linkage program.	
Course Contents	More and more data is available using various sources like databases, social media or the Internet of Things (IoT). Information sharing requires the integration of these systems. The ability to execute and process data is a key capability required by public and private organizations. The large variety of heterogeneous data, applications and business processes results in a complex and fragmented landscape that need to be dealt with when architecting and engineering new systems. Path dependencies and legacy block the easy integration, whereas at the same time new technologies appear that need to be integrated in the complex systems landscape. The purpose of this course is to teach the architectural design of innovative and large-scale ICT infrastructures and services in the light of the challenges imposed by the requirements from the systems physical, economic and social environment. Emphasis will be put on the concepts and role of ICT-architecture and modelling in order to properly design ICT solutions within a multi-actor context. For this purpose application and data integration technologies and information and system quality theories will be addressed.	
Study Goals	-The student is familiar with the state-of-the-art knowledge of ICT-architecting, design and governance within the field of large-scale ICT-systems within a multi-actor context -The student is able to describe basic concepts related to architecting and designing large and complex ICT-infrastructure and service systems within a multi-actor context. -The student should master architecture theories, methods and tools with the ability to combine and switch between them when dealing with complex ICT-problems. -The student is able to structure and analyze problems and identify dilemmas arising during the design process with regard to designing large ICT-systems within a multi-actor context. -The student is able to apply system engineering and architecture-based approaches, methods and tools to deal with problems with regard to designing large ICT-systems. -The student is able to report about the use of architectural concepts, methods and tools for translating business needs into ICT-designs within constellation of public and private actors.	
Education Method	· Lectures. The lectures are aimed at providing an overview of the knowledge of this course. During the lectures exercises are given to practice to internalize the knowledge. · Guest lectures (obliged). The guest lectures are aimed at giving an overview of the state-of-the art in practice and provide insight into practical challenges when using ICT in a multi-actor domain. · Exercises During some of the lectures assignments are given to practice what is learned. · Assignments. Various assignments are given which together create the final report. This includes developing various architectural models. · Presentation. Students should present their results during a lectures. · Report. The results should be reported in a concise report including the argumentation of design choices, the resulting architecture and an evaluation	
Computer Use	Architecture modelling in Archimate tool (http://www.archimatetool.com/) BPMN modelling using Eunomia process builder (http://www.eunomia-process.com/), Bizagi (http://www.bizagi.com/) or LucidChart (https://www.lucidchart.com/)	
Literature and Study Materials	slides book (open access e-book) Nitesh Bharosa, Remco van Wijk, Niels de Winne, Marijn F.W.H.A. Janssen (2015). Challenging the chain. Governing the Automated Exchange and Processing of Business Information. ISBN: 978-1-61499-497-8 (open access). This book covers a range from governance to technical issues and integrates them. Open access book: http://ebooks.iospress.nl/book/challenging-the-chain-governing-the-automated-exchange-and-processing-of-business-information Reader providing an outline of the area and focus on some aspects which is complemented by papers which goes in more depth.	
Assessment	Group assignment is 60% of the grade. The students work together in groups of 4-6 persons. The assignment is assessed by a report which should adhere to the APA reference standards. The report cannot be updated after handing it in. Groups that will not pass will be given the opportunity to write a new report. The exam is 40% of the grade. The exam will be digital and held in a computer room at the university. Each grade should be 5.75 or higher If one part is not passed and the other part is then the other part will remain valid for the next year.	
Elective	Yes	
Tags	Information & Communication Modelling Prototyping Technology	
Targetgroup	SEN I&C track Information Architecture (IA) students Electives for MOT, EPA, Computer science students, and non-TBM students having the required knowledge in this field are welcome	

SEN1622	I&C Service Design	5
Module Manager	Dr. Y. Ding	
Instructor	Dr.ir. G.A. de Reuver	
Instructor	Dr. Y. Ding	
Co-responsible for assignments	Dr.ir. G.A. de Reuver	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	Mobile apps, Internet-of-Things, cloud edge architectures and sensors are enabling a range of new I&C services. These services are typically offered in ecosystems of interdependent actors. However, designing services that add value for users as well as ecosystem stakeholders is challenging. In this course, you learn how to design practical ICT services and develop a valorization plan. The course contains two main parts. In the first part, it is all about designing innovative services that meet the needs of users. You will design service mockups and interview users to test your ideas. In the second part, it is about bringing that service idea to reality. You will design a valorization plan that specifies how to deal with external technologies, stakeholders and revenues. The course is practical and hands-on: you will create, test and plan your service ideas. At the same time, all is theory-informed and you will learn how to support design choices using relevant kernel and design theories.	
Study Goals	After the course, you are able to: - Analyze practical ICT services by applying concepts of design science research, action design research, agile development and service engineering - Evaluate practical requirements and make informed choices on supporting technologies and platforms, with specific attention to autonomous driving, medical IoT, AI, XR, and edge computing technologies - Evaluate an ICT-enabled service concept through semi-structured interviews - Design and illustrate a value-adding service concept driven by ICT - Design a valorization plan that explicitly covers how and when to involve external stakeholders in a design process and integrate the ICT service with value network	
Education Method	Lectures, design project, peer review, presentations, guest lecture.	
Assessment	Group essay = 3 EC; Group mockup = 1.25 EC, Individual reflection document = 0.75 EC All three marks should be >5.75. Participation in peer reviewing, guest lecture and presentations is mandatory.	

SEN163B	Responsible Data Analytics	5
Module Manager	Dr. N. Metoui	
Instructor	Dr.ir. J. De Stefani	
Instructor	S. Azimi Rashti	
Co-responsible for assignments	Dr. J. Ubacht	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	In recent years data has become the fuel of most digitalized organizations. Data Analytics is a powerful tool to observe, predict, diagnose issues within the organization and inform decision-making to solve these issues. It also helps explore, identify opportunities and assist in improving organizational processes. But with great power comes great responsibility. Data is seldom objective. It often represents a biased view of the world. When we base our observations and decisions upon this data, we need to be aware of this bias to avoid undesirable impacts. Besides, the way we frame, interpret, and act upon data analytics results can have negative externalities. A responsible data analytics project should account for and mitigate these harmful impacts and externalities. In this course, you will master important tools and techniques used in data analysis projects with a particular focus on the data scientist competencies. Data science can be defined as the computational skills of extracting meaningful insights from raw data and then effectively communicating those insights to generate value. You will apply data science tools and methods to answer real-life questions. You learn how to select data, preprocess data, and analyze and interpret results using basic statistics and machine learning techniques implemented in Python. You will be invited to think critically about the field of Data Analytics and Machin Learning and understand the potential harmful externalities of the tools you are using in this course. You also will practice methods to identify and mitigate some of these externalities.	
Study Goals	At the end of this course, you will be able to: LO1 Explain different roles and steps involved in a responsible data analytics project. LO2 Select the adequate data science tools and methods to address a real-life sociotechnical problem. LO3 Interpret the output of a data analytics project. LO4 Implement scripts (collecting, cleaning, and preprocessing data) and models (statistical and basic ML) in python to carry out a data analytics project. LO5 Implement fairness measures to identify and mitigate harmful societal and ethical externalities of data analytics (bias, inclusiveness, framing results, designing models, etc.)	
Education Method	In this course, you will have alternating lectures and and hands-on labs	
Literature and Study Materials	Materials (slides, tutorials, books, articles, references to software packages, etc.) will be made available via brightspace.	
Assessment	50% Groups Assignments 50% individual written exam In case of unforeseen circumstances or measures resulting from COVID-19, The written exam will be replaced by timed take-home exam (open book). Group assignments will remain the same.	

SEN1641	Digital Platform Design	5
Module Manager	Dr. J. Ubacht	
Instructor	Dr.ir. G.A. de Reuver	
Instructor	W. Agahari	
Instructor	C. Figueroa	
Responsible for assignments	Dr. J. Ubacht	
Co-responsible for assignments	Dr.ir. G.A. de Reuver	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Required for	Mandatory course for first year CoSEM students in the I&C track. Students from other master programmes and international visiting students are welcome to join too, please contact the module manager to discuss the match between your interests and the course.	
Course Contents	Digital platforms offer innovative ways to organize business and social processes. The rise of digital platforms is transforming entire industries (such as market places for data sharing, AirBnB in the hospitality industry, data sharing solutions in logistics). This leads to new and potentially disruptive business models. But these platforms also influence societal and business processes such as the effects of the sharing economy on the way we interact and the way in which businesses share data within their ecosystem) In this module we explore the design variables of digital platforms from a designer perspective: if you were to develop a new digital platform, which are the aspects/elements that you need to decide upon for this type of highly dynamic socio-technical complex systems throughout their life cycle? A tentative list of aspects that we explore are: the technical architecture of the platform, (distributed) platform control & governance, data platforms, the ecosystem and business models, generativity and openness of platforms, user centered design, user motivations for using the platform, and platforms mechanisms. The bottom line is: how to develop an integrated platform design that takes these aspects into account?	
Study Goals	After taking this module the student should be able to 1) formulate technical design principles to develop a digital platform/architecture; 2) identify and evaluate the mechanisms that are used in platform design 3) develop a business model for a new or improved digital platform concept 4) analyze digital platforms and their influence in the context of their application domain in depth and explain differences in platform design; 4) use academic literature in the field of platform design to develop an analytical framework for conducting analyses of digital platforms 5) create a high-level architecture design for a digital platform for a sustainable development goal (SDG).	
Education Method	Lectures & interactive workshops	
Computer Use	Some workshops require the use of a laptop in class.	
Literature and Study Materials	Literature and study material will be provided via the e-learning environment. Students will also search for and use self-selected literature.	
Assessment	The assessment of the module consists of 3 parts: 1. writing an in-depth case analysis (report) within a small team: 30% 2. presenting the case analysis, including Q&A: 20% 3. an individual assignment on a digital sharing platform design: 50%. The calculation of 1 and 2 requires a score of at least 5.75. The score for 3 needs to be at least 5.75 In case of a sufficient score for either of the 3 parts, no retake is permitted.	
Elective	Yes	
Tags	Business Design Information & Communication Integrated Technology	

Year

Organization

Education

2023/2024

Technology, Policy and Management

Master Complex Systems Engineering and Management

Track T&L	
Director of Studies	Dr.ir. G. Bekebrede
Director of Studies	Dr.ir. I. Bouwmans

SEN171A	Advanced Evaluation Methods for Transport Policy Decision-making	5
Module Manager	N. Mouter	
Instructor	Dr. J.A. Annema	
Instructor	J. Rezaei	
Instructor	N. Mouter	
Instructor	Dr. R.M. Shortall	
Co-responsible for assignments	J. Rezaei	
Contact Hours / Week x/x/x/x	2/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	Transport policies are almost always controversial. Several methods are applied worldwide to support decision-making in transport. Nowadays, societal cost-benefit analysis (CBA) and Multi Criteria Decision Methods (MCDM) are the most prominent and well-known transport policy appraisal methods. In this course you will learn about the newest scientific developments in these appraisal methods. Additionally, you will learn about new appraisal methods such as Participatory Value Evaluation.	
Study Goals	Describe and explain traditional evaluation methods (CBA and MCDM) and non-traditional evaluation methods. Explain the pros and cons of different methods for different evaluation circumstances Describe the ethical critiques for different types of evaluation methods Define/pick a research gap (to be used in the paper) Write an academic paper, based on completed case study exercise, where: A research gap that the student can fill/contribute to is identified ; Pros and cons of ONE chosen method are critically reflected upon; The suitability of the method is convincingly argued; The results are analysed and interpreted, criticised with particular attention to ethical issues; Policy recommendations are made on the use of the results	
Education Method	In formal lectures the methods and theories will be explained. By two exercises students will apply the methods and theories. Finally, by analysing a pre-defined real-world controversial transport policy case and at least ten scientific papers (which they have selected themselves) students will learn to critically reflect on appraisal outcomes. They have to write a concise essay to report their findings and reflections.	
Assessment	The exam is an individual paper of 3500 words (excluding appendices and list of references). (70%) A CBA assignment (15%) A MCDM assignment (15%)	

SEN1721	Travel Behaviour Research	5
Module Manager	M. Kroesen	
Instructor	Dr. E.J.E. Molin	
Instructor	M. Kroesen	
Instructor	B. Pudane	
Responsible for assignments	M. Kroesen	
Co-responsible for assignments	Dr. E.J.E. Molin	
Contact Hours / Week x/x/x/x	0/0/6/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Knowledge about basic statistics and data-analysis (probability density functions, hypothesis testing, regression) is absolutely required as prior knowledge. Without this knowledge it will be very difficult to follow this course.	
Course Contents	In this course students learn how to perform travel behaviour research and how to translate the findings from such research into recommendations for transport policy design. The students will acquire knowledge about six modelling paradigms in travel behaviour research, namely the econometric, psychological, geographical, biographical, mobility style and sociological paradigm. The strengths and weaknesses of each paradigm will be discussed. In addition, three state-of-the-art methods to analyse travel behaviour (structural equation models, latent class choice models and latent class clusters models) will be treated. These methods figure prominently within the six modelling paradigms. For each method there will be a group assignment (so 3 in total). These group assignments will count for 50% of the grade.	
Study Goals	After this course the students will be able: (1) to carry out travel behavior research using three state-of-the-art statistical methods, namely structural equation modeling, latent class cluster analysis and latent class choice models. (2) to translate the study findings to recommendations for transport policy design.	
Education Method	Lectures Computer practicals Group project (4 persons)	
Computer Use	In the course the following statistical softwarepackages will be used: SPSS, Latent Gold, Apollo and AMOS	
Assessment	Knowledge about the modelling paradigms and three methods will be tested in an individual written exam in week 10, which accounts for 50% of the final grade. In case of unforeseen circumstances or measures resulting from COVID-19, the prescribed assessment will be an online "open book" Mobius exam. The group project will consist of three assignments which together account for 50% of the final grade. Both grades should be >5.75 to pass the course. Both parts can be separately passed.	

SEN173A	Analysis and Design of Freight and Logistic Systems	5
Module Manager	Prof.dr.ir. L.A. Tavasszy	
Instructor	Dr.ir. A.J. van Binsbergen	
Instructor	Prof.dr.ir. L.A. Tavasszy	
Instructor	Dr. S. Fazi	
Instructor	Dr.ir. M. Saeednia	
Responsible for assignments	Prof.dr.ir. L.A. Tavasszy	
Co-responsible for assignments	Dr. S. Fazi	
Contact Hours / Week x/x/x/x	0/0/0/x	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	- Global and local megatrends in freight transportation - Recap of logistics and freight transport policy - Introduction to freight transport demand modelling - Design of multimodal freight networks including terminals - Lectures on theory and a practical modelling assignment - External speakers: Panteia and Significance on freight models in practice	
Study Goals	At the end of the course, the student can: - Reproduce and explain the analysis and design framework for the freight transport system, including its background in supply chain management (demand), basic design elements (network, policy) and related quantitative models. - Select and combine relevant parts of this framework for various freight transport analysis and design problems. - Provide functional and mathematical specification of a freight transport demand and network design model. - Implement and apply an empirical freight demand and supply model for a practical freight network design problem.	
Education Method	Lectures: > Four lecturers covering different connected aspects of the freight transport system > Guest lectures: expert on policy applications of freight models > Modelling assignment on EU multi-modal freight network (group size: 3-5) > Self-study	
Assessment	Written exam with open questions (70% of the final grade) Group assignment with written reports (30% of the final grade) A minimum grade of 5.0 is necessary for all parts.	

SEN1741	Innovations in Transport and Logistics	5
Module Manager	Dr. M. Comes	
Instructor	Dr. S. Balakrishnan	
Co-responsible for assignments	Dr. J.A. Annema	
Contact Hours / Week x/x/x/x	0/x/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	The transport and logistics systems are continuously changing due to the introduction of innovative concepts in vehicles, infrastructure, and services, examples being the introduction of the car, e-bike, drones, or high speed train (vehicles), ride-sharing applications, GPS tracking, or satellite navigation (services), or charging facilities (infrastructure). But many innovations fail. Others may undermine the resilience of the system or create unwanted cascading effects for other infrastructures, social or environmental systems or services. Therefore, policy-makers need to understand the implications of such innovations.	
Study Goals	In this course you will learn about factors important to understand the impact of transport and logistics innovations. By the end of this course, you should be able to LO1: Explain the main frameworks that describe the impact of innovations. LO2: Explain the main pathways that link transport and logistics innovations and their societal impacts within the system (e.g., accessibility/equality, resilience) and cascading effects. LO3: Analyse the success and failure factors of candidate transport and logistics innovations using a combination of frameworks from the literature. LO4: Design and develop a conceptual model that illustrates the impacts of candidate transport and logistics innovations.	
Education Method	Interactive Lectures (frameworks; impacts), Group projects, resulting in a presentation and a short essay Interactive group presentations	
Assessment	1. Group essay and group presentation 2. Individual exam (open book, take home) Each part has to be graded with at least 5.75. The final mark is composed as follows: part 1 40%; part 2 60%. In case of unforeseen circumstances or measures resulting from COVID-19, the prescribed individual assessment will be a timed open-book assignment, and it will be written at home.	

Year	2023/2024
Organization	Technology, Policy and Management
Education	Master Complex Systems Engineering and Management

2e jaar CoSEM 2023	
Director of Studies	Dr.ir. G. Bekebrede
Director of Studies	Dr.ir. I. Bouwmans

SEN232A	Master Thesis Research Design	5
Module Manager	Dr. J. Ubacht	
Instructor	Dr. A.F. Correlje	
Instructor	Dr. J.A. Annema	
Instructor	Dr. T. Hoppe	
Responsible for assignments	Dr. J. Ubacht	
Co-responsible for assignments	Dr. J.A. Annema	
Contact Hours / Week x/x/x/x	2/2/2/2 4 workshops of 2 lecture hours in week 1, 3, 5, 7	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	1 2 3 4	
Course Language	English	
Required for	This course is a formal prerequisite course for the starting the Master Thesis project. The student cannot start their master thesis project before this course has been marked with at least 5.75 for the research proposal (excluding the bonus).	
Expected prior knowledge	The TPM Graduation Portal provides all information on the master thesis project requirements, procedures, forms etc. https://www.tudelft.nl/studenten/faculteiten/tbm-studentenportal/onderwijs/master/graduation-portal/ Student that enter this course are expect to: 1. know how to find scientific articles using e.g. Web of Science or Scopus (these are library information skills, you can take the TU Library modules Information Literacy 1 and 2 (see digital study guide, select MSc and select the courses with TPM in the name); 2. know how to store references, e.g. by making use of a reference database such as EndNote, Zotero or Mendeley; 3. know how to include a bibliography (in which a reference database can support); 4. know how to include references to other people's work correctly (citations or paraphrasing) 5. know how to acknowledge other people's input. (for upgrading your knowledge on 1-5 see: http://tulib.tudelft.nl) 6. be proficient in writing in academic English (for polishing/upgrading your English writing skills: see classes by the Centre for Languages and Academic Skills) Students are encouraged to have chosen the subject of their master thesis project to work on during the course, but this is not obligatory.	
Course Contents	Students will orient themselves on a thesis subject that has societal relevance by performing a literature review in order to discover an academic knowledge gap. On the basis of this, the student develops a solid research proposal, which includes formulating a main research question, selecting an appropriate research approach and formulating sub questions, selecting research methods and/or modelling techniques and data analysis tools; presenting a research flow diagram and making a thesis project planning. During the course the student will read and analyze approx. 10 scientific articles and conducts a peer-review of partial draft chapters and the final research proposal of peers in the course. Final deliverables: A research proposal in a predefined format. In order to pass and get access to the Master Thesis Project this deliverable needs to be graded as sufficient (≥ 5.75). It will be graded according to a fixed list of criteria; Peer reviews as formative assignments that will be assessed by peers; All assignments during the module have to be handed in before the deadline, and have to comply with the assessment criteria.	
Study Goals	After concluding this course the student should be able to: * write a research proposal, that demonstrates: * societal relevance * scientific relevance based on a state of the art literature review * coherent and consistent scientific argumentation * program specific research(able)questions and research approach * a do-able research plan including a conscious and justifiable choice of research methods and an accompanying time schedule * design research typical for their MSc program * do serious and high quality peer reviewing	
Education Method	Four two hours interactive workshops that support students to discuss and learn about writing a research proposal and doing thesis research. Peer review is an important learning activity in this course. It is obligatory to meet the peer review and final submission deadlines: if not, students are excluded to continue the course.	
Literature and Study Materials	Learning materials are provided on the platform Great Library, which also supports the peer review process.	
Books	All materials are available electronically. Students will search and use their own sources for developing their research proposal.	
Assessment	All assignments need to be fulfilled. The grade will be based on the final versions of - a research proposal of ~3500 words (grade) - peer assessment of reviews of formative assignments: plus or minus bonus The course is successfully passed with at least 5.75 for the research proposal(so excluding the bonus). We do not allow retakes of proposals that have been graded as sufficient (>5.75).	

Special Information	This course is offered each quarter. This is a demanding course. Because of its structure (the peer-review process) strict deadlines are maintained; full commitment is expected. The course is open for EPA and CoSEM students without required prior permission (but being a second year MSc student is strongly recommended). All other students: please contact the module manager in advance to check expectations and requirements. EPA students with questions on the course: please contact the EPA module manager Nihit Goyal: nihit.goyal@tudelft.nl
Elective	Yes
Tags	Challenging Intensive Involved Project planning / management Research Methods
Targetgroup	2nd year COSEM students; 2nd year EPA students who cannot take the regulator EPA course in Q3; Other students needs to contact the module manager to check eligibility.
Category	MSc level

SEN2331	CoSEM Master Thesis	30
Module Manager	Dr.ir. G. Bekebrede	
Co-responsible for assignments	Dr.ir. I. Bouwmans	
Contact Hours / Week x/x/x/x	0/0/x/x	
Education Period	3 4	
Start Education	3	
Exam Period	none	
Course Language	English	
Summary	The Master's Thesis Project covers a period of 21 weeks and consists of an individual academic research project.	
Course Contents	Within this project, students must demonstrate their capacity for academic analysis, synthesis, design, reflection and written communication on a particular issue in the field of complex systems engineering and management. The project starts with an orientation phase that leads to a detailed project proposal. In the fourth week of the project, the student presents this proposal to the graduation committee during a kick-off meeting. The proposal should give information on the goal of the project, the main research/design question, a set of derived sub-questions/goals that give meaning to the project, data to use/collect, methods to be used, and a detailed project planning. After a successful kick-off meeting the student starts with the actual execution. During the project the student works towards documenting the results of the thesis project in a master thesis report. The graduation committee or the student may organize a mid-term meeting to guide the progress of the project. When the project is almost finished, a green-light meeting is held for which the students presents a complete master thesis report. During this meeting the student can get a green light to organize the final thesis project defence. This requires a formal approval by the graduation committee of the contents of the report and approval by the administration office for graduation (100% of the courses completed except for the graduation project). The master thesis report should at least be sufficient to pass without any adjustment, but expected improvements to both the thesis project results and the master thesis report as suggested by the thesis committee should be considered to be incorporated into the final version. Students are required to make the final thesis report available in digital format to the administration as well as the TU Delft repository two weeks before the defence. During the final defence meeting the student gives a presentation on the project and defends the final project results in discussion with the committee members. If all results, including the defence and presentation, are sufficient, the diploma and final grade for the thesis project as a whole are handed out. A standard master thesis assessment matrix is used by the committee as a guide for the final grade. More detailed and essential information about the Master Thesis Project can be found at the TPM Graduation Portal, where all administrative procedures, requirements and forms for the Project are available.	
Study Goals	The aim of the CoSEM master thesis project is to design solutions for large and complex contemporary socio-technical problems. This requires the consideration of technical, institutional, economic and social aspects. CoSEM students employ methods for creatively designing technical solutions in organisations and assessing their impact. The solutions contain system engineering approaches as well as effective management strategies to deal with the technological complexity and stakeholders with widely diverging interests. Thesis projects take into account both public and business values and look at the physical system as well as the actor network, confronting not only technical challenges, but also management and ethical choices. During and after the CoSEM thesis project a student should be able to: - develop a relevant scientific contribution to a CoSEM-related subject; - determine the societal relevance of the thesis topic to which the master thesis will contribute; - apply the knowledge and skills learned during the CoSEM programme; - analyze the complexities of the socio-technical system and its institutional and/or managerial aspects; - develop an integrated research design including research approach and methods, tools and data requirements to design, evaluate or simulate an innovation in a well-delineated part of a complex socio-technical system; - critically evaluate the consequences of an intervention within a complex socio-technical system; - formulate informed recommendations and future research topics based on the project results; - defend and justify the research outcomes; - independently manage, set up, carry out, report and present a research project. During the Master thesis project, the student needs to demonstrate a thorough command of the knowledge and skills learned during the study programme and to show they are able to apply them to the master thesis project. This entails knowledge, insight and skills on an academic level for: 1. independently setting up and carrying out a research project; 2. project planning and management; 3. cooperating and communicating with others; 4. documenting, reporting and making a presentation; 5. the specific graduation subject. General criteria for graduation are: - the work contains an analytical component - the work is multidisciplinary in nature - the work focuses on a technical domain or application	
Education Method	Students carry out the project individually. The thesis project can be conducted as part of a research/design project being carried out at the faculty, or during an internship at an external partner organisation in either the public or the private sector.	
Course Relations	The CoSEM Master Thesis Preparation course is a prerequisite for starting the master thesis project (see prerequisites) and prepares the student for this master thesis project in developing a research plan.	
Literature and Study Materials	The students are expected to consult their own information sources.	
Practical Guide	The TPM Graduation Portal provides all information on the master thesis project requirements, procedures, forms etc. https://www.tudelft.nl/studenten/faculteiten/tbm-studentenportal/onderwijs/master/graduation-portal/	
Prerequisites	Starting conditions for the master thesis project are formulated in the Teaching and Examination Regulations (TER). See https://www.tudelft.nl/en/student/faculties/tpm-student-portal/education/rules-and-guidelines	
Assessment	The faculty of TPM uses a standard assessment matrix as a guide for the assessment of master thesis projects: the MSc Thesis Assessment Guide TPM. The core components of the assessment are:	

	A. Research Quality; B. Research Skills; C. Reporting Quality; D. Quality of Oral Defence. In addition, the Graduation Committee uses the Final Assessment Form to motivate the final grade. Both the Assessment Guide and the Final Assessment form can be found on the TPM Graduation Portal: see the execution tab or the link to the forms on the left hand column of the portal.
Enrolment / Application	Graduation coordinators assist students in finding a master thesis project, finding supervisors, etc. A full list with contact details and affiliations can be found on the TPM Graduation Portal. https://www.tudelft.nl/studenten/faculteiten/tbm-studentenportal/onderwijs/master/graduation-portal/
Targetgroup	All second-year CoSEM students.
Category	MSc level

Year	2023/2024
Organization	Technology, Policy and Management
Education	Master Complex Systems Engineering and Management

Track Electives CoSEM (10 EC)

TPM593A	Internship TPM	10
Module Manager	Dr. J.M. Duran	
Contact Hours / Week x/x/x/x	x/x/x/x	
Education Period	1 2 4	
Start Education	1 2 3 4	
Exam Period	none	
Course Language	English	
Course Contents	The external project or internship, TPM593A, is an elective part of the TPM Master Programmes MoT, EPA and CoSEM. Enrolling for the course always starts with contacting the module manager. MoT and EPA students are permitted to do the external project instead of the Elective Courses. CoSEM students are permitted to do the external project as Track electives (in which case the subject must be related to the Track) or as part of their free electives. The internship may not be part of your master thesis study. The execution of the internship as part of the MSc programme may only start after complying with the following requirements: The student is registered as master student during the duration of the external project (the work may be done part-time), The required TPM593A Initiation Document is completed and signed by the module manager. This document can be applied for to the module manager. The document checks whether: - the work is done in a professional environment: in a company or an authority (e.g. a government) outside the university, - the external supervisor and the MoT/EPA/CoSEM lecturer are known and committed, - the content is of sufficient quality Authorization of the MSc programme, including the signed TPM593A Initiation Document, is given by the Board of Examiners Contents wise, the work must have the following characteristics: the work contains research on a MoT/EPA/CoSEM-relevant subject and/or design of a MoT/EPA/CoSEM-relevant object, the work has clear design and/or engineering component(s), the focus should be on socio-technical system design of preferably large-scale technological systems, problems should deal with a multi-actor environment, the work should result in an individually written report, with clear argumentation, with a clear structure containing problem description and boundaries, problem statement, objective, research question(s), research and/or design process, research and/or design results, conclusions and recommendations and a list of references. The report will be assessed. Students are encouraged to organize their own external project. The number of available projects is limited. The availability cannot be guaranteed. The authorisation of the Board of Examiners is not required, when TPM593A is a course supplementary to the MSc programme.	
Study Goals	The aim of the external project is: to get familiar with MoT/EPA/CoSEM practices by working in a professional environment, to apply the knowledge you learned from the university in a practical problem, to deal with complex issues both systematically and creatively and to demonstrate functioning in a professional, preferably multi-disciplinary team	
Education Method	Practical work in a professional environment outside the University, supervised by an external supervisor and a MoT/EPA/CoSEM-lecturer.	
Assessment	The internship and the report will be assessed by the external supervisor as well as the TPM supervisor by applying standard assessment forms. In case of exceptional differences in assessment results, the module manager will align both supervisors. The module manager will determine the final mark.	
Elective	Yes	
Category	MSc level	

Year	2023/2024
Organization	Technology, Policy and Management
Education	Master Complex Systems Engineering and Management

Master Specialisaties

Year	2023/2024
Organization	Technology, Policy and Management
Education	Master Complex Systems Engineering and Management

Emerging Technology-based Innovation and Entrepreneurship	
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MSc Coordinator	Dr. G. van de Kaa
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TPM029A	Responsible Innovation	3
Module Manager	A.R. Gammon	
Responsible for assignments	A.R. Gammon	
Co-responsible for assignments	Dr.ir. U. Pesch	
Contact Hours / Week x/x/x/x	x/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Required for	This course is part of the master specialisation Emerging Technology-based Innovation and Entrepreneurship	
Course Contents	Technology has become a main determinant of the quality of life of individuals and the quality of society. New technologies contribute to human well-being, but they may also introduce considerable risks to humans, the environment and future generations. We therefore have every reason to ascertain that the new technologies we develop as a society respect the values we hold dear. Various governments, companies and research funding agencies have now recognized this need for responsible innovation. Responsible innovation can, as a concept, be understood in a more substantive and in a more procedural sense. As a procedural notion, responsible innovation refers to a process of innovation that meets certain procedural norms, like anticipation, reflexivity, responsiveness and inclusiveness. As a substantive notion, responsible innovation refers to a process of innovation which results in certain products, i.e. innovative technologies, which reflect important moral values. This includes values like, health, safety, human welfare, sustainability, justice, inclusiveness, democracy, privacy, trust, and autonomy. In this course we will discuss the concept of responsible innovation and how it might be understood. We will discuss tools the approach of Value Sensitive Design (VSD) as a way to put in practice the more substantive notion of Responsible Innovation. We discuss each of the four procedural dimensions: anticipation, reflexivity, responsiveness and inclusiveness. For each, we will look at possible approaches, possibilities, and pitfalls. Lectures The following topics will be covered in the lectures: - The notion of Responsible Innovation (RI) - Notions of moral responsibility and for individuals and corporations/organizations - The approach of value Sensitive Design (VSD) - The four procedural dimensions of RI: anticipation, reflexivity, responsiveness and inclusiveness - How stakeholder publics are formed and how to engage publics in participatory processes for RI	
Study Goals	After this course, students should be able to: - Provide and justify specific definitions of the notion of responsible innovation - Explain the approach of Value Sensitive Design (VSD) and apply it to concrete cases and examples - Relate responsible innovation to moral notions of responsibility - Identify opportunities and challenges for responsible innovation in concrete cases with respect to the four dimensions of responsible innovation: anticipation, reflexivity, responsiveness and inclusiveness - Develop strategies for implementing responsible innovation - Work in groups to develop, over the duration of the course, a group project/case on responsible innovation and reflect on this process of group working and learning	
Education Method	Weekly class meetings, preparation of mini-lectures and required reading, and group work on projects, including in-class presentations.	
Literature and Study Materials	All materials are posted on Brightspace.	
Assessment	Assignments There is no exam for this course. Instead, assessment takes the form of several required assignments throughout the course. These are described and the weights of each assignment are indicated below. (A) 25% - weekly quizzes (in class, no option to retake missed quizzes. Best 5 out of 6, graded individually) (B) 15% - participation (engagement in class meetings, feedback on other groups' presentations, graded individually) (C) (graded per small group: 4-5 students) 20% - small group presentation 30% - final presentation (group poster) & writeup 10% - peer feedback Final Grades The final grade is calculated: $((A) \times .25) + ((B) \times .2) + (\text{small group presentation} \times .2) + (\text{final group presentation \& writeup} \times .3) + (\text{peer feedback} \times .1)$ To pass the course, the final grade has to be 5.75 or higher. All partial grades have to be 5.0 or higher. There is no resit because there is no exam. However, partial grades are valid for A, B, and C, and can be carried from one year to the next if students do not pass or complete all parts of the course. (Parts of C cannot be carried over: this means that students need to complete all parts of C in one year.) This is only allowed for insufficient or incomplete grades, not to improve passing grades.	
Remarks	For ethics teaching at TU Delft check https://www.tudelft.nl/ethics/ethics/teaching-activities/ethics-teaching/	
Targetgroup	Master students EPA, CoSEM and MOT.	
Category	MSc level	

TPM033A	Sustainable Innovation and Transitions	3
Module Manager	Dr.ir. J.N. Quist	
Instructor	Dr. L.M. Kamp	
Co-responsible for assignments	Dr. L.M. Kamp	
Contact Hours / Week x/x/x/x	2/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1	
	2	
Course Language	English	
Required for	TPM specialisation on Emerging Technology-based Innovation and Entrepreneurship	
Course Contents	The course Sustainable Innovation and Transitions takes as starting points: (1) that technology and innovation on the one hand and society and users on the other hand mutually influence one another and evolve in a co-evolutionary way; (2) that sustainable innovations, as well as system innovations and transitions towards sustainability, are strongly needed to bring about sustainability, and; (3) that stakeholder involvement and participatory intervention instruments are needed and required to enable and realise implementation in a socially responsible way. Main (lecture) topics are: 1. Sustainable and Social Innovation 2. Innovation Systems (FIS/TIS) 3. Transition theory, Multi-Level Perspective & Strategic Niche Management 4. Transition Governance, Transition Contexts & Transition Pathways 5. Sustainable Business Models 6. Intervention instruments: Backcasting and Transition Management 7. Sustainable Innovation Policy The course consists of two parts: (1) A theoretical part consisting of lectures, readings and discussions, which is assessed by a written exam. (2) A group assignment in groups of 4-5 students on a self-chosen sustainable or social innovation. It can be a technical innovation (e.g. recycling, EVs, biogas), an organisational innovation (e.g. car sharing, eco-industrial parks), or a bottom-up grassroots or social innovation (e.g. energy cooperatives, urban agriculture) in both industrialised and developing countries.	
Study Goals	By the end of the course students have:	
	A. Knowledge of (i) main sustainable innovation, innovation system & transition theories; (iii) main concepts of sustainable business models; (iv) Main multi-actor intervention instruments at the level of socio-technical systems like transition management, and backcasting.	
	B. Understanding of: strengths and limitations of the above mentioned theories, frameworks, tools, methods and instruments.	
	C. Experience & skills in the application of the frameworks to a self-chosen topic in a step-by-step group assignment. Academic skills that are trained in the course include (i) development of research questions & proposal; (ii) writing coherent, well-structured and well-argued academic reports; (iii) dealing with user and social aspects of innovation, and (iv) stakeholder analysis and plurality in stakeholder perspectives.	
Education Method	The course consists of two parts: a theoretical part and a group assignment, while there is a voluntary bonus assignment. Theoretical part contains lectures and paper discussions. Lectures are not obligatory but attendance is strongly recommended. It is important that students are present during the scheduled meetings and have studied the papers that are scheduled for that particular lecture. Students have to collect reading material themselves, which avoids copyright costs. Most papers can be obtained through the links provided in the course description. Otherwise, readings will be uploaded. Group assignment is done in groups of 4-5 students. The assignment is split into several steps. The group assignment consists of applying different sustainable innovation frameworks and perspectives to a chosen sustainable or social innovation. For more information on the assignment, see the Group assignment document. Bonus assignment consists of 6 short written assignments based on 2 core readings every week, and one individual review of the mid-term group work by another group. A grade of 0.8 on the exam grade is possible (till a maximum exam grade of 10.0).	
Literature and Study Materials	Readings consist of selected journal papers and book chapters on the topics relevant to the course; a reading list of articles and chapters will be provided.	
Assessment	Summative assessment consists of: (1) an individual open book written exam using open questions (that counts for 50%), (2) a group project on a chosen sustainable or social innovation (also counting for 50%). (3) A bonus assignment of 0.8 on the exam grade is possible (till a maximum exam grade of 10.0). The bonus assignment enabling a weekly 0.1 bonus during 6 weeks + 0.2 bonus point for the group work peer review (0.1 if the peer review is sufficiently well done, 0.2 if the peer review is well done). The final grade is calculated as follows: (1*written exam + 1*bonus assignment+1*group project)/2. Final grades are expressed by means of a figure between 1 and 10, rounded to the nearest half. Both exam (complemented with the bonus assignment point) and group project must be completed with at least 5.5. So, the course is passed if the final grade is at least 5.75, while each graded part in the course needs to be at least 5.5. Evaluation criteria and grading details of the group project and report are presented in the group assignment document. Formative assessment takes place through group work peer review as part of the group assignment. If the group project assessment is not sufficient to pass (grade at least 5.5) the group can update the project by revising the report during Q2. The exam is based on the core readings and the lectures (including slides). There is one resit possibility at the end of Q2 if the exam grade does not meet the minimum requirement (at least 5.5, including the bonus point). Review of the exam will be organised within 14 days after the grades are published (if online reviewing is not possible). Students can contact the module manager after the announcement if they would like to make use of this opportunity.	
Targetgroup	Partial grades are valid only for the current academic year and are not transferable to the next academic year.	
	TPM 2nd year Master students, including CME. This course CANNOT be taken by Industrial Ecology students and SET students, due to considerable overlap with Transitions, Innovation & Governance and Sustainable Energy Innovation and Transitions courses, respectively.	

TPM959A	Technology Battles	3
Module Manager	Dr. G. van de Kaa	
Instructor	Dr. J.R. Ortt	
Instructor	Dr. G. van de Kaa	
Responsible for assignments	Dr. G. van de Kaa	
Co-responsible for assignments	Dr. J.R. Ortt	
Contact Hours / Week x/x/x/x	x/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	The course is organized into regular sessions. Each of the regular sessions consists of a mixture of lecturing and discussions. For each session you study the literature. Through class discussion we will further deepen our understanding of the topics and themes of the class. Every student is expected to participate in a class-wide discussion.	
Study Goals	After the course students are able to understand the theoretical background of standardization.	
Education Method	1 group assignment	
Assessment	The grade for this course is based on the group assignment. A minimum of 5.75 must be obtained to pass the course. Right to inspect the results (report) is possible on request.	

Year	2023/2024
Organization	Technology, Policy and Management
Education	Master Complex Systems Engineering and Management

Electives

TPM406A	Corporate Entrepreneurship and Start-ups	6
Module Manager	Dr.ing. V.E. Scholten	
Responsible for assignments	Dr.ing. V.E. Scholten	
Co-responsible for assignments	Dr. L. Hartmann	
Contact Hours / Week x/x/x/x	0/x/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	No prior knowledge is required.	
Summary	This course examines how large corporations engage startups to tap into new markets, technologies and business models. You will develop knowledge and skills to analyse strategic situations of existing corporates and how they innovate through engaging startups. The course provides an eye on corporate startup engagement from both the corporate and startup perspective.	
Course Contents	Large international organizations often find themselves in a luxurious position compared to small start-ups. They have more financial capital, access to markets, and more technological knowledge. However, large international organizations are confronted with increasing external turbulence, technological advancement and globalization, which puts their luxurious position in jeopardy. At the same time the luxurious position constrains them to depart from present activity and focus on radical new innovations. Hence, corporates try to renew themselves through internal entrepreneurial behaviour, but also by initiating start-ups and acquiring start-ups to experiment with new technology. Many times, these start-ups originate in academic environments. Therefore we pay specific attention to the emergence of academic start-ups and how corporate organizations provide incubation facilities and venture capital support to engage startups to renew themselves. This course examines the various forms of corporate entrepreneurship and the relationship with corporate strategy in particular. You will develop knowledge and skills to analyze strategic situations and design the most appropriate solutions in terms of corporate entrepreneurship strategy and organization.	
Study Goals	Understand the concept of corporate entrepreneurship; the contexts, the forms, and the relationship with strategic management Analyze strategic situations and design appropriate corporate entrepreneurship strategies and organizations Analyse the contribution of internal and external ventures in the renewal of large corporates Understand the emergence of academic spin-offs and how they relate to the technology development of large established organisations Identify the various key actors that facilitate and support the emergence of new business ventures and academic spin-offs	
Education Method	In case of Covid-19 restrictions for on-campus education we will offer this course in 14 online sessions. Each session consists of a mixture of lecturing, and discussions of cases, literature and assignments, and presentations. Each student will be expected to be present during lectures (two times a week), actively participate during the lectures, complete assigned readings, read posted lectures, listen to thought leader presentations and submit written assignments when due.	
Literature and Study Materials	Journal articles: Will be posted in Brightspace Slides presented in class: Will be posted in Brightspace	
Practical Guide	Given the interactive nature of this course, attendance at classes is mandatory. Only very serious grounds for absence (such as illness accompanied with doctors note) are acceptable. Holidays and (paid or voluntary) work obligations do not count as acceptable reasons for absence. This is a full-time course with scheduled holidays. Unjustified non-attendance will result in expulsion from this course.	
Assessment	The assessment is Covid-19 proof and includes three components. 1) The first component is a team level online presentation of a case analyses of an existing company. Each team will analyse a practical company case provided to you and based on literature of a specific lecture session you will present your findings online in class (40%) 2) The second component is a written report of a large firm in which you will analyse with your team the startup engagement activities this firm has put in place to renew itself. The report is an analyses of the company activities w.r.t. corporate venturing and an reflection of these activities by making use of literature. The report accounts for 40% of the final grade. 3) The third component is an individual analyses of a startup that is involved in the corporate startup engagement activities of your team case company. It reflects on the motivation for the startup engagement from both startup and corporate perspective. It accounts for 20% of the final grade.	
Enrolment / Application	All students across the various faculties of the Delft University of Technology can take this course by directly enrolling into Brightspace. Brightspace enrollment will be activated by education support a couple of weeks before the course starts. No other means of registration, such as the app MyTuDelft are needed.	
Special Information	Each student team will present during the lectures, actively participate during the lectures, complete assigned readings, read posted lectures, listen to thought leader presentations and submit written assignments when due.	
Remarks	For more information about Entrepreneurship courses: THINK - ACT - START entrepreneurial, please check https://www.tudelft.nl/dce/education.	
Targetgroup	Master students from all master programs can apply.	
Category	MSc level	

TPM416A	Turning Technology into Business	6
Module Manager	Dr. L. Hartmann	
Responsible for assignments	Dr. L. Hartmann	
Co-responsible for assignments	Dr.ing. V.E. Scholten	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Course Contents	Turning Technology into Business (TTiB) aims to equip participants with a strong conceptual foundation to actively understand the dynamic process of technology-based entrepreneurship. Participants learn how business strategies are best formulated and how (through entrepreneurship) technology can create value. TTiB consists of seven lectures, introducing the theoretical backgrounds of technological, market and business analyses. A unique aspect of TTiB is that existing technologies (developed and patented by the TU Delft) are used as case subjects. The patent project is the focal point of the curriculum. Each group of 4-5 students will be assigned an original patent, and is expected to evaluate the commercial potential of this technology. This includes choosing promising applications (products) for the technology, and making recommendations for the most suitable business model to commercialize the technology. The patent project should provide a coherent and structured answer to the central question: which strategy is most likely to generate business from this patent? In addition to equipping participants with a thorough knowledge of the course subject, the section Technology, Strategy & Entrepreneurship (TSE) hopes that this course will be the starting point for a variety of university spin-offs in which the students will participate. To accommodate this, TSE offers the course TPM420A- Ready to Startup!	
Study Goals	This course aims to equip students with a strong conceptual foundation to an active understanding of two domains: (1) The dynamic process of technological innovation through concepts such as technology life-cycles, dominant design, disruptive technologies, Schumpeterian competition and the diffusion of innovations (2) How business strategies are formulated and, through entrepreneurship, technology can create value. This multi-faceted process of technology commercialization process is addressed in terms of assessing technology position, discovering market opportunities, competitive analysis, appropriability and the various modes of entrepreneurship.	
Education Method	7 interactive lectures, participant-centered case studies, homework assignments, classroom assignments, individual group coaching. Final Presentations in February.	
Literature and Study Materials	Will be provided during the course	
Assessment	Each group writes a report on their findings, analysis and recommendations for the technology (patent) they used as a case for this course. On the Final Day, the group has to give a 10 minute oral presentation. The report makes up 80% of the final grade. This team grade will be modified according to the individual student's class contributions and his/her performance within the group (peer review).	
Enrolment / Application	MAXIMUM CAPACITY is limited to 75 participants. Register on BrightSpace from October 1. Because of the huge popularity of this course, there is a selection procedure which consists of pre-course assignments. One of these assignments is a letter of motivation: why do you want to participate in this course? Based on the motivation and the other pre-course assignments, the final selection of 75 participants will be made.	
Remarks	For more information about Entrepreneurship courses: THINK - ACT - START entrepreneurial, please check https://www.tudelft.nl/dce/education .	
Targetgroup	Advanced Master students, PDEng and PhD students, and staff researchers from all faculties within the TU Delft.	
Category	MSc level	

Year

Organization

Education

2023/2024

Technology, Policy and Management

Master Complex Systems Engineering and Management

ICT Design and Management	
MSc Coordinator	Prof.dr. Y. Tan
MSc Coordinator	Dr.ir. G.A. de Reuver

SEN1611	I&C Architecture Design	5
Module Manager	Prof.dr.ir. M.F.W.H.A. Janssen	
Instructor	N. Bharosa	
Instructor	Dr. F. Kleiman	
Instructor	Dr.ir. J. De Stefani	
Responsible for assignments	Prof.dr.ir. M.F.W.H.A. Janssen	
Co-responsible for assignments	G.A. De Reuver	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	· Basic knowledge of databases, including Entity-Relationship Diagram (ERD) · Basic programming skills (such as JavaScript) · Basic knowledge of Unified Modelling Language (UML) and software engineering (agile development) · Basic understanding of multi-actor system and stakeholder analyses methods This prior knowledge are part of the linkage program.	
Course Contents	More and more data is available using various sources like databases, social media or the Internet of Things (IoT). Information sharing requires the integration of these systems. The ability to execute and process data is a key capability required by public and private organizations. The large variety of heterogeneous data, applications and business processes results in a complex and fragmented landscape that need to be dealt with when architecting and engineering new systems. Path dependencies and legacy block the easy integration, whereas at the same time new technologies appear that need to be integrated in the complex systems landscape. The purpose of this course is to teach the architectural design of innovative and large-scale ICT infrastructures and services in the light of the challenges imposed by the requirements from the systems physical, economic and social environment. Emphasis will be put on the concepts and role of ICT-architecture and modelling in order to properly design ICT solutions within a multi-actor context. For this purpose application and data integration technologies and information and system quality theories will be addressed.	
Study Goals	-The student is familiar with the state-of-the-art knowledge of ICT-architecting, design and governance within the field of large-scale ICT-systems within a multi-actor context -The student is able to describe basic concepts related to architecting and designing large and complex ICT-infrastructure and service systems within a multi-actor context. -The student should master architecture theories, methods and tools with the ability to combine and switch between them when dealing with complex ICT-problems. -The student is able to structure and analyze problems and identify dilemmas arising during the design process with regard to designing large ICT-systems within a multi-actor context. -The student is able to apply system engineering and architecture-based approaches, methods and tools to deal with problems with regard to designing large ICT-systems. -The student is able to report about the use of architectural concepts, methods and tools for translating business needs into ICT-designs within constellation of public and private actors.	
Education Method	· Lectures. The lectures are aimed at providing an overview of the knowledge of this course. During the lectures exercises are given to practice to internalize the knowledge. · Guest lectures (obliged). The guest lectures are aimed at giving an overview of the state-of-the art in practice and provide insight into practical challenges when using ICT in a multi-actor domain. · Exercises During some of the lectures assignments are given to practice what is learned. · Assignments. Various assignments are given which together create the final report. This includes developing various architectural models. · Presentation. Students should present their results during a lectures. · Report. The results should be reported in a concise report including the argumentation of design choices, the resulting architecture and an evaluation	
Computer Use	Architecture modelling in Archimate tool (http://www.archimatetool.com/) BPMN modelling using Eunomia process builder (http://www.eunomia-process.com/), Bizagi (http://www.bizagi.com/) or LucidChart (https://www.lucidchart.com/)	
Literature and Study Materials	slides book (open access e-book) Nitesh Bharosa, Remco van Wijk, Niels de Winne, Marijn F.W.H.A. Janssen (2015). Challenging the chain. Governing the Automated Exchange and Processing of Business Information. ISBN: 978-1-61499-497-8 (open access). This book covers a range from governance to technical issues and integrates them. Open access book: http://ebooks.iospress.nl/book/challenging-the-chain-governing-the-automated-exchange-and-processing-of-business-information Reader providing an outline of the area and focus on some aspects which is complemented by papers which goes in more depth.	
Assessment	Group assignment is 60% of the grade. The students work together in groups of 4-6 persons. The assignment is assessed by a report which should adhere to the APA reference standards. The report cannot be updated after handing it in. Groups that will not pass will be given the opportunity to write a new report. The exam is 40% of the grade. The exam will be digital and held in a computer room at the university. Each grade should be 5.75 or higher If one part is not passed and the other part is then the other part will remain valid for the next year.	
Elective	Yes	
Tags	Information & Communication Modelling Prototyping Technology	
Targetgroup	SEN I&C track Information Architecture (IA) students Electives for MOT, EPA, Computer science students, and non-TBM students having the required knowledge in this field are welcome	

SEN1622	I&C Service Design	5
Module Manager	Dr. Y. Ding	
Instructor	Dr.ir. G.A. de Reuver	
Instructor	Dr. Y. Ding	
Co-responsible for assignments	Dr.ir. G.A. de Reuver	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	Mobile apps, Internet-of-Things, cloud edge architectures and sensors are enabling a range of new I&C services. These services are typically offered in ecosystems of interdependent actors. However, designing services that add value for users as well as ecosystem stakeholders is challenging. In this course, you learn how to design practical ICT services and develop a valorization plan. The course contains two main parts. In the first part, it is all about designing innovative services that meet the needs of users. You will design service mockups and interview users to test your ideas. In the second part, it is about bringing that service idea to reality. You will design a valorization plan that specifies how to deal with external technologies, stakeholders and revenues. The course is practical and hands-on: you will create, test and plan your service ideas. At the same time, all is theory-informed and you will learn how to support design choices using relevant kernel and design theories.	
Study Goals	After the course, you are able to: - Analyze practical ICT services by applying concepts of design science research, action design research, agile development and service engineering - Evaluate practical requirements and make informed choices on supporting technologies and platforms, with specific attention to autonomous driving, medical IoT, AI, XR, and edge computing technologies - Evaluate an ICT-enabled service concept through semi-structured interviews - Design and illustrate a value-adding service concept driven by ICT - Design a valorization plan that explicitly covers how and when to involve external stakeholders in a design process and integrate the ICT service with value network	
Education Method	Lectures, design project, peer review, presentations, guest lecture.	
Assessment	Group essay = 3 EC; Group mockup = 1.25 EC, Individual reflection document = 0.75 EC All three marks should be >5.75. Participation in peer reviewing, guest lecture and presentations is mandatory.	

TPM409A	Digital technology entrepreneurship and management	5
Module Manager	Dr.ing. V.E. Scholten	
Instructor	J. Gartner	
Co-responsible for assignments	H. Khodaei	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Parts	The expected workload is 5 ECTS.	
Summary	This course offers the fundamentals of digital entrepreneurship and the management of emerging (digital) technologies. In this course, participants choose an upcoming digital technology of interest and assess it with different methods from multiple angles to evaluate its ability to create a competitive advantage or disrupt an industry or market. This course helps participants prepare for various career paths, from launching their product/service as an entrepreneur to making informed technology adoption and rejection decisions as an intrapreneur.	
Course Contents	Throughout the course, participants will learn to distinguish between emerging and disruptive technologies, understand influential dynamics such as innovation diffusion, hypes and emotions, and develop the methodical competencies to assess and evaluate an emerging technology in an environment of high information asymmetry. The goal is to learn to leverage the potential of emerging and digital innovations within a specific industry.	
Study Goals	Upon finishing the course, participants will be able to: LO 1 Describe the difference between emerging and disruptive technologies. LO 2 Understand current social and market dynamics that influence the diffusion and success of emerging technologies. LO 3 Develop a critical perspective on conflicting stakeholder interests. LO 4 Analyse an emerging technology's potential for a specific market/industry/company. LO 5 Adapt a business plan to the realities of digital technology. LO 6 Make informed decisions in the context of high information asymmetries.	
Education Method	The education includes: - Lectures - Self-learning study material - Formative Group assignments and hands-on exercises to apply the understanding of relevant material (contributing to 20% of the final grade). - One individual summative assignment to learn how to analyse and create an informed recommendation for action (contributing to 80% of the final grade).	
Assessment	Group work assessments Students will participate in formative group activities throughout the lectures contributing to 20% of the final grade. These activities may include minor in-class and home assignments, presentations, and collaborative projects. Depending on enrollment, group sizes will range from 2 to 6 members. To be considered successful, each assignment requires a grade of 5.0 or above. Individual assessment Students are expected to submit an individual research report at the course's conclusion, comprising 80% of the final grade. This report will be assessed based on the following criteria: - Depth and quality of analysis - Proper utilisation of theoretical concepts - Scope and quality of research and data collection - Strength and coherence of arguments and conclusions - Accuracy and comprehensiveness of the bibliography To pass, the final report must be submitted by November 10th, 2023 latest and achieve a grade of 6.0 or above. It is not possible to repeat or resubmit the report.	
Enrolment / Application	This course is part of the MOT specialisation ICT Design and Management and is also open for students from across the Delft University of Technology faculties. You can directly enrol into Brightspace. Brightspace enrollment will be activated by education support a couple of weeks before the course starts. No other means of registration, such as the app MyTuDelft, are needed.	
Special Information	Students are asked to register/enroll on Brightspace. Students are also asked to be present and active in the first lecture session, to facilitate the proper planning of the course.	
Remarks	For more information about Entrepreneurship courses: THINK - ACT - START entrepreneurial, please check https://www.tudelft.nl/dce/education.	
Targetgroup	Master students from all master programs can apply.	

Year

Organization

Education

2023/2024

Technology, Policy and Management

Master Complex Systems Engineering and Management

Economics and Finance	
MSc Coordinator	Dr. S.T.H. Storm

TPM021A	Economics and Finance	6
Course Coordinator	Dr. C.W.M. Naastepad	
Module Manager	Dr. S.T.H. Storm	
Co-responsible for assignments	Dr. S.T.H. Storm	
Contact Hours / Week x/x/x/x	0/x/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	The course Intermediate Economics.	
Summary	The two decades preceding the recent global financial and economic crisis saw an exceptionally strong growth of (global) financial markets, including a large (unregulated) shadow banking system. While the stock of global financial assets increased manifold, steeply raising the ratio of financial assets to world output, household and private corporate debt also showed unprecedented increases. The profitability of financial-sector firms (banks, investment firms) increased to a multiple of that of firms producing goods and services. What does this process of financialisation mean for the economy? What is the shadow-banking system and which purposes does it serve? Are high rates of financial return (for shareholders) conducive to economic growth and technological progress? Does the short-run (profit) orientation of shareholders square with the longer-term strategic interests of firms? What are the economic consequences of firms' increased reliance on stock market finance and volatile global financial markets? What is securitization? What are derivatives markets and how do they work? The recent economic crisis raises important questions concerning the principles on which our financial system is founded. From various sides proposals have been made for a restructuring of our financial system to reduce its instability. Is financial market regulation desirable? Which principles govern the origination and destination of financial capital in our current financial-economic setting? Are other principles of financial-economic organization available? These are the central questions addressed in this course.	
Course Contents	financialisation and shareholder value maximization; asset markets; private and public debts; liquidity and the liquidity trap; credit risk transfer instruments (securitization and structured finance); derivatives markets; options; interest rate swaps; credit default swaps; mortgage-backed securities (including cdos); asset price inflation; debt deflation; the global financial crisis; the crisis in the European Union; debt crisis; crisis economics; monetary financing; the future of finance; financial regulation.	
Study Goals	At the end of this course, students will be able to: analyse and interpret developments in financial markets from the point of view of financial investors, firms, consumers, governments and (central) banks understand the structural causes of the recent financial crisis including its persistence and evolution into secular stagnation interpret and assess proposals to arrest the spread of the crisis contribute to discussions about a new financial architecture that will bring transparency and stability to the financial-economic system and its composing elements.	
Education Method	Lectures and assignments.	
Literature and Study Materials	To be announced.	
Assessment	The assessment for this course consists of two parts; (1) the exam; and (2) a mandatory assignment. (1) The exam is a closed-book electronic (in-person, on campus) examination of 3 hours. The on-campus exam will be made in Ans. The passing grade for the exam is 5.8. Students will be offered two opportunities to do the exam in an academic year. (2) For the assignment the student has to co-author a critical book review of an economics book of choice; the critical book review will be written by a group of 4 students. The book review will be graded "pass" or "fail"; in case of a fail, the students will have to revise/rewrite their essay in order to bring it to the required level. The grade for the course TPM021A is the grade obtained in the exam, under the condition that the grade for the book review is "pass". In case of unforeseen circumstances (such as due to COVID-19), the exam will be given in the form of an on-line (remote) open-book & timed multiple-choice examination of 3 hours. Students may review their exam answers. The exam review will be held physically (and on-paper) according to the time scheduled by the lecturer(s).	
Permitted Materials during Tests	None.	
Category	MSc level	

TPM022A	Intermediate Economics	5
Module Manager	Dr. C.W.M. Naastepad	
Module Manager	Dr. S.T.H. Storm	
Co-responsible for assignments	Dr. S.T.H. Storm	
Contact Hours / Week x/x/x/x	x/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Required for	Specialisation "Economics & Finance"	
Expected prior knowledge	A completed introductory course in economics (e.g. MOT1421, EPA1223)	
Summary	This course introduces students to the basic theories and models underlying macroeconomic policymaking including New-Classical and New-Keynesian economics and Keynesian economics. Special attention will be given to central bank policy, based on NAIRU economics. We will look into causes and consequences of the rapid growth of the financial sector (financialisation) in the U.S. and Europe. Special attention will be given to the causes and consequences of the Global Financial Crisis of 2008 and its aftermath. Using this theoretical understanding we turn to an analysis of the origins of the Eurozone crisis and an evaluation of the costs and benefits of expansionary fiscal consolidation (or austerity economics). What role did the financial sector play in the Eurozone crisis? Why did Germany manage to recover from the financial crisis so quickly? Why did the crisis hit the Eurozone periphery (Greece, Italy, Portugal and Spain) so hard? How can growth in Europe be restored? To understand the role of finance in all this, we need to understand what money is, what it does and where it comes from. Therefore, the course concludes with a thorough analysis of the money creation process, and the roles played by the central bank, the state and the commercial banks in this process.	
Course Contents	NAIRU; money creation; central bank behaviour; lender-of-last-resort; fiscal and monetary policy; austerity; output gap; unemployment; regulation; labour market regulation; stabilization policy; financial markets; financialisation; global financial imbalances; Eurozone crisis; competitiveness; relative unit labour cost; liberal market economies; co-ordinated market economies; secular stagnation.	
Study Goals	At the end of this course, students will: understand how money is created and what this means for central bank policy. be able to appraise crucial differences between the main economic theories and what these mean for macroeconomic policy-making. be capable of diagnosing the origins of the Global Financial Crisis of 2008 and of evaluating policy responses to this crisis. have a keen understanding of the origins of the Eurozone crisis, as well as of how difficult it is to resolve this crisis. be able to identify key determinants of (technological) competitiveness.	
Education Method	Lectures and weekly exercise sets.	
Literature and Study Materials	to be announced	
Assessment	In normal circumstances, the exam is a closed-book electronic (in-person, on campus) examination of 3 hours. The on-campus exam will be made in Ans. The passing grade for the exam is 5.8. Students will be offered two opportunities to do the exam in an academic year. In case of unforeseen circumstances (such as due to COVID-19), the exam will be given in the form of an on-line (remote) open-book & timed multiple-choice examination of 3 hours. Students may review their exam answers. The exam review will be held physically (and on-paper) according to the time scheduled by the lecturer(s).	
Category	MSc level	

TPM023B	Cost-Benefit Analysis: Theory and Application	4
Module Manager	Dr.ir. Z. Roosenboom-Kwee	
Instructor	N. Mouter	
Responsible for assignments	Dr.ir. Z. Roosenboom-Kwee	
Co-responsible for assignments	N. Mouter	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Summary	The course TPM023B focuses not only on private (e.g. project, firm, industry) costs and benefits but also on social costs and benefits. While the private CBA only considers financial aspects, the social CBA considers both financial and non-financial aspects. For instance, in the construction of new highway, besides the financial aspects such as toll revenues or construction costs, we also assess the non-financial aspects such as safety benefits (lives saved, accidents avoided). Hence, the course actually has a much broader view than the one of the private CBA. This offers students a higher level of practical applications.	
Course Contents	Cost-benefit analysis (CBA) has become a widely used (often mandatory) tool for public policy decision making on infrastructural projects or environmental and climate policies. Through CBA, the potential current and future gains and losses of a proposed policy are identified, and then converted into monetary units in order to make them comparable with policy alternatives; certain decision rules of CBA help determine whether the project is desirable from society's standpoint. Because CBA evaluates policy and/or project proposals from the point of view of the public interest, the market prices used in CBA are corrected for possible market distortions or failure (for example, un-priced environmental damage). Nevertheless, the application of CBA is fraught with difficulties and limitations that need to be taken into account. What rates of interest are appropriate for determination of present and future costs and benefits of a policy proposal? Why and how do the private benefits and costs of a policy proposal differ from the social benefits and costs? How can we express e.g. environmental damage, or human health and life risks, in monetary units? How can we adjust market prices to account for external effects? How can we use alternative measures of welfare to identify the benefits and costs of a policy proposal? How can we ensure that politicians who aspire to use the results of CBA in their decisions can actually do so?	
Study Goals	At the end of this course, students will be able to: understand why CBA can be used to appraise projects with an objective to improve social welfare understand the strengths and limitations of CBA in project appraisal criticize and defend the CBA methodology from an ethical perspective explain the advantages and disadvantages of applying a CBA methodology in a political process with multi-stakeholders gain an understanding of the fundamentals of CBA perform steps or approaches to evaluation financial and economic values of a project interpret the results of a cost-benefit analysis	
Education Method	The lectures are on campus. In case of any unforeseen circumstances (e.g. the previous COVID-19 measures), further arrangement will be announced.	
Literature and Study Materials	* Textbook: Anthony E. Boardman, David H. Greenberg, Aidan R. Vining & David L. Weimer: Cost-Benefit Analysis: Concepts and Practice, 4th International Edition, Pearson, 2013. * Selected articles (available in Brightspace)	
Assessment	There are two types of the course assessment: 1. An individual essay (30%) 2. A written exam (70%) Both types of assessment are mandatory and the passing grade for each assessment is 5.5. Only the final grade will be entered in OSIRIS. Please register for the exam in OSIRIS on time (see: https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-examinations). For both types of assessment, students who do not pass in the first attempt are given the second attempt. Students may review their works for each type of assessment. This will be held physically according to the time scheduled by the lecturer(s). Rules for the individual essay only: For the individual essay, please be well informed that the maximum grade for the second attempt for the essay is a 6.0. Note that the second attempt for the essay is only meant for students who do not pass and NOT intended for students who have passed nor new students who would like to enrol in the course after the first attempt. Further details about the individual essay will be provided later in Brightspace.	
Category	MSc level	

Year

Organization

Education

2023/2024

Technology, Policy and Management

Master Complex Systems Engineering and Management

Advanced Modelling Gaming and Design	
MSc Coordinator	Dr.ir. I. Nikolic

SEN9110	Simulation Masterclass	5
Module Manager	Prof.dr.ir. A. Verbraeck	
Co-responsible for assignments	Y. Huang	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	Experience with a simulation environment such as Arena, Simio, Tomas, NetLogo, GAMA or MESA. Basic knowledge about probability theory and statistics.	
Course Contents	System Theory, Object Orientation, Discrete Event System Specification, Multi-Formalism Simulation, Distributed Simulation, Real-Time Simulation, Multi-Paradigm Simulation, Multi-Resolution Simulation, and Simulation Language selection will be the core topics of the course. After an introduction to system theory, the inner working of simulation environments will be illustrated on the basis of the DEVS, DESS, and DTSS formalisms. Then, possible integrating of the different formalisms will be shown. Several special topics will be taught, based on the latest research in simulation. This material will be illustrated in intensive and interactive courses. In addition to the lecture topics, several other simulation topics will be studied by groups of students, who will write a scientific paper, and present their findings in class. These topics can be focused on the MSc program that the students participate in; special topics to study are available for TIL, CoSEM, EPA, Computer Science, and other students. Finally, groups of students will study a simulation package in-depth and discuss the commonalities and differences with other packages. Again, the package chosen can be targeted at the MSc program of the students. TIL students can, e.g. study a package that is more aimed at logistics and transport, CoSEM students can focus on a package that is used in systems design, whereas EPA students can focus on a package used in policy analysis. Most packages can be made available for use from home. This course requires active participation and contains group work.	
Study Goals	After taking this course the student will have knowledge about: - systemic principles behind simulation, and clear understanding of key terms, such as state, time, transition, event, process, and activity; - internal working of different kinds of discrete event simulation languages and environments, with a focus on state transitions over time; - underlying theories and formalisms of discrete event simulation, such as DEVS, DTSS and DESS; - the DEVS paradigm, including state trajectories and transition functions; quantized DEVS (QDEVS); discretized time; - Dynamic Structure DEVS (DSDEVS), and other DEVS variants; - transformation between simulation formalisms; the formalism-transformation graph; - object-oriented simulation environments; service-based simulation; - structure and abilities of distributed simulation; the concept of HLA to manage time and state in distributed simulations; classes of functions in a distributed simulation environment; - latest research activities in the field of simulation, with research topics like web-based simulation, real-time control using simulation, agent based simulation, interactive simulation and gaming, and simulation in special domains; - examples of successful and less successful simulation studies and the learning experiences of those studies; - important similarities and differences between simulation languages, and selection methods for simulation packages. Practical skills will have been developed with respect to: - usage of at least one simulation package that was previously unknown; - writing of a scientific paper on the topic of simulation. Insight will have been gained on: - comparing and contrasting simulation languages; - the importance of time and state relationships for any type of simulation modeling; - the formal foundations of the simulation domain.	
Education Method	Lectures about state-of-the-art simulation theory Computer use for studying one discrete-event or agent-based simulation package in-depth per group Paper writing by student groups on a special topic Lectures by student groups on a special topic.	
Computer Use	Each group of students will study and evaluate a discrete-event or agent-based simulation package. Examples are Simio, Arena, Enterprise Dynamics, Plant Simulation, AnyLogic, Extendsim, Flexsim, Jaamsim, DSOL, Salabim, MESA, GAMA, Netlogo, and Repast Symphony. Also here, the simulation package to study can be specifically adapted to the group of students. Most packages can be made available for use from home.	
Course Relations	The course contains very little overlap with other simulation courses such as EPA1324, EPA1352, TB231/TB233, etc.	
Literature and Study Materials	Set of scientific papers and book chapters that will be made available through Brightspace. Typically, every class will have on average 2 papers to study for the exam. The set of papers is also available as a Reader that can be obtained on paper.	
Reader	A paper-based or PDF version of the Reader is available. The content of the Reader is also electronically available in Brightspace.	
Assessment	The mark for this course will be based on the result of two summative group assignments (paper with presentation, lab study of simulation package consisting of 4 sub-assignments that all count equally heavy) and on a summative individual written exam. The individual written exam will count for 50%, the group paper (including the class presentation) for 30% and the study of the simulation package for 20%. Each of the three marks has to be 5.75 or higher to be included in the final mark. Partial marks will stay valid for the study year in which they were obtained and the study year thereafter. The written exam takes place at the end of Q1 with a retake at the end of Q2. When a group assignment is insufficient or missing, the group will be given the opportunity to repair the insufficient or missing assignment.	
Permitted Materials during Tests	All written materials are allowed during the exam: papers, class notes, slides.	
Elective	Yes	
Tags	Challenging Intensive Modelling Programming concepts Software Stochastics	
Targetgroup	Students in EPA, CoSEM, TIL, TEL, PEL, Computer Science, and any other MSc student with some prior knowledge about	

	computer simulation.
Extra Skills Trained	Writing a high-quality scientific paper; exploring a new simulation language / package.

SEN9115	Participatory Systems	5
Module Manager	Prof.dr. F.M. Brazier	
Instructor	Prof.dr. F.M. Brazier	
Instructor	Dr. P.H.G. van Langen	
Responsible for assignments	Prof.dr. F.M. Brazier	
Co-responsible for assignments	Prof.dr. M.E. Warnier	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Course Contents	Addressing the grand challenges of our time demands participation. The objective of this course is to experience the value-based process of designing a participatory system, enabling individuals and organisations to take responsibility and act. Participatory systems are designed to foster trust, local engagement and empowerment of individuals and organisations to support self-organisation and emergence. In an iterative process with stakeholders, students (1) analyse stakeholder needs, desires and values to define their system's mission, (2) identify social and technical requirements, (3) design and evaluate one or more system concepts, and (4) propose a single solution for the challenge they address.	
Study Goals	The course focuses on the design of participatory systems combining principles from design thinking and systems thinking. Students will learn to apply their learning in a project that is centred around solving a complex problem through a bottom-up approach. Given a challenge encountered in today's society one or more knowledgeable stakeholders support students in this course. In groups of 4-5, students design a participatory system in interaction with relevant stakeholders, and implement a prototype (interactive website) that supports coordination of their participatory system. Students will show they are able to - analyse a complex context: actors, roles, information needs, relations - acquire and analyse values and requirements - formulate a value-based mission statement - explore and evaluate potential designs including explicit consideration of requirements and structures for governance and coordination.	
Education Method	Interactive lectures combine theoretical and methodological perspectives on participatory systems, with a concrete design challenge in practice. In the spirit of participatory systems, students are required to work together throughout the course.	
Assessment	The final grade of the course is based on written assignments (70%) and implementation of a supporting platform (30%). Assignments will be at individual and group level. All assignments must be completed with a passing grade (at least 5.75) to pass the course. A partial grade for the written assignments or the prototype implementation is only valid within the same academic year and cannot be compensated with a partial grade from a previous year. If not all parts are passed within the same academic year, a retake of the entire course is thus required.	
Elective	Yes	
Tags	Design Group work Projects	
Category	MSc level	

SEN9235	Game Design Project	5
Module Manager	Dr.ir. G. Bekebrede	
Instructor	Prof.dr.ir. A. Verbraeck	
Instructor	Dr.ir. G. Bekebrede	
Instructor	Dr. L.J. Kortmann	
Instructor	Dr. M. Freese	
Responsible for assignments	Dr.ir. G. Bekebrede	
Co-responsible for assignments	Dr. L.J. Kortmann	
Contact Hours / Week x/x/x/x	8/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	none	
Course Language	English	
Required for	The project is mandatory for the Advanced Modeling, Gaming and Design specialization, and open to students from all other faculties inside and outside the university.	
Expected prior knowledge	Bachelor degree	
Summary	The simulation game design project offers conceptual insights as well as hands on experience with simulation-gaming (SG). SG is an established field of practice with proven value for instance in the field of spatial and urban planning, ecology, engineering and design, public administration, business management, learning, research and consultancy. The staggering growth and success of the video gaming industry has triggered the interest in simulation games from paper based games to video games for learning and policy making - even more. Besides creativity and enthusiasm, there are no entry requirements. In this course, students acquire advanced knowledge in, and experience with, gaming-simulation for serious use, i.e. for policy- and decision support, organization and management and professional learning.	
Course Contents	Experienced speakers, from consultancy organizations (s.a. Accenture), game designers and universities, will give lectures on history, future, design, typology and facilitation of SG. (We will discuss the role of computers and video games, but emphasis is NOT on technology!) We will have ample opportunity to try out a number of (role-playing, board etc.) games that hold powerful messages about decision-making and management. Most of the work will be done in small teams, in which students will develop their own (non-digital) game to support a real-life or mock up case of policy/decision-making or organization and management. In a small group, you will also work on a scientific question related to game design.	
Study Goals	After completion of this course the student will have acquired knowledge and insights about: The history, backgrounds, key-concepts, formats and applications of simulation games. After completion of this course the student will be able to: 1. Design a (prototype) of a simulation game to be used for learning, research or intervention 2. Apply a game design cycle (or systematic design steps) 3. Define, conceptualize and construct the various game components 4. Facilitate simple simulation games	
Education Method	(Guest) lectures, workshops, game-play, game design, writing an essay Intensive workshops and group meetings Student group work	
Computer Use	Not likely; dependent on selected case. Possibility playing a computer game	
Course Relations	SEN9110 Simulation Masterclass (5 EC) Advanced Agent Based Modeling (5 ECTS) in4302 Building Serious Games (5 EC)	
Literature and Study Materials	To be announced (books and articles on gaming/simulation and game design). Reading materials will be made available via Brightspace.	
Assessment	Your final grade will be based on the following points: - Attendance and active participation during classes and workshops (required, no grade) - Grade for the game design assignment (group work 4-6 persons) counts for 80% of the final grade - Grade for the essay assignment (group work 2 persons) counts for 20% of the final grade - Facilitation of the warm-up game (same group as game design assignment) have to be done (required, no grade) Final grade of the course is $0,8 \times \text{grade game design} + 0,2 \times \text{grade essay assignment}$, and student have to active present and facilitated a warm-up game. To sufficiently finish the course the weighted average have to be 5,75 or higher. Retake: - For practical reasons, there is no retake opportunity for the game design assignment - The essay can be partly repaired, the essay can be improved, maximum grade of the essay is a 6 - If a student can not join the assigned date for the facilitation of the warm-up game, we will look for an additional opportunity during the course weeks.	
Enrolment / Application	Registration through Brightspace enrollment	
Special Information	Geertje Bekebrede (g.bekebrede@tudelft.nl)	
Remarks	Attendance during the class meetings is required.	
Elective	Yes	
Tags	Challenging Design Group work Intensive Project	
Targetgroup	Master students of the modelling, simulation and gaming specialization(MSG) Elective course for students CoSEM, EPA, MOT, other faculties such as Architecture, Industrial Design, EWI, Civil Engineering etc.	
Category	MSc level	

Year	2023/2024
Organization	Technology, Policy and Management
Education	Master Complex Systems Engineering and Management

Supply Chain Management	
MSc Coordinator	Ir. M.W. Ludema
Contact for students	M.W. Ludema phone 81885 m.w.ludema@tudelft.nl
Program Title	Supply Chain Management
In association with the University of	na
EC Program	3 time 5 ects = 15 ects
Program Start (Study Year)	2017-2018
Specialisation	This specialization Supply Chain Management consists out of 3 courses and is intended to teach students in operational, logistical and supply chain management with the use of several techniques like: optimization models, analysis and design frameworks and gaming. The Supply Chain Management specialization provides an integrated framework for understanding the product and information flows in and between companies that are part of global demand/supply chains in a rapid changing world. It shows how changes in the economy lead to changes in industries and local policies (with a focus on the Netherlands). It focuses on the understanding of the integration of enterprise functions as strategic and tactical management, forecasting, planning and scheduling, and on the co-operation and alignment between companies in supply chains these enterprises are part of, and its needs for positive policies and facilitating infrastructures. It goes into the most important innovations in the world of supply chain management. In general the specialization offers insight in a variety of proven management approaches in industries that produce high-quality products that focus in reducing costs and improving services on all relevant levels of an enterprise and their supply chain partners. It also focusses on the multi-actor planning and assessment of possible choices in an economic and societal context. Next to this it focusses on several innovations available in the field of logistics and supply chain management and possible benefits of application. A special focus will be on the big changes that are visible in our economy and how the Dutch local, regional and national governments should act to make logistics one of the TOP-sectors.
Administration by the Faculty of	TBM
Intended for	MOT Specialization and as electives for other students.

SEN9720	Logistics and Supply Chain Innovations	5
Module Manager	Ir. M.W. Ludema	
Instructor	Ir. M.W. Ludema	
Responsible for assignments	Ir. M.W. Ludema	
Co-responsible for assignments	Dr. J.H.R. van Duin	
Contact Hours / Week x/x/x/x	6/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Summary	Supply chain management is about the exchange of product, information and money between companies or organizational entities that are part of a supply chains or demand/supply network. Supply chain management contributes to the competitive strength of companies in the market place. The supply chain management function co-ordinates and controls operational business value adding processes in and between companies. For a large extent the supply chain management function determines the flexibility of companies to respond to market changes and determines the efficiency of the internal processes within companies, as well as the inter-company processes. Logistical and supply chain decision-making frequently relates to situations in which conflicting wishes and desires from various parts of companies or organizations have to be brought together cohesively. Gaining insight into the performance indicators of a company/organization, its logistical costs, customer expectations and the opportunities to improve these aspects is of vital importance for survival of companies/organizations. All engineers will encounter to a larger or lesser extent logistics and supply chain management in the business and institutional environment in where they will be working in, or in where they will be carrying out their assignments.	
Course Contents	This course contains logistical, supply chain and innovations in logistics and supply chain topics. Logistics Topics: - The significance of the logistics system of companies in supply chains; - Methods for analyzing logistics systems of companies; - The role of inventories in logistics systems of companies; - The role of transport between logistics systems of companies; - Connecting purchasing and customer service within companies; - Value added logistics, outsourcing and third party logistics; - Performance Measurement in logistics systems of companies; - Logistics control and information systems; - Design of logistics networks and locations for distribution centers. Supply Chain Topics: - Global trade lanes, industrial clusters and supply chains; - Strategy issues and the supply chain; - Organizational issues and supply chains; - Supply Chain Mapping, Benchmarking and the SCOR-reference model; - Physical and Virtual Value Systems (MRP-I, MRP-II, ERP and APS); - Supply Chain Value Adding Processes; - Lean and Agile Supply Chains and the strategic role of decoupling points; - Supply Chain Collaboration (Supply Chain Partnerships & Alliances); - Supply Chain Analysis; - Supply Chain Design. Innovations in Logistics and Supply Chain Management: - The Last-Mile of Supply Chains; - Multi-modal and Synchro-modal Logistics and Supply Chains; - Customs and Trade Compliance; - Supply Chain Integration and the use of Cross Chain Control Centers (4C); - Service Logistics (Asset Management, Maintenance Management and spare parts logistics); - Sustainable, Green and Closed-Loop Supply Chains within a sustainable economy, - Smart Supply Chains and the Internet of Things; - Supply Chain Finance; - Smart Contracts & Block Chains Technology and the Supply Chain.	
Study Goals	This course covers the supply chain management function in and between companies. This course will provide the student insight in the theoretical background of the supply chain, its current use and expected developments and a wide set of technical and informational innovations that companies consider to implement. The student learns and puts into practice the basic theoretical skills of a supply chain analyst and engineer and can show the contribution of relevant logistics and supply chain innovations. Theoretical knowledge and practical skills will be explained and exercised by means of a set of relevant cases.	
Education Method	The course contains 17-24 lectures (2 or 3 time 2 hours each week), in-class discussions about presented topics and distributed articles and cases. During the course assignment groups of 4-5 students will work on a logistics and supply chain analysis/engineering assignment. In the final weeks of the course the assignment groups will compile their results in a final report. Furthermore the course contains self-study and some group assignment feedback sessions. The draft report of the course assignment groups will discussed with the groups prior to final hand-in.	
Literature and Study Materials	This course works with a virtual reader. Suggested Course Materials is continuously under construction (see manual on Bright Space), relevant articles will be related to weekly lectures.	
Practical Guide	See Bright Space for the updated course and assignment manual and course assessment issues.	
Prerequisites	This course builds on the knowledge gained during courses like operations management, production logistics, logistics management or equivalent courses. Students without (sufficient) knowledge and background are advised to spend additional study time during the first two to three weeks of this course. Please consult the module manager / lecturer during the first lecture or earlier. During the first two lectures an overview of relevant operational and logistics management issues will be provided by means of intensive lectures and a few in-class assignments.	
Assessment	Final deliverable and group assessment: Students will work in groups on a large group assignment in one of the following example domains:	

Transportation Equipment Supply Chain;
Construction Supply Chain;
After-sales service /spare parts supply chain for the maintenance of durable goods;
Supply chain of conditioned goods for the retail market;
Supply chain of the inter city last mile;
Care supply chain of the Medical Equipment;
Other issues in respect to the interest of the student.

A final report including analysis of the current supply chain, impact assessment of a chosen logistics and supply chain innovation and a design of a feasible future supply chain.

SEN9725	Supply Chain Gaming	5
Module Manager	Ir. M.W. Ludema	
Instructor	Ir. M.W. Ludema	
Responsible for assignments	Ir. M.W. Ludema	
Co-responsible for assignments	Dr. J.H.R. van Duin	
Contact Hours / Week x/x/x/x	0/6/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Required for	Required (as a specialization elective) for :	
	MOT Specialization Supply Chain Management (to be taken after SEN9720 (Q1) or after sufficient other logistics (fundamental) courses.	
Expected prior knowledge	This course builds on the knowledge gained during courses like operations management, production logistics, logistics management, supply chain management or equivalent courses.	
	Students have ideally followed the course SEN9720 Logistics and Supply Chain Innovations (first quarter). Understanding of supply chain management trade-offs is necessary to be able to understand, analyze and develop supply chain game sessions.	
	Students without this prior knowledge have to spend additional study time prior to this course. Please consult the module manager before the start of the course.	
	During the first two lectures an overview of relevant operational, logistics and supply chain management issues and trade-offs will be discussed. By means of small in-class assignments students discuss possible serious (board) game development projects and themes for their group assignment.	
Summary	Supply chain management considers product and information exchange between companies in supply chains. It determines a company's flexibility to respond to market changes and determines the efficiency of the internal processes within that company, as well as the inter-company processes. Logistical and supply chain decision-making frequently relates to situations in which conflicting wishes and desires from various parts of supply chain organizations have to be brought together cohesively. By means of analyzing existing games and developing supply chain game sessions more awareness and insight in these conflicts can be gained. If supply chain stakeholders play games they will improve their awareness about supply chain mechanisms and possible means of actions to improve the alignment of supply chain processes.	
	A serious game is a suitable tool to create awareness about relevant supply chain issues. It levels the experience of participants (either novice or experienced from the field). After playing a serious logistics or supply chain game it contributes to discussions and facilitates the exchange of knowledge and personal experience. Short introduction to some relevant supply chain concepts, playing a game, and a clear introduction and participating in relevant discussions together are part of what is called a game sessions.	
	Playing games is a good learning exercise, but designing and developing helps the game developer to understand the essence of the trade-off from the real world. One of the approaches that can be applied for design and development of a serious design session is the Triadic Game Design (TGD)-approach of Hartevelde (2011). Games generally include three worlds, namely reality, meaning and play. These worlds must be in balance to provide a serious game which will be fun to play and also to learn from. The world reality is needed so the players of the game can relate to elements of the real world, this will make experience in the game intuitive and understandable. Meaning is needed, because the game should have a learning purpose. At last the world of play is needed to make a game playable. The game should have clear goals and rules. The play factor will guide players in the right direction of the learning goals of the game. The three worlds are equally important, therefore they should be balanced. Balancing of these three worlds, is what makes designing a game session complex. Other approaches and techniques will be added to this TGD-approach.	
	During the design, development and execution of a game session the players have to gain knowledge about a certain logistics or supply chain issues and/or trade-offs or the effect of changes in their part of the supply chain relative to changes in other parts of the supply chain. This could for example be the implementation of technological innovations to improve traceability in the in the logistic supply chain, the optimization of warehouses, the best collaboration approach, implementing IoT within the supply chain, the understanding of the benefits of synchro-modality. In general playing and designing games helps to create a better awareness about logistics and the supply chain in general and certain addressed problem areas in particular. In some cases games can help to gather data for analysis and research purposes.	
Course Contents	Common trade-offs in Logistics and Supply Chain Systems:	
	1) Fundamentals of logistics and supply chain systems;	
	2) Methods for analyzing logistics and supply chain systems on multiple levels;	
	3) Strategic, Tactical and Operational Trade-Offs in logistics and supply chain systems;	
	4) Game Concepts to facilitate improvement of the awareness of trade-offs in logistics and supply chain systems.	
	5) Discussing trade-offs and ideas of game concepts.	
	Game Essentials:	
	- Sharing (board, card and role-playing and computer) game experiences and discussing fundamentals;	
	- Flow experiences (Mihaly Csikszentmihalyi);	
	- Game Concept Decompositions and Construction;	
	- Systems and Process modeling and analysis for game design;	
	- Triadic Game Design approach;	
	- Design Methods (from established needs to workable results);	
	- Playing and Learning:	
	- Lemonade Tycoon;	
	- Rise of Industry;	
	- Playing MSc-Students developed games.	
	Game Design Assignment:	
	- Towards understanding a (chosen) supply chain problem situations;	
	- Developing Game Session Objectives;	
	- Setting Boundaries;	
	- In Depth Analysis of the problem situation and value of possible solutions;	
	- Setting Game Design Requirements;	
	- Developing the Game (Session) Concept;	
	- Generating Solutions;	

Study Goals	- Evaluating Solutions; - Game Session and Play (Delivery, Evaluation); - Reporting about the problem situation, gamed design, game development, game demonstration, game evaluation and improvements and final game deliverable. This course gives students knowledge and understanding in theoretical and practical issues concerning the design, development and execution of supply chain gaming sessions. The fundamentals of game session design and its application area logistics and supply chain analysis and management will be discussed and practiced. Important and problematic logistics and supply chain issues and opportunities for innovations will be explained and related to game sessions design alternatives. Upon completion of this course the student should be able to: - Elaborate on how companies can develop and maintain their position as an adequate supply chain member. - Elaborate on the inter-relationship between processes in the silos of business logistics systems and the need to align them. - Understand important logistics and supply chain trade-offs that are suitable to incorporate in supply chain games. - Suggested and develop supply chain (re)design directions for the future of the companies in a supply chain that need awareness between supply chain partners that can be facilitated by means of game sessions. - Demonstrate that they can decompose logistics and supply chain activities in meaningful game elements. - Understand the fundamentals of strategic game design approaches and elaborate on reality, meaning and play aspects and their relationships. - Develop and apply a practical descriptive and prescriptive design method for game session design. - Design, development, demonstrate and evaluate a functional supply chain game session including an playable logistics / supply chain game.
Education Method	The course contains 14 lectures (1 time 2 hours each week and once 3-4 hours each week), in-class discussions about presented topics and of distributed articles, cases and reviewing and playing games. The first two course weeks will be class room lectures, the third, fourth and fifth course weeks will be a combination of class room lectures and working on the group assignment, and the sixth and seventh week will be working on the group assignment. The last lecture will be playing and assessing each others game session. At the the end of the educational period students have to hand in the developed game and accompanying report and game session manual.
Assessment	During the course assignment groups will work on the design, the development and the execution of a supply chain game sessions on a relevant supply chain issue. Assessment: Game design consisting of the design project progress and the final delivered supply chain game and accompanying report (including a manual), [90% of 100%]. The game demonstration and deliverable will be assessed by the students playing the demonstrated game, [10% of 100%].

TPM028A	Decision Making in Multimodal Transport Systems	5
Module Manager	M.Y. Maknoon	
Instructor	M.Y. Maknoon	
Responsible for assignments	M. Farhani	
Co-responsible for assignments	J. Rezaei	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	- Knowledge in operations research (mathematical modelling, integer programming) - Basic knowledge of optimization solver (e.g. CPLEX) and programming In case of deficits, students need to invest additional time to acquire these knowledges.	
Course Contents	Coordinating physical, information and decision flow is a focal point in managing multi-modal freight transportation systems. By having a global view on the entire supply chain - from raw material acquisition to the end customer delivery multimodal systems aim to provide a rapid, efficient and low-cost solution for freight transportation. On the other hand, social and environmental concerns have a central role in political and economic decision making. These concerns result in laws and regulations that affect the production of goods and services. Therefore, in addition to provide efficient, reliable and green services multimodal systems should be flexible enough to respond to short and long-term environmental, social and political changes. To address these challenges, effective, powerful and intelligent methods and techniques are required to analyse, plan and manage multi-modal systems. Operations research is at the heart of this intelligence. Operational research models and methods, integrated with enterprise information management systems, analyse and transform data into useful information for planning and management. This course focuses on the multimodal freight transportation. In particular design, planning and operation of multimodal network. The models and techniques taught in this course deal with a wide range of problems. From the long-term physical design and operating structure to the transport route selection, service mode and timetabling.	
Study Goals	The course focuses on multimodal transportation network. It presents the main models and methods to analyse and solve problems arises in this context. The course aims at knowledge acquisition and development of problem solving skills. The goal is to enhance students analytical and modeling capacities in research and practice.	
Education Method	The course is based on active participation. In case of unforeseen circumstances or measures resulting from COVID-19, student participation will be facilitated by on-line platforms (e.g. Microsoft Teams, Skype). Readings will be given for each session and presentations will be supplemented by discussions with the students.	
Assessment	The course is assessed based on two mid-term take-home exams (15% each adding up to 30% of the total grade) and one closed-book final exam (70% of the total grade). In case of unforeseen circumstances or measures resulting from COVID-19, the final exam will be take-home and open-book. To pass this course, students need to get more than 5.5/10 for the final exam. The results of mid-term exams are valid only for the current year and are not transferable to the next academic year. In addition, re-take is not allowed for the mid-term exams.	
Elective	Yes	
Tags	Design Mathematics Modelling Optimalisation Programming Transport & Logistics	
Category	MSc level	

TPM040A	Logistics Systems Engineering	5
Module Manager	Ir. M.W. Ludema	
Responsible for assignments	Ir. M.W. Ludema	
Co-responsible for assignments	I. Dursun	
Contact Hours / Week x/x/x/x	0/x/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Summary	This course teaches the student to facilitate a multi-actor and societal environment by specifically designing simple and complex (service)systems in the Transport, Infrastructure and Logistics domain, dealing with substantive as well as procedural aspects of the systems (logistics) engineering process. The focus is on Transport & Logistics Systems from the perspective of Systems Engineering and Integral Systems Design. The course considers a wide range of tools and methods that guides policy-making, opportunity statements, the asset acquisition process, need-analysis, requirement statements/analysis, conceptual design, functional design, behavioural design, physical design and Life cycle engineering of the Transport & Logistics Systems Engineering that can be used during several engineering phases. Transport & Logistics System typically have a long life of type and often are (re)designed to meet changing demands. Attention is given to the context in which a Transport and Logistics System must perform (in past and future) often in respect to conflicting situations. Tools and methods will be applied to a wide range or TIL-Systems, like public rail transport, high way projects, container shipping, container handling systems, logistics distribution systems, freight delivery services, etc. Attention will be given to the Management of Service Systems that together with the physical Transport and Logistics Systems fulfill mobility demands for mainly goods, such as decision support systems, financing systems, event management systems, tracking and tracing systems and steering instruments to promote sustainable good mobility. Special attention will be given to asset management, logistics support and X-ability factors, like: availability, reliability, maintainability, risk, sustainability, constructability and producibility related design criteria. During the course, students practice activities and tasks of individual phases of a logistics systems engineering process. The course is a mixture of lectures and assignments.	
Course Contents	The course starts with a quick review of the origin, fundamentals, and qualities of the field systems logistics analysis and engineering design. An overview of the role of meta engineering approaches in respect to more disciplinary (often more detailed) engineering approaches is given from a project management perspective. During the lectures the student will be involved in small classroom assignments. The overall course is practical in nature. In working assignments the students will be working in small groups and will be engaged in a Systems Logistics Engineering Process and work on their own (group) design/engineering two-faced actor/stakeholder project setting. Theoretical Line: Short recap on engineering design, system engineering fundamentals and the role of stakeholders in logistics engineering projects. Design and development of product in production logistics environment. Identification of analysis and design activities within the engineering of Supply and Transport - Logistics systems. Functional Discovery and Analysis of a range reference Transport and Logistics Systems (e.g. distribution network, urban logistics, container shipping services, cargo tracking en tracing system, high-way covering and connections, transport system interaction in new development area, airtaxi-service, warehouse systems, cross-dock centers etc.) as a preparation for the systems logistics engineering exercises. The interrelation between Systems Engineering, Asset Management, Maintenance Management within supply chain and logistics systems. Introducing a Practical Framework for Logistics Systems Engineering Life Cycle Engineering, RAMS, Service Logistics, RAMS and Design-to-X-ability. Practical Line: Formalizing the Practical Framework for Systems Logistics Engineering Opportunity Statements and Need Analysis Concept Descriptions Mission Analysis Operational Concept Supportability Concept Informal Stakeholder Requirements Formal Design Requirements Behavioral/Temporal Design Functional Design Structural/Physical Design Interface Design and the role of Interface Management Configuration Management within Systems Logistics Engineering Processes Verification and Validation within Systems Logistics Engineering Processes Detailed design of one of more sub-systems and/or chosen component Re-engineering of the design from a specific Design-to-X-ability perspective Presentation of the final group assignment results Excursion(s) when initiated en co-organized with students	
Study Goals	The course aims at providing knowledge and skills to the student regarding the specific design of complex and dynamic systems in the domain of transport, infrastructure and logistics in essence logistics systems. The course focuses on the integral design of logistics systems (as part of supply chains) for (transport) of goods in respect to a design/engineering assignment, including the necessary infrastructure and its interfaces with the existing systems (that are not part of the systems logistics engineering assignment). On completion of the course, students are knowledgeable of and partly able to: Diversify notions, approaches, design principles and assessment techniques in the design of complex systems in the Transport Infrastructure and Logistics domain, from a perspective of integral design and systems (logistics) engineering. Understand and indicate the complexity and dynamics of an open and structured Transport Infrastructure and Logistics Systems architecture, focusing on the layered nature and interface issues between these layers and the operational environment. Incorporate the characteristics of Transport Infrastructure and Logistics Systems such as incremental or conceptual change processes, technological innovation, multi-actor participation, open planning process decision environments and project management configurations. Understand and being able to work on constraints of a societal, legal, operational nature in accepting and sustainably nature in accepting and sustainably implementing in a socio-economical and spatial environment under conditions of societal constraints and requirements. At the end of this course the student will be capable of: Assessing the scope and applicability of the most important methods and techniques applied in the various phases of the design process and assessment of the design products in Logistics Systems Engineering Processes;	

	Addressing the most important steps in the different phases of a generic systems logistics engineering process; Understanding of both the consumer and designer perspective in a large system logistics engineering process; Dealing with conflicting interests and design aspects at the various phases of the design process for new innovative systems engineering processes; Assessment and integration of constraints in the integral design of systems engineering systems, such as environmental, safety, spatial and sustainability requirements; Participating in the mainstream of the design of aspect oriented or functional systems of the integral design, such as city logistics, decision support systems, special transport systems, control and management systems, container handling, regional distribution networks, cargo (air-)ports, road and waterway infrastructures, to name a few; Drafting and assessing system performance indicators of a legal and procedural nature, operating envelopes, safety cases, impact assessment and incident handling strategies; Participating in decision making processes relevant to the design processes for Systems Logistics Systems; Develop functional, temporal and physical design and be able to evaluate the interrelationships between these designs. Perform final verification and validation of delivered designs against earlier stated requirements; Understand the implication of designing and the resulting design when taking a Design-to-X-ability perspective.
Education Method	A mix of lectures, assignments, discussions two times a week, 2 hours each. A large part of the course is aimed at playing and experiencing a systems logistics engineering process for a dedicated project from both a consumer as a designer perspective. The course can be seen as preparation for (interdisciplinary/Design) Projects and/or Design focused Thesis Project or a career in an engineering (consultancy) firm or company engaged in the production or acquisition of durable (logistics) assets. The two-weekly process leads the design/engineering of a logistics systems for each group from both the the client and producer perspective. Groups will request logistics system from another group. Will designing their own system the will be supervised by another group and will supervise the design process of another group.
Reader	The course applies Bright Space facilities and electronic documentation. A virtual reader containing the study material chapters from books and journal articles or references to them will be made available for this course.
Prerequisites	This course builds on the content and knowledge gained during logistics analyses/management and other (domain) engineering courses. It should be preceded by logistics courses and engineering and/or design courses. If a student has not a design/engineering or logistics background additional self-study may be required during the half of the course. In the case both are missing considerable extra pre-learning should be expected of it may be advised to take this course after other courses. Please consult the module manager before the start of this course regarding pre-knowledge issues.
Assessment	The assessment is a final report contains three parts. The first is about the logistics system engineering process followed (build upon the lectures and course material. (25%) The second contains the results of the assignment given and the guidance and feedback during the development process of other groups progress and outcomes. (25%) Third the result of the groups logistics systems engineering process and its (intermediate) results. (50%) Minimum results for each of these parts should be $\geq 5,0$, final results should be $\geq 5,75$.

Year	2023/2024
Organization	Technology, Policy and Management
Education	Master Complex Systems Engineering and Management

Cyber Security	
MSc Coordinator	S.E. Parkin
Minor Content 1	https://www.tudelft.nl/studenten/faculteiten/tbm-studentenportal/onderwijs/master/specialisation-electives

Year

Organization

Education

2023/2024

Technology, Policy and Management

Master Complex Systems Engineering and Management

Compulsory (three courses of the following)	
MSc Coordinator	S.E. Parkin

TPM010A	Cyber Crime Science	5
Module Manager	Dr. R.S. van Wegberg	
Instructor	Dr. R.S. van Wegberg	
Instructor	F.E.G. Miedema	
Co-responsible for assignments	Prof.dr. M.J.G. van Eeten	
Contact Hours / Week x/x/x/x	0/0/3/0	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	- Introduction to crime science (i.e., how do scientists study crime), - Socio-technical research skills (i.e., research design, data collection, data analysis), - Studying cybercrime from a socio-technical perspective (e.g., evaluate the effectiveness of phishing detection).	
Study Goals	The student will acquire: - A good understanding of the theoretical principles of crime science, - An appreciation of the whole spectrum of different cybercrimes, - Skills necessary to empirically research cybercrimes.	
Education Method	The course runs from Q3 till Q4. In Q3 the course consists of both lectures and feedback sessions. In Q4 students are expected to work on their research paper, give and receive (peer)feedback and present their results in a conference.	
Assessment	1. Research proposal (groups of 3 students) pass/no-pass; 2. Team presentations pass/no-pass; 3. Reviews of other papers pass/no-pass; 4. Final conference grade - Research paper - Presentation	

TPM020B	Economics of Cybersecurity	5
Module Manager	S.E. Parkin	
Instructor	Prof.dr. M.J.G. van Eeten	
Responsible for assignments	S.E. Parkin	
Co-responsible for assignments	Prof.dr. M.J.G. van Eeten	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	With a significant increase in high-profile data breaches and cybersecurity threats in the last couple years, it is critical for businesses to learn about the costs and investment decisions around securing their online systems. While many businesses think of cybersecurity as a technical problem, this course broadens that view and shows that security failures are caused as often by bad business decisions and incentive systems as by bad technical design. This course provides an introduction to the field of the economics behind cybersecurity. It will provide you with the economic concepts, measurement approaches to make better security decisions, while helping you to understand the forces that shape the security decisions of other businesses, products and services.	
Study Goals	The student will: -Gain a sound understanding of the economics of cybersecurity as a systems discipline, from security policies (modelling what ought to be protected) to mechanisms (how to implement the protection goals). -Obtain foundational skills in identifying and assessing data on complex decisions around information security issues. -Gain insights into the design of effective policies to enhance and maintain cyber security must take into account a complex set of incentives facing not only the providers and users of the Internet and computer software, but also those of potential attackers. -Learn to apply economic analysis and analytic approaches to the open issues and pending activities in cybersecurity.	
Education Method	Structured lectures, background reading, group discussions, and regular non-graded quizzes.	
Assessment	Group report and presentation (60%); written individual assignment (40%). There is not a resit option for the group assignment; the resit option for the individual assignment is an oral exam.	

TPM025A	User-Centred Security	5
Module Manager	S.E. Parkin	
Instructor	S.E. Parkin	
Responsible for assignments	S.E. Parkin	
Co-responsible for assignments	Prof.dr. M.J.G. van Eeten	
Contact Hours / Week x/x/x/x	0/2x2/0/0 This course is shifted to Q2 and is replaced by TPM030a in Q4	
Education Period	2	
Start Education	2	
Exam Period	2	
Course Language	English	
Course Contents	Cybersecurity attacks on organizations and services increasingly target the people who are involved, such as employees in companies or home computer users. Solutions to address this problem need to be feasible for individuals, alongside other user priorities (such as completing paid work). Without consideration of user skills or needs, solutions such as security training or browser warning pop-ups add effort and burden to the user, and encourage less secure behaviours as workarounds (such as writing down difficult-to-remember passwords to ensure system access). In this course, students will learn about the user perspective of security technologies. This will leverage key human factors concepts, around security usability and its connections to decisions in policy, planning, and technology investment. This will include how to assess a security solution from the perspective of different kinds of technology users and their tasks. By assessing the strengths and weaknesses of particular security mechanisms for users in practice (policies, training, monitoring, etc.), security implementation and management decisions can be made which better fit the context in which mechanisms are used. This can ensure long-term security which better matches the requirements of a particular user organisation or community.	
Study Goals	The student will: -Gain a sound understanding of security usability as a discipline, from assessing the context of use alongside primary tasks, to identifying the time and effort costs to users. -Obtain foundational skills in matching security technologies and processes to user abilities, motivations, and perceptions of security-related technologies. This will include examination of authentication and identity technologies, employee security training, trust in online contexts, privacy-related evaluations of personal data disclosure, etc. -Gain insights into the design of effective security technologies and their deployment from a human-centred perspective, to inform interface and policy design and ensure compliance with security expectations. This will include management of security behaviour change activities, and identifying how to position expert support to aid users, e.g., effective communication of policy, use of persuasive design in security.	
Education Method	Structured lectures, background reading, problem-driven group discussions	
Assessment	Individual assignment on assigned reading material (20%); a final individual assignment is an essay with feedback moment (80%). The resit option for both assignments is an oral exam.	
Elective	Yes	

TPM027A	Cyber Risk Management	5
Module Manager	Prof.dr.ir. P.H.A.J.M. van Gelder	
Instructor	Prof.dr.ir. P.H.A.J.M. van Gelder	
Responsible for assignments	Prof.dr.ir. P.H.A.J.M. van Gelder	
Co-responsible for assignments	Dr. A.P. Afghari	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	MOTIVATION: The challenge of selecting the optimal technical, organisational, legal, and other preventive and repressive measures to reduce cyber risks to acceptable levels can only be understood in the context of the application of Cyber Risk Management. Risk Management is about analysing the relationships between threats, incidents and risks (here in the complex world of cyberspace), based on which an adequate set of countermeasures can be designed. SYNOPSIS: Risk (= the potential lo losing something of value) can manifest itself in cyberspace in all kinds of ways: values at stake are financial wealth, health, physical condition (of people, materials, goods, infrastructures, etc.), well-being, reputation, privacy, trust, etc. Based on a conceptualisation of cyberspace and its various sub-domains (discussed in the project week of year 1), we introduce risk assessment approaches, both of qualitative and quantitative manner, illustrated with case studies. CONTENT: Cyberspace Risk concepts Probabilistic and adversarial risk System and attacker models Requirements and policies Risk assessment methods Qualitative and quantitative methods Measuring security and effectiveness	
Study Goals	After this course, students are able to: Understand and apply risk concepts to (possible) incidents in cyberspace Understand the theoretical principles of cyber risk management Explain the role of adversarial risk models, system models, and attacker profiles in cyber risk assessment Analyze and apply state-of-the-art cyber risk assessment methodologies to (complex, multi-step) cyber incidents Analyze the (expected) effect of measures that help to prevent the occurrence of cyber incidents Justify investments in cyber security	
Education Method	LECTURES: There are two lecture slots per week EXAM: There is a written exam at the end of the course. LANGUAGE: The course is taught in English. LECTURERS: Prof. dr. ir. Pieter v. Gelder (TUD/TPM) and guest lecturers.	
Assessment	#### Assessment Grading will be based on the individual written exam at the end of the course. Final mark of this course is determined by the grade of the individual exam (100%). The minimum grade for the individual exam to pass this course is 6.0. The grade for the individual exam is 2 years valid. The individual exam is summative. Detailed feedback on the individual exam can be reviewed by students by contacting the teaching staff of this course. #### Resit The students are allowed one re-sit per examination. It is not allowed to re-sit an examination or assignment for which they have received a pass (6,0 or higher). It is allowed to re-sit an examination which they haven't done during the first occasion. The re-sit format needs to be discussed with the teacher of the course in line with examination regulations. In case the student is granted an extra re-sit by the Board of Examiners, this re-sit has to take place within the study year 2023-2024.	
Elective	Yes	
Targetgroup	MSc	
Category	MSc level	

TPM030A	Introduction to Cloud as Infrastructure: The effects of the new business of computing on practice	5
Module Manager	Dr. F.S. Gürses	
Instructor	Dr. F.S. Gürses	
Co-responsible for assignments	Prof.dr. M.J.G. van Eeten	
Contact Hours / Week x/x/x/x	0/0/0/2x2 This course is shifted to Q4 and is replaced by TPM025a in Q2.	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	What if intelligence is not the main driver of AI? What if data is not the primary resource of the tech sector? What if mobile phones are not actually personal devices? What if software as a product no longer exists? And what if the dominance of the major tech companies remains a political problem even after we reduce surveillance and discrimination, or apply privacy or fairness by design? These are some of the questions we will ask in this course as we explore the technical, business, and political implications of a specific form of contemporary computational infrastructure (CI [0]): cloud + mobile. The thesis of the course is that CI is an environment for (experimentation with) software production, i.e., an environment for the creation of value through the engineering of software based services. We will study how CI, developed and controlled by a handful of corporations, is vying to become the default environment for a movement that will transform relationships of production --be it digital or physical production-- across the globe. We begin the course by examining common assumptions about the business of computing. We will consider the reigning belief that cloud is somebody else's computer or another node on the internet. We will also unpack the wildly popular take that tech companies generate revenue from personal data and surveillance through personal devices, with privacy as the obvious antidote to individual harms and power asymmetries resulting from the accumulation of information. The second part of the course presents a hard challenge to these assumptions by comparing the past to the present. We will study concrete shifts in the production of software that gave rise to CI such as the turn from personal computers to (mobile) devices attached to the clouds; waterfall methods to agile/lean methodologies; monolith architectures to (micro)services; instruction-based programming to AI. In part three we explore whether and how the mega-corporations that control CI aim to become the environment for software production. We will also look at how the resulting modes of software production go hand-in-hand with transformations, like flattening of the Internet or the move from general-purpose to specialized chips. The final section of the course explores how software production, passing through CI, promotes what we call agile production. Together, we will explore how agile software production enters other industries to become a more general model of production made possible by the rise of CI. We will end the course by examining the long-term objectives of the companies that currently dominate the business of computing. By the end of the course, you will understand why CI is one of the most significant drivers of (geo)political, economic, and technological developments in our day by looking at the way it shapes European digital policies. We will also unpack why CIs hold over software production places important limits on data protection and privacy engineering. [0] A mainframe, server rack, or even a personal computer can serve as any entity's computational infrastructure. In this course, however, we will use the term CI to refer to the dominant and corporate form of materially organizing computing for the development and global delivery of digital services, for example AWS by Amazon, Azure by Microsoft, or mobile devices offered by Apple or Google. Using popular terms, CI is cloud infrastructures plus end- or mobile devices as their accessories.	
Study Goals	In this course you will discover how the way we produce software impacts how society is organized by reaching the following learning-objectives: Demonstrate a basic understanding of the history of computing through exemplary shifts in software production Develop an understanding of how corporate computational infrastructures serve as a powerful environment for software production Analyze how privacy or data protection by design are weakened in their protective role due to the rise of computational infrastructures Speculate and predict possible futures of software production in CI and its ambition to transform global economic production Interpret strengths and weaknesses of current EU digital policy vis a vis computational infrastructures	
Education Method	Lectures, work groups, case studies, presentations, self-study, writing.	
Prerequisites	It is advantageous to have some understanding of or an interest to learn about software engineering and computing to attend the course.	
Assessment	10% engagement 50% 5 assignments 40 % oral exam The total of the assignments as well as the oral exam must be graded with at least 5.0. The final grade consists of the average of the three parts (engagement, assignments and exam), and the rounded grade must be at least 6.0 (5.75 rounded up).	
Elective	Yes	
Targetgroup	This course is listed as part of the Cybersecurity Specialization and is open to students participating in that program TU Delft and Twente. The course is also open to Master Students at other TBM programs, including specializations listed https://www.tudelft.nl/en/tpm/education . If you are not in any of these programs and would like to attend the course, please contact the course teacher per email.	
Category	MSc level	

Year

Organization

Education

2023/2024

Technology, Policy and Management

Master Complex Systems Engineering and Management

AI Specialisation	
MSc Coordinator	A. Ghorbani

Year	2023/2024
Organization	Technology, Policy and Management
Education	Master Complex Systems Engineering and Management

EPA and CoSEM students

SEN9110	Simulation Masterclass	5
Module Manager	Prof.dr.ir. A. Verbraeck	
Co-responsible for assignments	Y. Huang	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	Experience with a simulation environment such as Arena, Simio, Tomas, NetLogo, GAMA or MESA. Basic knowledge about probability theory and statistics.	
Course Contents	System Theory, Object Orientation, Discrete Event System Specification, Multi-Formalism Simulation, Distributed Simulation, Real-Time Simulation, Multi-Paradigm Simulation, Multi-Resolution Simulation, and Simulation Language selection will be the core topics of the course. After an introduction to system theory, the inner working of simulation environments will be illustrated on the basis of the DEVS, DESS, and DTSS formalisms. Then, possible integrating of the different formalisms will be shown. Several special topics will be taught, based on the latest research in simulation. This material will be illustrated in intensive and interactive courses. In addition to the lecture topics, several other simulation topics will be studied by groups of students, who will write a scientific paper, and present their findings in class. These topics can be focused on the MSc program that the students participate in; special topics to study are available for TIL, CoSEM, EPA, Computer Science, and other students. Finally, groups of students will study a simulation package in-depth and discuss the commonalities and differences with other packages. Again, the package chosen can be targeted at the MSc program of the students. TIL students can, e.g. study a package that is more aimed at logistics and transport, CoSEM students can focus on a package that is used in systems design, whereas EPA students can focus on a package used in policy analysis. Most packages can be made available for use from home. This course requires active participation and contains group work.	
Study Goals	After taking this course the student will have knowledge about: - systemic principles behind simulation, and clear understanding of key terms, such as state, time, transition, event, process, and activity; - internal working of different kinds of discrete event simulation languages and environments, with a focus on state transitions over time; - underlying theories and formalisms of discrete event simulation, such as DEVS, DTSS and DESS; - the DEVS paradigm, including state trajectories and transition functions; quantized DEVS (QDEVS); discretized time; Dynamic Structure DEVS (DSDEVS), and other DEVS variants; - transformation between simulation formalisms; the formalism-transformation graph; - object-oriented simulation environments; service-based simulation; - structure and abilities of distributed simulation; the concept of HLA to manage time and state in distributed simulations; classes of functions in a distributed simulation environment; - latest research activities in the field of simulation, with research topics like web-based simulation, real-time control using simulation, agent based simulation, interactive simulation and gaming, and simulation in special domains; - examples of successful and less successful simulation studies and the learning experiences of those studies; - important similarities and differences between simulation languages, and selection methods for simulation packages. Practical skills will have been developed with respect to: - usage of at least one simulation package that was previously unknown; - writing of a scientific paper on the topic of simulation. Insight will have been gained on: - comparing and contrasting simulation languages; - the importance of time and state relationships for any type of simulation modeling; - the formal foundations of the simulation domain.	
Education Method	Lectures about state-of-the-art simulation theory Computer use for studying one discrete-event or agent-based simulation package in-depth per group Paper writing by student groups on a special topic Lectures by student groups on a special topic.	
Computer Use	Each group of students will study and evaluate a discrete-event or agent-based simulation package. Examples are Simio, Arena, Enterprise Dynamics, Plant Simulation, AnyLogic, Extendsim, Flexsim, Jaamsim, DSOL, Salabim, MESA, GAMA, Netlogo, and Repast Symphony. Also here, the simulation package to study can be specifically adapted to the group of students. Most packages can be made available for use from home.	
Course Relations	The course contains very little overlap with other simulation courses such as EPA1324, EPA1352, TB231/TB233, etc.	
Literature and Study Materials	Set of scientific papers and book chapters that will be made available through Brightspace. Typically, every class will have on average 2 papers to study for the exam. The set of papers is also available as a Reader that can be obtained on paper.	
Reader	A paper-based or PDF version of the Reader is available. The content of the Reader is also electronically available in Brightspace.	
Assessment	The mark for this course will be based on the result of two summative group assignments (paper with presentation, lab study of simulation package consisting of 4 sub-assignments that all count equally heavy) and on a summative individual written exam. The individual written exam will count for 50%, the group paper (including the class presentation) for 30% and the study of the simulation package for 20%. Each of the three marks has to be 5.75 or higher to be included in the final mark. Partial marks will stay valid for the study year in which they were obtained and the study year thereafter. The written exam takes place at the end of Q1 with a retake at the end of Q2. When a group assignment is insufficient or missing, the group will be given the opportunity to repair the insufficient or missing assignment.	
Permitted Materials during Tests	All written materials are allowed during the exam: papers, class notes, slides.	
Elective	Yes	
Tags	Challenging Intensive Modelling Programming concepts Software Stochastics	
Targetgroup	Students in EPA, CoSEM, TIL, TEL, PEL, Computer Science, and any other MSc student with some prior knowledge about	

computer simulation.

Extra Skills Trained

Writing a high-quality scientific paper; exploring a new simulation language / package.

TPM034A	Machine learning for socio-technical systems	5
Module Manager	Ir. S. van Cranenburgh	
Instructor	Dr. A.P. Afghari	
Instructor	Dr. N. Metoui	
Co-responsible for assignments	Dr. N. Metoui	
Contact Hours / Week x/x/x/x	0/6/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Students require prior knowledge of linear algebra, statistics, probability theory and data structures. Furthermore, intermediate Python programming knowledge is required. It is recommended (but not mandatory) to have taken the course "EPA122A Spatial Data Science"	
Course Contents	Machine Learning (ML) is increasingly seen as a crucial part of the puzzle to solving socio-technical challenges of today's networked and urbanised knowledge-driven societies. Successful adoption of ML does, however, not only require skilled computer scientists that do hard-core programming. Also, professionals are needed that have both the domain knowledge of socio-technical systems and a profound understanding of ML. This course aims to provide students in the socio-technical domain with a profound understanding of ML. It prepares students for the challenges and questions ML will pose to them in their later careers. To this end, the course consists of three parts: 1. Fundamentals of ML 2. Explainability of ML 3. Applications of ML for socio-technical challenges In part 1, students learn about ML fundamentals and methods. These weeks a critical technical foundation is laid for grasping the strength and weaknesses of ML for the analysis and design of socio-technical systems. Part 2 is devoted to the explainability of ML. For public decision-making as well as for decision-making in high-stake contexts, such as, e.g. autonomous vehicles, legal systems transparency and explainability of the models are of critical importance. Students learn several popular explainability techniques and discuss their value for applications in socio-technical systems. In part 3, a rotating group of TPM scholars provides exemplary applications of ML in socio-tech systems. This serves two purposes: (1) to show where and how ML is applied for analysis and application in socio-technical systems and (2) to deepen reflection on the impact of ML-based solutions and interventions on individuals, organisations, and society. Consecutively, students work in a group on a final project, building forth on one of the presented applications. This project brings together the three parts of this course: Fundamentals of ML, Explainability of ML, and application of ML for socio-technical challenges. Students need to apply ML models and techniques on real-world data in a notebook, and interpret and communicate their results, taking into account the socio-technical setting through a presentation. The course consists of oral lectures and lab sessions. The aim of the lab sessions is to show and reinforce how the ML models, explainable ML techniques, and ML ideas presented in the oral lectures are put into practice. Also, they help students gather hands-on machine learning skills. The lab sessions involve a series of exercises in the form of Jupyter notebooks. The course consists of one or two oral lectures per week and, in some weeks, a lab session. Attendance of the lectures and lab sessions is highly recommended to keep up with the course but is not mandatory.	
Study Goals	After the course, students can: 1. explain the fundamental concepts of machine learning (ML). 2. conceptually explain the workings of a selected number of ML models and explainable AI (XAI) techniques, and apply these to empirical data. 3. identify applications of ML and XAI techniques in real-world socio-technical systems. 4. examine the impact of ML-based solutions and interventions on individuals, organisations, and society through XAI. 5. conduct an in-depth analysis of a real-world socio-technical challenge by applying ML and XAI to empirical data. 6. reflect on the strengths and limitations of ML and XAI in real-world socio-technical systems.	
Education Method	The education methods used in this course are: 1. Oral lectures with discussions 2. Lab sessions (formative) 3. Assignments 4. Group work	
Computer Use	Jupyter notebooks	
Assessment	The final grade is computed by taking the weighted average across the three summative parts: 1. Individual exam, Written, 50%, Minimum grade: 5.0 2. Individual assignments, Jupyter notebooks, 25%, Minimum grade: 5.0 3. Group project, 25%, Presentation and notebook, Minimum grade: 5.0 For students to pass the course, the final grade must be equal to or larger than 5.8. The grade of each part is individually registered in OSIRIS and transferable to the next year(s). For the Individual exam, a regular exam and a resit exam will be offered within one academic year. There is no resit offered for the Individual assignments and the Group project. After each individual exam, an inspection moment will be organised for the students.	
Elective	Yes	
Tags	Artificial intelligence Modelling Numeric Methods Programming Research Methods	

Year	2023/2024
Organization	Technology, Policy and Management
Education	Master Complex Systems Engineering and Management

Electives

TPM014B	Ethics of AI	5
Module Manager	Dr. J.M. Duran	
Co-responsible for assignments	Dr. J.B. van Grunsven	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	This course aims to learn to argue about moral, social, and epistemic values that students involved with AI (computer scientists, engineers) will encounter in their professional practice. The course focuses on contemporary pressing topics such as: 1. Introduction to theories of normative ethics: utilitarianism, deontology, virtue ethics. The distinction between descriptive and normative claim, epistemology and ethics, values, norms, and virtues. 2. Ethical and epistemological issues in AI: identification of six different sources of ethical and epistemological concerns in AI. 3. Bias in AI: Automated-decision making systems are polluted with different forms of bias that can be found at a different levels. Here we address the different forms of bias and what can be done about it. 4. Explanatory AI and the right to explanation: The GDPR grants the so-called "right to an explanation." But what does this mean, and how far have we got into explanatory AI? In this unit, we explore the scope and limits of current approaches to explainable AI and how this affects users' rights 5. Privacy: there are different ways of studying and understanding privacy. In addition, this unit discusses the value of privacy in AI in comparison with other values, such as explainability, accuracy, bias, and fairness. 6. Trustworthy AI: What does it mean to trust an AI system? is there more than one way to trustworthy AI? can we design trust? In this unit, we will address questions such as these. We will also present and discuss the two major approaches to trustworthy AI: transparency and computational reliabilism. Lights and shadows of both. Students taking the course will develop the following academic skills: reading philosophical texts, critical analysis, oral presentation and discussion, and essay writing.	
Study Goals	The learning objectives of the course are: 1. Students will recognize and dissect current epistemic, normative, and societal issues emerging in the context of the design and use of AI. 2. Students can identify strengths and weaknesses in epistemological and normative claims about AI. 3. Students are able to critically assess ethical, societal, and epistemological arguments related to AI 4. Students are able to recognise modes of reasoning in ethics.	
Education Method	The course will have lectures and tutorials. Students will be involved in individual as well as group activities. The methodology used is philosophical analysis, reflection, and discussion.	
Assessment	Assignments in tutorials (ca. 20% of the final grade), a final exam (ca. 40% of the final grade), and a final paper (ca. 40% of the final grade). In case of unforeseen circumstances or measures resulting from COVID-19, the prescribed assessment will be a timed take-home exam (open book).	
Remarks	For ethics teaching at TU Delft check https://www.tudelft.nl/ethics/ethics/teaching-activities/ethics-teaching/	
Elective	Yes	
Tags	Artificial intelligence Ethics Philosophy Technology	
Category	MSc level	

TPM038A	Artificial Intelligence and Cybernetic Justice	5
Module Manager	R.I.J. Dobbe	
Instructor	S.J.J. Nouws	
Instructor	I. Martinez de Rituerto de Troya	
Co-responsible for assignments	A. Ghorbani	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	The student has successfully finished at least one graduate level course in machine learning, artificial intelligence or a related topic, in which they learned to design and develop data-driven and algorithmic systems. The student checks the prerequisite proactively with the module manager, ideally before the start of the semester and the latest in the first week of Q1.	
Course Contents	Recently, a range of critiques, audits and empirical accounts of algorithmic and AI systems has identified a broad variety of social and environmental harms. Think about: - the use of statistical learning models in the Dutch childcare benefit scandal, - the role of algorithms in spreading hate speech and amplifying polarization on social media, - the ways that deep learning is introducing new safety hazards in self-driving cars leading to new kinds of fatal accidents, - the necessity of algorithmic approaches to ensure the energy transition, creating new forms of inequity. Most mainstream responses to harmful AI systems tend to focus on technical fixes (like fairness metrics or explainable AI), product standards, ethical frameworks or policy instruments, which are often implemented a priori through a checklist procedure or as ad hoc design fixes. While these efforts all may have their merits, their application does not suffice the prevention of algorithmic harm and injustice. An impressive history in software-based automation and information systems shows that harm or injustice emerge in the broader social, political and institutional context in which AI systems are built and integrated. Harm then should be understood as the result of technology interacting with social actors, political processes, infrastructural arrangements and institutions. Understanding such sociotechnical and cybernetic dynamics is notoriously hard but therefore worth studying. TPM students and scholars are uniquely positioned to make sense of it and develop new conceptual and practical frameworks to help practitioners, policy makers, civil society and those affected by AI systems properly anticipate, prevent and respond to algorithmic harms and injustice. In this course we therefore ask: - How do we understand whether an algorithmic or AI system is "safe", "sustainable", "fair" or "just"? - What variables and system complexities do we include in our frame of analysis to properly understand and mitigate forms of algorithmic harm? - What can policy makers, system developers, civil servants and civil society actors do (differently) to ensure AI systems are safe and used responsibly, both in development as well as in governance? In this course we will cover: - various emerging methods and practices to understand and assess algorithmic and AI systems as socio-ecological-technical systems - normative dimensions and challenges of AI systems, looking at power relations and asymmetries and frameworks to overcome these through a.o. democratic mechanisms, legal protection and collective action - historical and contemporary methods for safeguarding AI functionality in sociotechnical systems, both through technical and institutional means - strategies from environmental and climate justice domains for working towards algorithmic justice across different actors in society The answers to these questions require the gathering of insights from systems engineering, as well as a variety of social sciences, humanities and engagement with real actors. Course material will draw primarily from recent work of the module manager that conceptualizes and explores these junctions: - Rakova, B., & Dobbe, R. (2023). Algorithms as Social-Ecological-Technological Systems: an Environmental Justice Lens on Algorithmic Audits. https://arxiv.org/pdf/2305.05733.pdf - Dobbe, R. (2022). System Safety and Artificial Intelligence. In Oxford Handbook of AI Governance. https://arxiv.org/pdf/2202.09292 - Dobbe, R., Krendl Gilbert, T., & Mintz, Y. (2021). Hard choices in artificial intelligence. Artificial Intelligence, 300, 103555. https://doi.org/10.1016/j.artint.2021.103555	
Study Goals	By the end of this course, the student will be ready to perform academic research at the intersection of algorithmic/AI systems and their justice/harm implications, through meeting these learning objectives: - Demonstrate basic understanding of the history of software-based automation in safety-critical systems - Apply mapping techniques to situate AI systems in their complex social, political, infrastructural and institutional ecosystem - Model algorithmic and AI systems as socio-technical-ecological systems - Analyze mapped and modeled AI systems for harm and safety hazards using principles and techniques from system safety - Analyze mapped and modeled AI systems for normative complexity and tradeoffs in systems engineering using the Hard Choices in AI (HCAI) Framework - Develop conceptual definitions and requirements for safety, sustainability and/or justice in an AI system application - Understand core strategies for attaining just outcomes from the history of environmental justice (EJ) - Propose different design and governance configurations to meet the requirements (based on system safety, HCAI, EJ and previous training in TPM or elsewhere) - Reflect and discuss possible implications and limitations of requirements and design and governance configurations - Draft a short report to turn your insights into recommendations and actions for policy makers, system developers, civil society and other relevant actors	
Education Method	Lectures, work groups, case studies, presentations, self-study, writing.	
Assessment	Exam (40%) Essay (20%) Group project (30%) Engagement (10%)	
Elective	Yes	

Tags	Algorithmics Artificial intelligence Information & Communication Software Engineering Sustainability
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Year	2023/2024
Organization	Technology, Policy and Management
Education	Master Complex Systems Engineering and Management

MOT students

EPA122A	Spatial Data Science	5
Module Manager	Dr. T. Verma	
Instructor	Dr. T. Verma	
Responsible for assignments	Dr. T. Verma	
Co-responsible for assignments	N.Y. Aydin	
Contact Hours / Week x/x/x/x	0/8/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 5	
Course Language	English	
Required for	All EPA Master Students	
Expected prior knowledge	Students should have some prior programming experience. It will be beneficial if you have dealt with a functional programming language like R or Python before. If you have never programmed before, I recommend doing a crash course in Python through Coursera or other online services before joining the class. This course is not about learning how to program. If you are not from the TPM faculty and still want to join the course, send the module manager an email with your motivation to join the course before the start of the course. Without receiving such a motivation email, we will not be able to accommodate you fully.	
Course Contents	This class will train students to gather, merge and clean data from multiple sources. We will focus on urban areas with a spatial lens to gain valuable insights into the reality of multiple societal problems like climate change, energy transition, and inequalities. With unprecedented growth in cities, policymakers strive to balance providing equal opportunity and benefit to its citizens and sustainable development. Data science will help us understand and estimate multiple implications of solutions and communicate results to a broad audience effectively. The course is divided into five major modules, each focusing on crucial steps in the lifecycle of a data science project. Obtain : Obtaining data from multiple [open] data sources. Scrub : Data cleaning, munging, and sampling to consolidate all information into a dataset that is manageable, informative and relates to your problem. Explore : Exploratory data analysis to make sense of what your data is trying to say. Model : Estimation and modelling based on statistical tools such as regression, classification and clustering. Interpret : Communicating results and reflections through visualisation, storytelling and interpretable summaries. Students will need their laptops only during labs. All classroom sessions will be a mixture of interactive lectures, discussions and labs. Labs will cater to a series of exercises to reinforce the skills and topics presented in class. A final project will allow students to run through the entire cycle of 5 modules by obtaining and synthesising data to explore a problem of choice and interpreting and communicating their results through an article. For students who have had statistical, math or computer programming courses in their bachelor's or elsewhere, this course will add to your skills by providing you with tools to become future policy-makers, academics, responsible data scientists, and supporters of open science. The course meets 8 hours a week, divided into two interactive and blended lecture hours, 4 lab hours and two extra wildcard hours weekly. Although attendance for all lectures is not mandatory at all, substantial self-study will be required to keep up with the course if you do not attend at least 4 hours a week.	
Study Goals	After successful completion of this course, you will be able to: interpret and discuss data sources usable and relatable for a problem presented. manipulate data and consolidate all information into a manageable, informative dataset that relates to your problem. describe and analyse the consolidated dataset(s) to support your problem with evidence. apply statistical and machine learning models to infer results in turning data into valuable information. report results and reflections through visualisation, storytelling and interpretable summaries, especially when faced with a new dataset. and hopefully, get great data-driven policy jobs in the future, or go on adventurous travels with an open and critical mind.	
Education Method	Out of class education methods involve 1. Reading materials and watching videos assigned before lectures, making notes and asking important questions in class. (Blended Learning) 2. Working on notebooks in Python (ungraded, and useful to stay on track). In class education methods involve 1. Interactive lectures and discussions 2. Labs / open hours to solve assignments	
Computer Use	The course is set in Python programming language. Students will be expected to use Python notebooks on their personal laptops. If this is not possible for you for any reason, please contact me before the course and we will try to figure an alternative solution while maintaining your privacy.	
Books	No books needed	
Reader	No reader needed	
Assessment	50% Individual assessment methods include Assignments (Summative) 50% Group assessment methods include (size: 4 students/group) Project (Summative) It is mandatory to score ≥ 5.8 in both parts separately to pass the course. Partial grades are valid. Only if students get < 5.8 in any part, they may retake that part next year in Q2. For assignments this means retaking all assignments, and for project, this means redoing the whole project with a new group. No exam	

Exam Hours	No exam
Elective	Yes
Tags	Adventurous Artificial intelligence Building & Spatial Development Group work Modelling Policy Analysis Programming Programming Software

EPA123A	Introduction to TPM Modelling	5
Module Manager	F.M. d'Hont	
Instructor	Prof.dr.ir. J.H. Kwakkel	
Instructor	F.M. d'Hont	
Co-responsible for assignments	Prof.dr.ir. J.H. Kwakkel	
Contact Hours / Week x/x/x/x	0/6/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Basic knowledge about differential equations, programming in Python, probability theory, and statistics.	
Summary	This module introduces two different methods for modelling and simulating complex grand challenges. The theory is discussed according to the modelling cycle: Problem articulation, Conceptualisation, Formulation, Evaluation, and Policy analysis. The course focusses on the basics of continuous modelling (i.e., System Dynamics) and of discrete modelling (i.e., Agent Based modelling).	
Course Contents	Continuous part: In the continuous part, students are first introduced, by means of lectures and practical computer labs, to the problem articulation, conceptualisation, formulation, evaluation, and use of System Dynamics (SD) models. Discrete part: The discrete part of this course teaches the student to create and use models based on Agent-Based Modelling (ABM).	
Study Goals	Upon completion of this course, students should be able to - explain the process of modelling for supporting decision-making on large societal challenges; - conceptualise, formulate, parameterise, test, and use a model of a complex, societal policy problem using SD and ABM formalisms; - analyse a complex societal policy problem using an SD model and ABM; - identify and test interventions for addressing the policy problem using the developed simulation model; - assess and argue whether SD and/or ABM are appropriate for analysing a policy problem; - Link system structure to possible dynamic behaviour of important indicators of a policy problem. For SD, this system structured is aggregated. For ABM, this system structure is microscopic. SD specific - Implement and use a medium complexity SD model in Vensim ABM specific - Implement and use a medium complexity ABM model in MESA.	
Education Method	This course uses two main methods of education: - lectures for the theoretical aspect of the course - lab sessions to develop skills for working with the tools Vensim and MESA	
Computer Use	Continuous part: The simulation models will be made with Vensim. Students can request a temporary Vensim license for Windows or iOS. This program is also installed in the computer rooms in the Faculty (TPM in Delft) and can be used via TU Delft's weblogin. Discrete part: The simulation models will be made with MESA. MESA is a Python 3 based framework for ABM. Student will need to have a working Python 3 installment before they install MESA.	
Assessment	Summative assessment (individual grade): This course is examined through an in-person, written, online, open-book computer exam. After grading, the grades and student answers will be made available online. EPA123A also offers two formative tests that serve as practice for the final, summative exam. Transitional rules 2023-2024: Students who participated in Epa1324 in previous years, can opt to do a partial exam (in Q2 and Q3), with only the Theory, Discrete, and/or Continuous questions. These categories will be indicated on the exam sheet. The final partial grade will be calculated by weighing the correct questions against the total obtainable in each category. This academic year (Q2 and Q3 in 2023-2024) offers the final opportunity to take resits for each partial exam.	
Enrolment / Application	Enroll through Brightspace.	

Year	2023/2024
Organization	Technology, Policy and Management
Education	Master Complex Systems Engineering and Management

Electives

TPM014B	Ethics of AI	5
Module Manager	Dr. J.M. Duran	
Co-responsible for assignments	Dr. J.B. van Grunsven	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	This course aims to learn to argue about moral, social, and epistemic values that students involved with AI (computer scientists, engineers) will encounter in their professional practice. The course focuses on contemporary pressing topics such as: 1. Introduction to theories of normative ethics: utilitarianism, deontology, virtue ethics. The distinction between descriptive and normative claim, epistemology and ethics, values, norms, and virtues. 2. Ethical and epistemological issues in AI: identification of six different sources of ethical and epistemological concerns in AI. 3. Bias in AI: Automated-decision making systems are polluted with different forms of bias that can be found at a different levels. Here we address the different forms of bias and what can be done about it. 4. Explanatory AI and the right to explanation: The GDPR grants the so-called "right to an explanation." But what does this mean, and how far have we got into explanatory AI? In this unit, we explore the scope and limits of current approaches to explainable AI and how this affects users' rights 5. Privacy: there are different ways of studying and understanding privacy. In addition, this unit discusses the value of privacy in AI in comparison with other values, such as explainability, accuracy, bias, and fairness. 6. Trustworthy AI: What does it mean to trust an AI system? is there more than one way to trustworthy AI? can we design trust? In this unit, we will address questions such as these. We will also present and discuss the two major approaches to trustworthy AI: transparency and computational reliabilism. Lights and shadows of both. Students taking the course will develop the following academic skills: reading philosophical texts, critical analysis, oral presentation and discussion, and essay writing.	
Study Goals	The learning objectives of the course are: 1. Students will recognize and dissect current epistemic, normative, and societal issues emerging in the context of the design and use of AI. 2. Students can identify strengths and weaknesses in epistemological and normative claims about AI. 3. Students are able to critically assess ethical, societal, and epistemological arguments related to AI 4. Students are able to recognise modes of reasoning in ethics.	
Education Method	The course will have lectures and tutorials. Students will be involved in individual as well as group activities. The methodology used is philosophical analysis, reflection, and discussion.	
Assessment	Assignments in tutorials (ca. 20% of the final grade), a final exam (ca. 40% of the final grade), and a final paper (ca. 40% of the final grade). In case of unforeseen circumstances or measures resulting from COVID-19, the prescribed assessment will be a timed take-home exam (open book).	
Remarks	For ethics teaching at TU Delft check https://www.tudelft.nl/ethics/ethics/teaching-activities/ethics-teaching/	
Elective	Yes	
Tags	Artificial intelligence Ethics Philosophy Technology	
Category	MSc level	

TPM038A	Artificial Intelligence and Cybernetic Justice	5
Module Manager	R.I.J. Dobbe	
Instructor	S.J.J. Nouws	
Instructor	I. Martinez de Rituerto de Troya	
Co-responsible for assignments	A. Ghorbani	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	The student has successfully finished at least one graduate level course in machine learning, artificial intelligence or a related topic, in which they learned to design and develop data-driven and algorithmic systems. The student checks the prerequisite proactively with the module manager, ideally before the start of the semester and the latest in the first week of Q1.	
Course Contents	Recently, a range of critiques, audits and empirical accounts of algorithmic and AI systems has identified a broad variety of social and environmental harms. Think about: - the use of statistical learning models in the Dutch childcare benefit scandal, - the role of algorithms in spreading hate speech and amplifying polarization on social media, - the ways that deep learning is introducing new safety hazards in self-driving cars leading to new kinds of fatal accidents, - the necessity of algorithmic approaches to ensure the energy transition, creating new forms of inequity. Most mainstream responses to harmful AI systems tend to focus on technical fixes (like fairness metrics or explainable AI), product standards, ethical frameworks or policy instruments, which are often implemented a priori through a checklist procedure or as ad hoc design fixes. While these efforts all may have their merits, their application does not suffice the prevention of algorithmic harm and injustice. An impressive history in software-based automation and information systems shows that harm or injustice emerge in the broader social, political and institutional context in which AI systems are built and integrated. Harm then should be understood as the result of technology interacting with social actors, political processes, infrastructural arrangements and institutions. Understanding such sociotechnical and cybernetic dynamics is notoriously hard but therefore worth studying. TPM students and scholars are uniquely positioned to make sense of it and develop new conceptual and practical frameworks to help practitioners, policy makers, civil society and those affected by AI systems properly anticipate, prevent and respond to algorithmic harms and injustice. In this course we therefore ask: - How do we understand whether an algorithmic or AI system is "safe", "sustainable", "fair" or "just"? - What variables and system complexities do we include in our frame of analysis to properly understand and mitigate forms of algorithmic harm? - What can policy makers, system developers, civil servants and civil society actors do (differently) to ensure AI systems are safe and used responsibly, both in development as well as in governance? In this course we will cover: - various emerging methods and practices to understand and assess algorithmic and AI systems as socio-ecological-technical systems - normative dimensions and challenges of AI systems, looking at power relations and asymmetries and frameworks to overcome these through a.o. democratic mechanisms, legal protection and collective action - historical and contemporary methods for safeguarding AI functionality in sociotechnical systems, both through technical and institutional means - strategies from environmental and climate justice domains for working towards algorithmic justice across different actors in society The answers to these questions require the gathering of insights from systems engineering, as well as a variety of social sciences, humanities and engagement with real actors. Course material will draw primarily from recent work of the module manager that conceptualizes and explores these junctions: - Rakova, B., & Dobbe, R. (2023). Algorithms as Social-Ecological-Technological Systems: an Environmental Justice Lens on Algorithmic Audits. https://arxiv.org/pdf/2305.05733.pdf - Dobbe, R. (2022). System Safety and Artificial Intelligence. In Oxford Handbook of AI Governance. https://arxiv.org/pdf/2202.09292 - Dobbe, R., Krendl Gilbert, T., & Mintz, Y. (2021). Hard choices in artificial intelligence. Artificial Intelligence, 300, 103555. https://doi.org/10.1016/j.artint.2021.103555	
Study Goals	By the end of this course, the student will be ready to perform academic research at the intersection of algorithmic/AI systems and their justice/harm implications, through meeting these learning objectives: - Demonstrate basic understanding of the history of software-based automation in safety-critical systems - Apply mapping techniques to situate AI systems in their complex social, political, infrastructural and institutional ecosystem - Model algorithmic and AI systems as socio-technical-ecological systems - Analyze mapped and modeled AI systems for harm and safety hazards using principles and techniques from system safety - Analyze mapped and modeled AI systems for normative complexity and tradeoffs in systems engineering using the Hard Choices in AI (HCAI) Framework - Develop conceptual definitions and requirements for safety, sustainability and/or justice in an AI system application - Understand core strategies for attaining just outcomes from the history of environmental justice (EJ) - Propose different design and governance configurations to meet the requirements (based on system safety, HCAI, EJ and previous training in TPM or elsewhere) - Reflect and discuss possible implications and limitations of requirements and design and governance configurations - Draft a short report to turn your insights into recommendations and actions for policy makers, system developers, civil society and other relevant actors	
Education Method	Lectures, work groups, case studies, presentations, self-study, writing.	
Assessment	Exam (40%) Essay (20%) Group project (30%) Engagement (10%)	
Elective	Yes	

Tags	Algorithmics Artificial intelligence Information & Communication Software Engineering Sustainability
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Year	2023/2024
Organization	Technology, Policy and Management
Education	Master Complex Systems Engineering and Management

Water and Delta Systems	
MSc Coordinator	Dr. L. Scholten
MSc Coordinator	Dr.ir. M.P.M. Ruijgh-van der Ploeg
Administration by the Faculty of	TBM

TPM003A	Water Ethics	5
Module Manager	Prof.dr.mr.ir. N. Doorn	
Co-responsible for assignments	Dr.ir. U. Pesch	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Course Contents	Water is essential for life. While we can in principle survive weeks without food, without water, we would not survive more than a couple of days. The aim of this course is to provide an introduction to philosophical water ethics. It will present the main philosophical issues in the water domain in a practical way by situating these issues within real cases. Questions addressed will include: Is it fair to drink water, knowing that the production of one cup of coffee requires 140 litres of water on average? Should we prioritize between different water uses, like agriculture, navigation, and recreation? Should we continue building hard flood defences at the expense of the environment? Should local citizens be given a voice in water-related decision making? These and other questions will be linked to ongoing discussions in the relevant other disciplines within philosophy, most notably in ethics of technology, philosophy and ethics of risks, environmental ethics, climate ethics, and global justice. This course will provide students with an understanding of approaches in water engineering and water policy. Students will get experience with discussing and evaluating ethical considerations in water engineering and water policy. Topics covered include: water and justice, water and economic valuation, water and human rights, water and responsibility, and water and engineering.	
Study Goals	At the end of this course you will be able to: - Describe the world's water challenges, also in view of climatic and geographic developments. - Describe and analyse the different water uses and explain how they may lead to competing demands; - Describe how the world's water challenges pose questions of justice and apply different concepts of justice to concrete water problems; - Recognize and critically reflect on the methodological and ethical aspects of economic valuation in the context of water; - Describe and apply different legal frameworks and rights concepts to the context of water; - Describe and apply different concepts of responsibility to the context of water; - Recognize and analyse the value-ladenness of water engineering and design; - Describe and analyse current water governance practices and identify opportunities to include ethical considerations in water governance; - Write a well-argued essay on a topic related to water ethics.	
Education Method	The structure of classes will combine lectures with seminars and debates. Dependent on the Corona restrictions, the course may take place on campus or online.	
Books	"Water ethics: An introduction", by Neelke Doorn. Publisher: Rowman & Littlefield. 2019. ISBN: 9781786609519 (Pb) / 9781786609526 (E-book).	
Assessment	This course has no formative assessments or exams. It will be assessed via written assignments. The assessment consists of four parts: - Case report (group assignment in groups of 2-4 students); - Review (individual assignment); - Reflection (individual assignment); - Essay (individual assignment). Students are not allowed to write the case report and the essay on the same topic. Calculation of the course grade: The course grade is calculated as follows: $0.3 * \text{case report grade} + 0.15 * \text{review grade} + 0.15 * \text{reflection grade} + 0.4 * \text{essay grade}$. The final grade will be rounded to the nearest half-integer. Students with a final grade of 6.0 or higher have passed the course. Validity: Grades for the different parts are valid only in the academic year in which the student participates in the tutorial sessions. Resit: In case of a grade lower than 6.0, students are allowed to write a new case report on a different case or a new essay on a different topic. This needs to be completed in the quarter following the quarter in which the tutorial sessions take place. Feedback: Students receive feedback on their assignments via Brightspace. Students receive detailed feedback on the draft version of their essay. Note: The assessment form does not depend on whether the course will be delivered online or on campus.	
Enrolment / Application	The course TPM002A is the 3 ECTS version of the water ethics course. There is also a 5 ECTS version of the water ethics course, with course code TPM003A. Students enrolled in the 3 ECTS course follow exactly the same study activities as the students enrolled in the 3 ECTS course, except that they don't write the additional essay that students enrolled in the 5 ECTS version of the course need to write as one of their final assignments. This essay is worth 2 ECTS. All students should enrol in the 5 ECTS version of the course, also the students who do not wish to be eligible for this additional 2 ECTS. In the first week of the course, students can indicate whether they wish to follow the 3 ECTS or the 5 ECTS version of the course. There is no separate active Brightspace site for the 3 ECTS course (TPM002A). Please enrol for the course via the Brightspace site of TPM003A, also if you want to enrol for the 3 ECTS version of the course. Enrolment is possible until 1 November of the academic year. On that day, the total number of available places is determined. After 1 November, enrolment takes place upon approval and may be denied if there are no more places available.	
Special Information	Attendance of the tutorial sessions is mandatory. Students are allowed to miss only one session and only if they have a valid reason.	
Remarks	For ethics teaching at TU Delft, check https://www.tudelft.nl/ethics/ethics/teaching-activities/ethics-teaching/	
Elective	Yes	
Tags	Ethics Sustainability Water Engineering Water management	
Targetgroup	All students enrolled in a MSc-program at TU Delft with an interest in water ethics.	
Category	MSc level	

TPM026A	System Reliability in Quantitative Risk Assessment	4
Module Manager	Prof.dr.ir. P.H.A.J.M. van Gelder	
Instructor	Prof.dr.ir. P.H.A.J.M. van Gelder	
Instructor	Dr. M. Yang	
Responsible for assignments	Prof.dr.ir. P.H.A.J.M. van Gelder	
Co-responsible for assignments	Prof.dr.ir. G.L.L.M.E. Reniers	
Contact Hours / Week x/x/x/x	2/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	This course presents the basic concepts of reliability analysis of complex systems, one of the corner stones of risk assessment and management of technological systems. It starts with a description of the fundamental procedural steps of a quantified risk assessment and identification of the scientific disciplines and available methodological tools to complete each step. A system (scenario-based) approach is employed, where the source of risk is considered as a system with constituent components or elements. Risk is then defined and determined as a set of triplets {si, pi, ci}: si are the scenarios (sequence of events) that lead to the undesired consequences, ci, and pi are the probabilities with which these scenarios are expected to occur. The course focuses on methodologies and corresponding tools that can provide the scenarios (si) and the corresponding probabilities of occurrence (pi). To this end, the complex system is modeled as a collection of interconnected subsystems or components among which there are functional interdependencies. Methods for modeling the functional interdependencies of the components will be presented both for two-state as well as for multi-state components. These methods model the states of the complex system as collection of states of its components. Next, the principles of probability are employed to assess the probability of the various system states in terms of the probability of the component states. Fundamental concepts and measures of reliability will be introduced at a components level. Then complex systems (multi-component) will be addressed. Static or on demand events will be examined. Methods to be studies include block-diagrams, fault-trees, event-trees. Techniques for quantification of these models including dependence analysis will be presented. Importance analysis will be examined providing methods for determining the relative contribution of each component into the proper function of a complex system, as well as, its contribution to the probability of system failure. The use of data (observed behavior of the system and its components) to derive the values of various reliability parameters will be presented. Throughout the course, practical examples of systems from the chemical, nuclear, electricity generation, water, energy and railway industry, as well as, examples from occupational risk and land use domain will be used to demonstrate the various methods and techniques.	
Study Goals	After successfully completing the course, students will be able to: - identify potential risk problems in complex systems - explain analytical probability formulas - apply different risk methods (block diagrams, fault tree analysis, event tree analysis, sensitivity analysis) for reduced scale problems - analyze a real life problem using combined methods	
Education Method	Lectures, exercises and project	
Assessment	#### Assessment Closed-book written exam or oral exam (depending on the number of students) and individual homework assignments. #### Weighing Final mark of this course is a weighted average of 25% for the individual homework assignments and 75% for the individual exam. The minimum grade for the homework assignments, as well as the individual exam, is 5.0. The grade for the homework assignments, as well as the individual exam, is 2 years valid. Both homework assignments and individual exam are summative. Detailed feedback on the graded homework assignment and individual exam can be reviewed by students by contacting the teaching staff of this course. #### Resit The students are allowed one re-sit per examination. It is not allowed to re-sit an examination or assignment for which they have received a pass (6,0 or higher). It is allowed to re-sit an examination or assignment which they haven't done during the first occasion. The re-sit format needs to be discussed with the teacher of the course in line with examination regulations. In case the student is granted an extra re-sit by the Board of Examiners, this re-sit has to take place within study year 2023-2024.	
Elective	Yes	
Category	MSc level	

TPM801A	Introduction to Water Governance in Deltas	5
Module Manager	Dr.ir. M.P.M. Ruijgh-van der Ploeg	
Instructor	Dr.ir. E. Minkman	
Instructor	Dr. L. Scholten	
Instructor	Prof.dr. F. Klijn	
Co-responsible for assignments	Dr. L. Scholten	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1A	
Start Education	1A	
Exam Period	1A 2A	
Course Language	English	
Course Contents	Water governance is concerned with the formulation and implementation of policies to achieve a fair distribution of water as a resource, reduction of the risks associated with water, and regulation of conflicting demands for water as a condition for life. In this course, students will learn to understand and value the context-specificity of water governance in deltas of large river systems. Students will acquire a framework of reference that enables them to analyse the major issues in water management in deltas: flooding, competition for scarce water resources, deterioration of the conditions for nature in water and land systems. Extra attention will be given to governance of adaptation to climate change. This course offers the basic knowledge that enables students to engage in research on issues related to water governance and water system modelling. With that in mind, the course also intends to stimulate development of personal skills such as curiosity, visual appraisal, engagement with stakeholders, conceptualization of cause-effect chains, etc.	
Study Goals	1. Explain and apply important concepts for governance of water in deltas in relation to different water-related problems and contexts. 2. Draw conceptual models - for water-related problems that are typical for deltas - that give insight in the cause-effect chains that explain the functioning of ecosystems and socio-economic systems, and that identify the main factors that steer system performance. 3. Discuss, and illustrate with examples, the main issues for water governance as related to flood risk management, water resources management, urban water management and water as a condition for nature and land-use and adaptation to climate change, including objectives and the means with which these can be achieved (technical, nature-based, rules, communication, planning etc.). 4. Reflect on dealing with spatial and temporal scales in governance of 'water in deltas.' 5. Question (Dutch) practices in water governance and comment on the factors that have shaped these practices. 6. Communicate and motivate your recommendation for governance of water-related issues in a delta of your choosing, with respect to the historical, present and future performance of the water system while making use of visual appraisal, stakeholder interviews and other resources of information.	
Education Method	Weekly meetings to lecture and to introduce and discuss reading materials. Two obligatory excursions, 0.5 day in city of Delft (by bike) and 1.0 day river excursion (by van). During the course students work on an individual assignment that is related to learning objective 6.	
Course Relations	This course is designed to be taken as an elective and in combination with one or more of the following courses: TPM802 Model-based assessment of Water systems (Q1), TPM803 Building with Nature & Beyond (Q2), TPM003A Water Ethics (Q2), TPM026A System Reliability in Quantitative Risk Assessment and/or TPM804 Adaptive planning for management of water systems	
Assessment	The final grade (>5,75) is based on the results of two partial grades that will be published in Osiris. Written exam (>5,5 and 70% of final grade) Individual assignment (>5,5 and 30% of final grade) Written exam is administered in week 1.10. Retake exam is administered in week 2.5. Assignment is to be submitted in week 1.9. Retake assignments must be submitted in week 2.5 and the maximum grade will be 6. The individual assignment is related to learning objective 6. There are two moments that students receive feedback in class (formative assessment), namely when they pitch the topic of their assignment (week 3) and when they present their assignment (80-90% version) in week 8. We grade the assignments (summative assessment, 1-10) and motivate the grade with criteria and quality indicators as published on Brightspace. The exam tests the learning objectives 1-5 and consists of short answer questions and essay questions. An exemplar for the exam is published on Brightspace. If the exam must be administered as an on-line exam (due to TU wide decision), the exam will be an open book exam.	
Elective	Yes	
Tags	Integrated Policy Analysis Water management	

TPM802A	Model-based Water Systems Assessment	5
Module Manager	J. Zatarain Salazar	
Instructor	Dr. J.E.A. Storms	
Instructor	Prof.dr. J.H. Slinger	
Instructor	Prof.dr.ir. M. Bakker	
Instructor	Dr. L. Scholten	
Instructor	J. Zatarain Salazar	
Co-responsible for assignments	Dr. L. Scholten	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1A	
Start Education	1A	
Exam Period	none	
Course Language	English	
Course Contents	This hands-on modeling course gives an introduction to overarching challenges of water and delta systems and their assessment when facing multiple sources of uncertainty, multiple-competing demands and challenging climate and demographic conditions. In addition, we tackle different temporal and spatial and administrative scales demonstrated in water systems from urban water supply, river basin management, water quality control, coastal environments and delta systems. Upon completion, the students are expected to have command of different water and delta systems and their modeling challenges for assessment and policy making.	
Study Goals	LO1. Students can frame policy-research questions in terms of factors that govern water system dynamics, and to choose appropriate geographical and temporal scales for system analysis and for the analysis of critical uncertainties. LO2. Students can give arguments for the selection of model type and where applicable modeling software suited for analysis of water system properties in relation to strategic, tactical, and operational water management objectives. LO3. Students can apply decision support and data-management tools that are typically used in water management, and develop insights into their appropriate use, and how to deal with relevant uncertainties. LO4. Students can apply the principles of strategic planning and implement it to a water policy problem and communicate the findings.	
Education Method	In-class lectures, lab sessions, reading assignments and discussion	
Assessment	Depending on the size of the group, the assignments will be developed individually or by groups of maximum 2 students. A minimum of 5.0 must be obtained for each partial assessment The average of all partial grades is at least 5.75 There are five total assignments and a final assignment. The final grade is calculated as = 0.17x(Assignment 1+Assignment 2+Assignment 3+Assignment 4+Assignment 5)+0.15x Final Assignment The assignments can be repaired to a maximum of 6.	
Elective	Yes	

TPM803A	Building with Nature and Beyond	5
Module Manager	Prof.dr. J.H. Slinger	
Instructor	H.S.I. Vreugdenhil	
Co-responsible for assignments	Dr.ir. L.M. Hermans	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2A	
Start Education	2A	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Masters students with prior knowledge of policy analysis, systems engineering, engineering design, aquatic ecology, coastal and river management, delta planning and environmental science are welcome. A STEM Bachelor is a prerequisite.	
Summary	Are you interested in Nature-based Solutions and how to design them? Join us in the Building with Nature & Beyond course!	
Course Contents	The course deals with the following subjects week by week: 1. Introduction to Building with Nature, 2. Engineering design principles, 3. Ecological design principles, 4. Integrated BwN design using the H-E principles, 5. Integrated design review, 6. Why beyond engineering?, 7. Stakeholder analysis and mapping, 8. Cooperative game theory, 9. Social design principles, the S-principles, 10. Towards coalition building. Parallel to this course content the students will work on a group design assignment, supported by discussions in the weekly ateliers.	
Study Goals	In this course students will learn the Building with Nature ecosystem-based design concept and its applications in water and coastal systems. More specifically, they will deepen their knowledge of hydraulic engineering principles (H), ecological principles (E) and social design principles (S) to develop more effective and sustainable nature based hydraulic infrastructure. Learning objectives include: 1. Evaluate whether a hydraulic infrastructure is meeting engineering, ecosystem and societal goals 2. Evaluate the suitability of potential stakeholder coalitions for designing and implementing a particular Building with Nature project 3. Design a BwN solution for a coastal or riverine challenge based on engineering design principles, ecological design principles, and social design principles relevant to the Building with Nature approach. These learning objectives clearly focus on designing nature based solutions in complex socio-technical systems, and on design for viable solutions in a multi-actor space. The additional complexity of taking interactions with and in the environment into account make this a challenging and relevant course.	
Education Method	This is a blended course, run in combination with MOOC material on EdX, namely: Engineering - Building with Nature 1x and Beyond Engineering - Building with Nature 2x. Discussions, tutorials and design assignments will be supported in weekly class meetings or ateliers.	
Computer Use	Standard TPM software.	
Books	e-Reader: Building with Nature & Beyond (Slinger, 2021). Freely downloadable from TU Delft Open Publishing.	
Prerequisites	STEM Bachelor degree.	
Assessment	The final grade is an average of the following 2 grades: 1. Evaluation of integrated BwN designs and designs for social engagement (by applying the H-E and S-principles in exercises in the MOOCs and interactive e-book). 2. Final BwN & Beyond Design Assignment (using the S-H-E principles). Students must attain a minimum of 5,5 for the two components of the course to pass the entire course. Only the final grade is entered in OSIRIS.	
Enrolment / Application	This course can be chosen as an elective.	
Special Information	This course distills and applies principles for Building with Nature design taking the ecological and social context into account in the process of developing nature based solutions. It differs from the iterative design cycle approach underpinning the course CIE9154 Building with Nature in Hydraulic Engineering, which is intended for students with a civil engineering bachelor study. This course is open to a wider student group and explores what Building with Nature is, and how nature based solutions can be developed for different stakeholder settings and diverse ecosystems. Students who complete the course will be able to apply the S-H-E principles in designing context appropriate nature based solutions.	
Elective	Yes	
Tags	Design Integrated Policy Analysis Project Sustainability Water Engineering Water management	
Targetgroup	The primary target group is those interested in Nature-based Solutions. This is formed by masters students in the Engineering and Policy Analysis master with an interest in Water and Delta Systems management as well as students from the CoSEM and MoT masters. The course will be open to students from other faculties, and is anticipated to attract attention from CiTG students, particularly masters students in Hydraulic Engineering, Coastal Modelling and Management, Water Management and Environmental Engineering. Students from other masters programmes who have systems engineering, environmental systems knowledge, or planning competence in coastal, wetland and/or riverine environments are also eligible.	

TPM804A	Adaptive Planning for Management of Water Systems in Deltas	5
Module Manager	Dr.ir. M.P.M. Ruijgh-van der Ploeg	
Instructor	Dr.ir. M.P.M. Ruijgh-van der Ploeg	
Instructor	Dr.ir. L.M. Hermans	
Co-responsible for assignments	Dr.ir. L.M. Hermans	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2A	
Start Education	2A	
Exam Period	2A 3A	
Course Language	English	
Expected prior knowledge	This course assumes that you are able to apply (1) skills in conceptual system modeling from a multi-stakeholder perspective and (2) insight in governance of water systems in deltas (as taught in course TPM801)	
Course Contents	Society is vulnerable to unexpected or difficult to predict events that can boost, but more often damage social and economic activities and/or the natural environmental. Can we prepare for these events given that their frequency, scope, and impact are uncertain and that climate change exacerbates the uncertainties? What measures may we take top down to protect the functionality, reliability, and/or the returns on investments of existing infrastructures and other built structures? And, while we plan for dealing with climate change (or other uncertain contextual factors such as the economy or political stability) what processes do we need to take into account in relation to the preferred locations for housing and production? Researchers and policy-makers are deeply engaged with the questions. In this course we study their contributions with regard to coping with water-related, deeply uncertain events with large scale, uncertain impacts. Examples in this course will relate to floods, droughts and sea level rise. You will delve into literature that helps you understand the concepts and tools that play a role in planning and decision-making under deep uncertainty. We will discuss how these tools differ from conventional planning methods and under what conditions they may be useful. One important tool that is being introduced in the water-domain is the adaptation pathway. You will learn to develop adaptation pathways. We will challenge you to critically reflect on the incentives that these pathways create for economic behavior. And you will learn how to set up an implementation monitoring plan that reflects the interests of the multiple stakeholders that are involved.	
Study Goals	After taking this course students are able to: 1. Discuss important concepts - such as uncertainty, scenarios, options, system failure, economic behavior, vulnerability, robustness, resilience, adaptive capacity, and adaptation - and explain how they are related 2. Articulate and analyse choices in long-term planning or decision-making under uncertainty 3. Apply a structured method to develop scenarios for plausible but uncertain futures 4. Design an adaptation pathway and evaluate its future performance from different stakeholder perspectives 5. Design a plan for monitoring of the implementation of adaptation pathways 6. Make recommendations on how to deal with the strengths and weaknesses of adaptation pathways during their implementation	
Education Method	Weekly lectures are used to discuss reading materials. Workshops are organized in support of the individual assignment and address practicing skills in: conceptual uncertainty analysis, creating adaptation pathways, testing the effectiveness of these pathways, and developing monitoring pathway implementation.	
Assessment	One written, close book exam with short answer and essay questions and one individual assignment. If the exam must be administered as an on-line exam (due to TU wide decision), the exam will be an open book exam. The final grade is based on the results of the exam (70%, grade >5,5) and the assignment (30%, grade >5,5). An exemplar of the exam and a rubric with assessment criteria for the assignment will be published on Brightspace.	
Elective	Yes	

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